Dissecting the Aggregate Market Elasticity*

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May 5, 2026

Abstract

We study aggregate stock market elasticity in a general equilibrium model with heterogeneous investors, passive demand, and financial constraints. Without frictions, the elasticity of the endowment claim is infinite, as interest rate and risk premium responses offset each other. With frictions, price impact for the endowment claim remains modest (about 0.7 in our calibration). In contrast, the equity (dividend) claim exhibits large price impact (above 8), consistent with empirical evidence, as frictions dampen interest rate responses while leverage amplifies risk premia responses to portfolio flows. We introduce a state-global perturbation method that yields closed-form, state-dependent elasticities. Calibrated to match the magnitude of portfolio flows, the model simultaneously matches the equity premium, return volatility, and the level and countercyclical dynamics of price impact.

Keywords: Aggregate market elasticity, risk misallocation, demand shocks, demand shifters, excess volatility, asset pricing.

JEL Classification: G12, G11, G21.

*We thank Andy Atkeson, Tim Johnson, Leonid Kogan, Ralph Koijen, Tyler Muir, Stavros Panageas, Yinan Su, Fabrice Tourre, and Pierre-Olivier Weill; and conference and seminar participants at AFA 2026 Annual Meeting, Liquidity in Macroeconomics Workshop at St. Louis Fed, UIUC, UCLA macro-finance lunch, and University of Zurich for valuable comments and suggestions. We thank Rotman-FinHub for computational resources. All errors are our own.

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1 Introduction

Why is the stock market so volatile? This question has long been central to asset pricing research. Classic tests of excess volatility attribute the puzzle to unobserved "dark matter" factors.[1] More recent evidence points instead to a demand-based channel: asset prices respond strongly to portfolio flows. Yet these flows are small relative to market capitalization, implying that markets must be highly *inelastic*, with even modest quantity shifts generating large price changes (Gabaix and Koijen, 2023). Moreover, the ratio of return volatility to flows—the "volatility multiplier"—is itself time-varying and countercyclical, rising sharply during periods of financial stress. This pattern points to a state-dependent relationship between prices and portfolio flows.

These facts raise a central question. How can small portfolio flows generate large price movements in equilibrium? More specifically, what determines the *aggregate* elasticity of the stock market, why does price impact vary over time, and which frictions are essential for matching its magnitude in the data?

To address these questions, we develop a general equilibrium model with heterogeneous investors, passive demand, and financial constraints. We study the *aggregate (macro) elasticity*, defined as the change in total stock market value in response to a \$1 flow between (short-term) bonds and stocks. Unlike micro elasticities, which capture substitution across individual stocks, aggregate elasticity reflects market-wide reallocations between risky and risk-free assets. As a result, understanding aggregate elasticity requires tracking how the risk-free rate and the risk premium respond jointly to portfolio flows.

This joint response is central for the elasticity of the *endowment claim*, the claim on aggregate output. We show that, in the absence of frictions, the aggregate market elasticity of this claim is *infinite*, regardless of how price-inelastic individual investors are.[2] Consider an outflow of funds from stocks to bonds. The resulting increase in the risk premium tends to lower stock prices. At the

[1]See LeRoy and Porter (1981), Shiller (1981, 1992), and Cochrane (1992); and Chen, Dou, and Kogan (2024) for a recent discussion.

[2]A portfolio-flow shock is an asymmetric shock to risky-asset demand, affecting only a subset of investors and triggering portfolio reallocation in equilibrium.

same time, the associated decline in wealth would reduce consumption. With a fixed endowment, this adjustment cannot occur in equilibrium. The interest rate therefore falls to clear the goods market. In frictionless economies, it adjusts until it exactly offsets the increase in the risk premium, leaving the price of the risky asset unchanged. This result holds regardless of how inelastic individual demand is. Aggregate elasticity is therefore infinite even when individual demands are highly inelastic.

This result highlights a central distinction between individual and aggregate elasticities. If cross-price effects are absent, so that demand for risky assets does not respond to the interest rate, aggregate elasticity reduces to a weighted average of individual elasticities, as in Gabaix and Koijen (2023). Once interest rates adjust endogenously, however, aggregate elasticity is governed by general equilibrium forces rather than by the average slope of individual demands. In particular, aggregate elasticity is governed by how portfolio flows affect saving incentives. When flows do not alter these incentives, movements in the risk-free rate fully offset movements in the risk premium, so asset prices remain unchanged.

Aggregate elasticity becomes finite when risk is misallocated. Portfolio reallocation then alters saving incentives, so interest rates can no longer fully offset changes in risk premia.[3] This generates a nonzero price impact, even in general equilibrium. Frictions thus reduce aggregate elasticity. Even with frictions—such as passive demand and leverage constraints—the near-cancellation between interest rates and risk premia remains quantitatively important, so the endowment claim exhibits only a modest price impact of 0.7 in our calibration. We establish these results first in a transparent two-period model, then extend them to a fully dynamic setting.

Studying aggregate elasticity in models with frictions is challenging because closed-form solutions are typically not available. We develop a tractable state-global perturbation method that yields closed-form, state-dependent elasticities. Elasticity is time-varying and shaped by both the wealth distribution and risk allocation across investors. Passive demand, preference heterogeneity, and

[3] Intuitively, risk misallocation means investors hold too much or too little risk relative to the efficient allocation. Investors who are already overexposed to risk are less willing to save in assets that further increase their risk exposure. As a result, interest rates do not need to adjust as much to restore equilibrium after a portfolio-flow shock.

leverage constraints each amplify price impact via risk allocation. Excess volatility results from the interaction of portfolio-flow shocks with finite (state-dependent) elasticity. Neither inelasticity nor flows alone generate excess volatility; both are required.

Our second main contribution is to provide a characterization of the aggregate elasticity of the equity claim. Dividends are modeled as a time-varying share of output, as in Menzly, Santos, and Veronesi (2004). In contrast to the endowment claim, the equity claim exhibits substantially larger price impact, with a multiplier above 8 in our calibration, consistent with Gabaix and Koijen (2023) and Koijen, Richmond, and Yogo (2024). This difference arises because the general equilibrium offset between the risk-free rate and the risk premium that renders the endowment claim highly elastic does not extend to the dividend claim. Dividends are a levered share of aggregate output, so the dividend risk premium responds more strongly to portfolio flows than the endowment risk premium. Consequently, changes in risk premia are not fully offset, so prices respond strongly to portfolio flows. This result highlights that modeling the dividend claim, rather than aggregate consumption as is standard in macro-finance, is essential for generating realistic price impact.

We provide a tractable way to incorporate the dividend claim in a dynamic heterogeneous-agent economy with frictions. We establish conditions under which a version of the mutual fund theorem holds in our economy with frictions, so investors optimally hold risky claims at market-capitalization weights. This implies that the stochastic discount factor from the aggregate (endowment-claim) economy can be used to price the dividend claim in a second step. As a result, the problem separates into an aggregate equilibrium and a simpler pricing problem, reducing its dimensionality.

For the quantitative analysis, we solve the full model using the deep neural network method of Duarte, Duarte, and Silva (2024) and calibrate it to Flow of Funds data. We target the magnitude of portfolio flows observed in the data. The model matches both the level and countercyclical dynamics of price impact, while reproducing key asset pricing moments such as the equity premium and return volatility. These results reflect the joint role of frictions, which dampen interest rate responses, and leverage, which amplifies risk premia responses to portfolio flows.

Related literature

Our work relates to the literature on macro demand elasticity. The papers closest to ours are Johnson (2006), who finds finite elasticity even in a frictionless Lucas economy by perturbing risky asset supply, and Gabaix and Koijen (2023), who introduce a behavioral element that fixes interest rates and generates large price impacts. We build on this literature by developing a GE model with heterogeneous investors and financial frictions, showing that both GE effects and constraints are central for obtaining inelastic markets.[4] This contrasts with the larger body of work on micro elasticity, which examines relative price changes across individual stocks (e.g., Shleifer, 1986; Harris and Gurel, 1986; Chang, Hong, and Liskovich, 2015; Pavlova and Sikorskaya, 2023; Schmickler, 2020) and finds much larger elasticities, consistent with stocks being closer substitutes for each other than for bonds.

We also connect to demand-system asset pricing (Koijen and Yogo, 2019) and its extensions incorporating cross-asset elasticities (Fuchs, Fukuda, and Neuhann, 2023; Haddad, He, Huebner, Kondor, and Loualiche, 2025a). Our framework shows that even small cross-elasticities between equities and bonds generate large effects on macro elasticities, and extends the analysis to dynamic settings where the persistence of shocks and the evolution of risk premia shape both short-run and long-run adjustments (Binsbergen, David, and Opp, 2025; He, Kondor, and Li, 2025; Davis, Kargar, Li, and Silva, 2026). Our model of passive investors also connects to the literature on intermittent rebalancing (Chien, Cole, and Lustig, 2012), where sluggish portfolio adjustments are themselves an important source of aggregate inelasticity. Finally, our paper relates to preferred-habitat models (Vayanos and Vila, 2021; Kekre, Lenel, and Mainardi, 2025; Ray, Droste, and Gorodnichenko, 2024), which emphasize intermediary frictions and market segmentation. By contrast, our framework highlights wealth effects as a distinct channel through which shifts in aggregate wealth alter portfolio demand and amplify price movements.

[4] For more on macro elasticity, see Johnson (2008, 2009), Deuskar and Johnson (2011), Li, Pearson, and Zhang (2020), Hartzmark and Solomon (2025), among others.

2 A Simple Model of the Aggregate Market Elasticity

In this section, we analyze the determination of the aggregate market elasticity in a simple general equilibrium, demand-based asset pricing model. To keep the discussion transparent, we specify investor demands directly, rather than deriving them from utility functions. A micro-founded version of these demands, embedded in a dynamic heterogeneous-agents model, will be presented in Section 3.

Environment. We consider a two-period economy with two assets: a risky asset and a riskless bond. There are two types of investors: passive (p) and active (a). A fraction ω_j of the population is of type $j \in \{a, p\}$. The risky asset pays a random dividend Y' in the second period. Its price is denoted by P, while the price of the riskless asset is R_f^{-1}.

The budget constraints of investor j are

$$PQ_j + R_f^{-1}B_j + C_j = W_j, \qquad C_j' = Y'Q_j + B_j,$$

where Q_j is the investor's holdings of the risky asset, B_j denotes holdings of the riskless bond, C_j is first-period consumption, and C_j' is second-period consumption. Initial wealth is given by $W_j = (P + Y)Q_{j,-1} > 0$.

We define investor j's portfolio share in the risky asset as

$$\alpha_j \equiv \frac{PQ_j}{W_j - C_j}.$$

Passive investors maintain a fixed risky asset share $\alpha_p = \overline{\alpha}_p \geq 0$. Active investors' portfolio share, in contrast, depends on the risk premium

$$\alpha_a = \overline{\alpha}_a(\pi),$$

for some increasing function $\overline{\alpha}_a(\cdot)$, where $\pi \equiv \log \frac{1}{R_f} \mathbb{E}\left[\frac{Y'}{P}\right]$ denotes the risk premium.

Consumption is specified as

$$C_j = \overline{c}_j(r, \pi) W_j,$$

where $r \equiv \log R_f$ is the log risk-free rate. The consumption-wealth ratio $\overline{c}_j(\cdot)$ depends on both r and π. If $\overline{c}_j$ decreases with r, investors save more when the interest rate is high (a substitution effect); if $\overline{c}_j$ increases with r, investors save less (an income effect). Similar logic applies to the risk premium. We assume that $\overline{c}_j(\cdot)$ does not depend on the passive portfolio share $\overline{\alpha}_p$.

The market clearing conditions are

$$\sum_{j\in\{p,a\}} \omega_j C_j = Y, \qquad \sum_{j\in\{p,a\}} \omega_j Q_j = 1, \qquad \sum_{j\in\{p,a\}} \omega_j B_j = 0,$$

with initial risky asset endowment $\sum_{j\in\{p,a\}} \omega_j Q_{j,-1} = 1$.

Market for risky assets. Let $p = \log \frac{P}{Y}$ denote the log price-dividend ratio, and let $\mu \equiv \log \frac{\mathbb{E}[Y']}{Y}$ denote the log dividend growth. The risk premium is

$$\pi = \mu - p - r. \tag{1}$$

Investor j's demand for the risky asset is

$$Q_j = \alpha_j \left[1 - \overline{c}_j(r, \pi)\right] \frac{W_j}{P}.$$

Using Equation (1) to eliminate π, we can express demand as a function of price of the risky asset, the risk-free rate, and, for passive investors, the exogenous portfolio share: $Q_j = \overline{Q}_j(p, r, \overline{\alpha}_p)$.

The market clearing condition for the risky asset is therefore

$$\underbrace{\omega_a \overline{Q}_a(p, r, \overline{\alpha}_p)}_{\text{active demand}} = \underbrace{1 - \omega_p \overline{Q}_p(p, r, \overline{\alpha}_p)}_{\text{net supply}}. \tag{2}$$

Equilibrium in the risky asset market requires that active investors absorb the net supply, defined as total supply minus the amount held by passive investors. Since active demand does not depend on $\overline{\alpha}_p$, changes in the passive portfolio share affect only the net supply curve.

Market for goods. The response of the consumption-wealth ratio to the interest rate and the risk premium is central to our analysis. In the dynamic model of Section 3, the average consumption-wealth ratio depends only on the sum $(r+\pi)$, so its sensitivity to interest rates equals its sensitivity to risk premia. Assumption 1 imposes the same structure in this two-period setting, ensuring consistency with standard micro-founded models.

Assumption 1. *The average consumption-wealth ratio is a function of the expected return on the risky asset,* $(r+\pi)$*,*

$$\sum_{j\in\{p,a\}} x_j \overline{c}_j(r,\pi) = \overline{c}(r+\pi),$$

for some function $\overline{c}(\cdot)$*, where* $x_j \equiv \frac{\omega_j W_j}{\omega_a W_a + \omega_p W_p}$ *denotes investor* j*'s wealth share.*

Intuitively, since bonds are in zero net supply, the return on investors' overall portfolios equals the return on the risky asset. From goods market clearing and Equation (1), we obtain

$$\overline{c}(\mu - p) = \frac{1}{1+e^p}, \tag{3}$$

using the fact that $r + \pi = \mu - p$ and $\sum_j x_j \overline{c}_j = \frac{Y}{Y+P}$.

Assumption 1 implies that the system of equations determining p and r has an important *recursive property*: condition (3) depends only on p, while condition (2) depends on both p and r.

2.1 General equilibrium implications of portfolio flows

We now examine how the price-dividend ratio p and the riskless return r respond to a portfolio flow from the riskless bond into the risky asset. Let $\overline{\alpha}_p^\star$ denote the passive portfolio share in the initial equilibrium, with $(p^\star, r^\star)$ the corresponding equilibrium prices.

For a small perturbation, $\overline{\alpha}_p = \overline{\alpha}_p^\star e^{\hat{\alpha}_p}$, we linearize demand around the initial equilibrium. Defining $\hat{p} \equiv p - p^\star$, $\hat{r} \equiv r - r^\star$, and $q_j \equiv \log(Q_j/Q_j^\star)$, we obtain

$$q_j = -\zeta_{j,p}\,\hat{p} - \zeta_{j,r}\,\hat{r} + f_j, \tag{4}$$

where the own-price and cross-price elasticities are given by $\zeta_{j,p} \equiv -\frac{\partial \log \overline{Q}_j}{\partial p}$ and $\zeta_{j,r} \equiv -\frac{\partial \log \overline{Q}_j}{\partial r}$. The *flow shock* is given by

$$f_j \equiv \frac{\partial \log \overline{Q}_j}{\partial \log \overline{\alpha}_p} \times \hat{\alpha}_p,$$

capturing an exogenous reallocation from bonds to stocks. Since active demand does not depend on $\overline{\alpha}_p$, we have $f_a = 0$. Formally, a flow shock is an asymmetric demand shock affecting some investors but not others, inducing equilibrium portfolio rebalancing.

An important implication of (4) is that demand for the risky asset depends on both the price-dividend ratio p and the bond price through the risk-free rate r. Since active investors respond to the risk premium $\pi = \mu - p - r$, their demand depends on both prices. Therefore, the prices of the risky and risk-free assets must be jointly determined in equilibrium.

Equilibrium. Let $s_j \equiv \omega_j Q_j^\star$ denote the equilibrium quantity share of the risky asset held by type j. Linearizing the market-clearing condition $\sum_j \omega_j Q_j = 1$ gives

$$s_a\left(-\zeta_{a,p}\,\hat{p} - \zeta_{a,r}\,\hat{r}\right) = s_p\left(\zeta_{p,p}\,\hat{p} + \zeta_{p,r}\,\hat{r} - f_p\right), \tag{5}$$

as $s_a q_a + s_p q_p = 0$.

The left panel of Figure 1 shows the equilibrium in the risky-asset market. The downward-sloping curve represents active demand, while the upward-sloping curve is net supply, i.e., total supply minus passive holdings.[5]

[5]Passive demand is $Q_p = \overline{\alpha}_p\left[1 - c_p(r,\pi)\right]\frac{W_p}{P}$, given $W_p = (P+Y)Q_{p,-1}$ and $\pi = \mu - p - r$, so its linearized contribution $-s_p q_p$ makes net supply upward sloping in p provided $\frac{\partial c_p}{\partial p}$ is not too large.

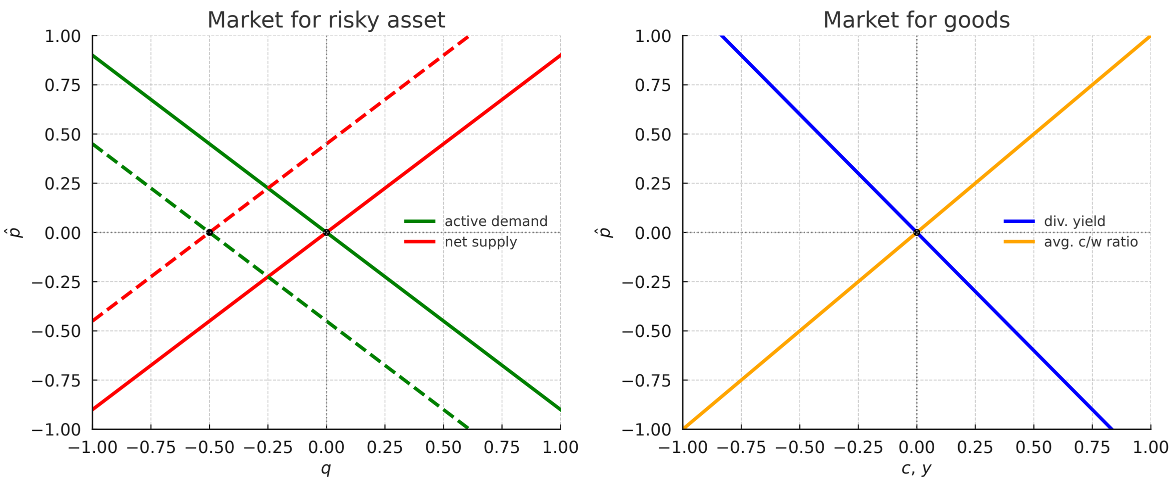


Figure 1. Equilibrium in the risky asset (left) and goods (right) markets.

Linearizing the goods-market condition $\bar{c}(\mu - p) = 1/(1 + e^p)$ around $p^\star$ yields

$$\chi_p\, \hat{p} \;=\; -\frac{e^{p^\star}}{(1 + e^{p^\star})^2}\, \hat{p}, \qquad \text{where} \qquad \chi_p \equiv -\bar{c}'(\mu - p^\star). \tag{6}$$

The right panel of Figure 1 shows the goods market. We focus on the case $\chi_p > 0$, so the average consumption-wealth ratio is increasing in p (investors save less when expected returns are low). The right-hand side in (6) is the (local) slope of the dividend-yield curve, which is downward sloping in p. By the recursive property, only p enters the goods market clearing condition.

The inelastic markets hypothesis. Consider a *partial equilibrium* version of the model in which we hold the interest rate fixed at $r = r^\star$ and ignore goods market clearing. Using the linearized risky asset market condition (5) with $\hat{r} = 0$, the price of the risky asset satisfies

$$\hat{p} \;=\; \underbrace{\frac{1}{\zeta_p}}_{\substack{\text{inverse market}\\ \text{elasticity}}} \times \underbrace{f}_{\text{flow shock}}, \tag{7}$$

where

$$\zeta_p \;\equiv\; s_a\, \zeta_{a,p} + s_p\, \zeta_{p,p}, \qquad f \;\equiv\; s_p\, f_p, \qquad s_j \equiv \omega_j Q_j^\star.$$

Equation (7) shows that the price-dividend ratio responds to flow shocks, with price impact given by the inverse of the (aggregate) market elasticity—an average of investors' own-price elasticities. This is the core implication of the *inelastic markets hypothesis* of Gabaix and Koijen (2023), who estimate large price responses to stock-market flows, and study a model where investors are relatively inelastic.

In the risky asset panel of Figure 1, an increase in the portfolio share of passive investors shifts the net supply curve to the left. The new equilibrium lies further up the active demand curve, implying lower expected returns and a higher price-dividend ratio. When demand is inelastic, even small inflows to stocks generate a sizable increase in $\hat{p}$.

The crucial role of cross-elasticity. The partial equilibrium analysis abstracts from cross-elasticity. In general equilibrium, however, both r and p adjust, and the aggregate market elasticity is fundamentally different, as the next proposition shows.

Proposition 1 (Infinite market elasticity). *Define the aggregate cross-elasticity $\zeta_r \equiv s_a\,\zeta_{a,r} + s_p\,\zeta_{p,r}$. If $\zeta_r \neq 0$, then a portfolio-flow shock has* no *price impact: $\hat{p} = 0$. The shock is absorbed by offsetting movements in the interest rate and the risk premium:*

$$\hat{r} = \frac{1}{\zeta_r} f, \qquad \hat{\pi} = -\frac{1}{\zeta_r} f,$$

where $\hat{\pi} \equiv \pi - \pi^\star$.

Proof. From Equation (6), we obtain $\hat{p} = 0$. If $\zeta_r \neq 0$, then $\hat{r} = \frac{1}{\zeta_r} f$ from Equation (5). Since $\hat{\pi} = -\hat{p} - \hat{r}$ with $\hat{p} = 0$, it follows that $\hat{\pi} = -f/\zeta_r$. $\square$

With any nonzero cross-elasticity $\zeta_r \neq 0$, the aggregate market becomes infinitely elastic, regardless of how inelastic individual demands may be. Even if $\zeta^q_{j,p} \approx 0$ for all j, a small but positive ζ_r is sufficient to make aggregate elasticity unbounded.

The disconnect between individual and aggregate elasticities arises from a general equilibrium effect. In the goods market (right panel of Figure 1), the price-dividend ratio is determined by consumption behavior. This mechanism is well known in the macro-finance literature, most clearly in models with unit EIS, where p is tied directly to the discount rate. Our framework generalizes this logic. To restore equilibrium in the risky asset market (left panel), the interest rate adjusts so that active demand equals net supply at the *original price*. Graphically, this appears as the dashed active-demand curve shifting to intersect the new net-supply curve at the initial price.

Consider a positive flow shock to the risky asset (e.g., funds shift toward stocks, so $f > 0$). Lower bond demand increases r, while increased demand for the risky asset reduces the risk premium. In equilibrium these effects offset exactly, leaving the price-dividend ratio unchanged. If the rise in r were smaller than the fall in the risk premium, there would be excess demand for goods (equivalently, excess bond supply). Simultaneous clearing of risky and risk-free asset markets therefore requires infinite aggregate elasticity in this model.

The need for a micro-founded demand system. The analysis underscores the central role of cross-price elasticity. When allowed to adjust, the interest rate has the potential to fully offset the impact of portfolio flows on the price-dividend ratio. The aggregate market can be infinitely elastic—even if individual investors are inelastic—so long as the average consumption-wealth ratio is unaffected by the flow shock f. When the flow shock shifts both the net-supply curve and the average consumption-wealth ratio, prices adjust and aggregate elasticity becomes finite.[6]

Understanding aggregate elasticity therefore requires a framework that links portfolio reallocation shocks to both risky asset demand and bond demand (i.e., savings behavior). To this end, we develop a dynamic heterogeneous-agent asset pricing model and introduce a new methodology for deriving investor demand within this setting.

[6]An analogous argument applies to uncertainty. With unit EIS, the consumption-wealth ratio is fixed, so higher uncertainty leaves prices unchanged. For uncertainty to reduce prices, EIS must exceed one so that higher uncertainty lowers the consumption-wealth ratio (see Bansal and Yaron, 2004).

3 A Dynamic Asset Pricing Model with Passive Investors

We now develop a dynamic heterogeneous-agent asset pricing model that provides a micro-founded demand system. The economy extends the framework of Section 2 by embedding passive and active investors in a full general-equilibrium setting.

Passive investors hold exogenously specified, stochastic portfolio shares in risky assets. Shocks to these portfolio shares generate portfolio flow (i.e., noise trading) shocks that are imperfectly correlated with cash flow shocks, so the economy is driven by both fundamental and non-fundamental demand shocks. Active investors, in contrast, optimally rebalance their portfolios subject to financial frictions. We model two types of active investors that differ in risk aversion: a high risk aversion (unconstrained) investor and a low risk aversion investor subject to a leverage constraint.

Unlike in the simple model in Section 2, investors now trade two distinct claims, a *dividend claim* (equity) and a *risky corporate bond*, whose payoffs sum to the aggregate endowment. Modeling the dividend process explicitly lets us study the response of equity prices, rather than of the aggregate claim, to portfolio flows. As we show below, this distinction is essential: the price impact of the dividend claim differs substantially from that of the aggregate claim and is the key to matching the data.

3.1 Environment

Financial markets. We begin by specifying the aggregate endowment process. The aggregate endowment Y_t follows a geometric Brownian motion:

$$\frac{dY_t}{Y_t} = \mu \, dt + \sigma \, dZ_t, \tag{8}$$

where $\mu \in \mathbb{R}$ and $\sigma \in \mathbb{R}^{1\times d}$ are constants.[7] The process $Z = \{Z_t\}_{t\geq 0}$ is a standard d-dimensional Brownian motion on the filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ satisfying the usual conditions. In the baseline model, $d = 3$, capturing three sources of risk: (i) aggregate endowment shocks, (ii)

[7]Throughout the paper, we adopt the convention that loadings on the Brownian motion are row vectors.

portfolio-flow shocks to passive investors, and (iii) shocks to the dividend share.

We consider a representative firm that issues two securities whose payoffs sum to the aggregate endowment. The first is a *dividend claim* (equity), with payoff flow $D_t = s_t Y_t$. Following Menzly et al. (2004), the dividend share $s_t \in (0, 1)$ follows the mean-reverting process:

$$ds_t = \kappa_s(\bar{s} - s_t)\, dt + s_t(1 - s_t)\, \sigma_s\, dZ_t, \tag{9}$$

given $s_0 \in (0, 1)$, with $\bar{s} \in (0, 1)$, $\kappa_s > 0$, and $\sigma_s \in \mathbb{R}^{1\times d}$. This specification ensures $s_t \in [0, 1]$ at all times and yields a stationary share process, so consumption and dividends are cointegrated. As a result, dividends inherit both aggregate risk and additional variation from movements in the dividend share. By Itô's lemma, dividends follow:

$$\frac{dD_t}{D_t} = \mu_{D,t}\, dt + \sigma_{D,t}\, dZ_t,$$

where

$$\sigma_{D,t} = \sigma + (1 - s_t)\, \sigma_s, \qquad \mu_{D,t} = \mu + \kappa_s\left(\frac{\bar{s}}{s_t} - 1\right) + (1 - s_t)\, \sigma\sigma_s^\top.$$

This specification captures several key features of the data. First, it generates the well-documented fact that dividends are more volatile than consumption (see, e.g., Abel 1999; Campbell 1999). Second, the dividend share predicts future dividend growth, consistent with Menzly et al. (2004). Third, dividend volatility increases after negative shocks to s_t, capturing a leverage effect.

The second claim is a *corporate bond* (risky debt), with payoff flow $B_t = (1 - s_t)Y_t$. Since $D_t + B_t = Y_t$, the bond payoff is smoother than aggregate consumption, reflecting the safer nature of debt and its lower exposure to aggregate risk. This decomposition allows us to distinguish between claims with different risk exposures, with important implications for price impact.

Let $P_{D,t}$ and $P_{B,t}$ denote the ex-dividend prices of the dividend and bond claims, with returns:

$$dR_{D,t} = \frac{dP_{D,t} + D_t\,dt}{P_{D,t}} = \mu_{R_D,t}\,dt + \sigma_{R_D,t}\,dZ_t,$$
$$dR_{B,t} = \frac{dP_{B,t} + B_t\,dt}{P_{B,t}} = \mu_{R_B,t}\,dt + \sigma_{R_B,t}\,dZ_t.$$

Since $D_t + B_t = Y_t$, the combined claim replicates the aggregate endowment, with price $P_{Y,t} = P_{D,t} + P_{B,t}$. The return on the aggregate claim follows:

$$dR_{Y,t} = \frac{dP_{Y,t} + Y_t\,dt}{P_{Y,t}} = \mu_{R_Y,t}\,dt + \sigma_{R_Y,t}\,dZ_t.$$

Investors also have access to a risk-free asset paying rate r_t and, to ensure markets are dynamically complete, a derivative in zero net supply that spans risks not directly traded through the dividend and bond claims. In particular, we assume the derivative does not load on dividend-share shocks. As in a futures contract, there is no initial cash exchange to enter into the derivative contract. The derivative provides risk exposure $\sigma_{R_F,t} \in \mathbb{R}^{1\times d}$, orthogonal to the aggregate claim, $\sigma_{R_F,t}\,\sigma_{R_Y,t}^\top = 0$, so that it provides independent risk exposure, with normalization $\|\sigma_{R_F,t}\| = \|\sigma_{R_Y,t}\|$.

Stochastic discount factor. Absence of arbitrage implies the existence of a stochastic discount factor (SDF), Λ_t, which prices all assets in the economy:

$$\frac{d\Lambda_t}{\Lambda_t} = -r_t\,dt - \eta_t\,dZ_t,$$

where $\eta_t \in \mathbb{R}^{1\times d}$ is the market price of risk vector.

Absence of arbitrage implies that risk premia satisfy $\pi_{k,t} = \sigma_{R_k,t}\,\eta_t^\top$ for each claim $k \in$

$\{D, B, F\}$. Let Σ_t denote the 3×3 matrix collecting the risk exposures of all traded assets:

$$\Sigma_t = \begin{pmatrix} \sigma_{R_D,t} \\ \sigma_{R_B,t} \\ \sigma_{R_F,t} \end{pmatrix},$$

and $\pi_t \equiv [\pi_{D,t}, \pi_{B,t}, \pi_{F,t}]^\top$, so that $\pi_t = \Sigma_t \, \eta_t^\top$.

In equilibrium, prices depend on an aggregate state vector $X_t \in \mathbb{R}^N$, so we write $r_t = r(X_t)$, $\eta_t = \eta(X_t)$, and return coefficients as functions of X_t. The state vector evolves as

$$dX_t = \mu_{X,t} \, dt + \sigma_{X,t} \, dZ_t,$$

with drift $\mu_{X,t} \in \mathbb{R}^N$ and exposure matrix $\sigma_{X,t} \in \mathbb{R}^{N \times d}$, both determined in equilibrium.

Demography and preferences. The economy consists of three investor types, indexed by $j = 0, 1, 2$, with population mass ω_j: one passive type ($j = 0$) and two active types ($j = 1, 2$). Investors die with Poisson intensity κ, and each instant a mass $\kappa \, \omega_j$ of type-j agents are born, so the total population remains constant and normalized to one. Newborns inherit parental wealth, so wealth is redistributed across cohorts without affecting aggregate resources. This mortality structure ensures a nondegenerate stationary wealth distribution.

Preferences are recursive as in Duffie and Epstein (1992), with type-j risk aversion γ_j and a common elasticity of intertemporal substitution (EIS) ψ. The aggregator is given by:

$$f_j(C, V) = \rho \frac{(1 - \gamma_j) V}{1 - \psi^{-1}} \left[\left(\frac{C}{\left((1 - \gamma_j) V\right)^{\frac{1}{1-\gamma_j}}} \right)^{1-\psi^{-1}} - 1 \right],$$

where ρ denotes the discount factor.

Each investor chooses consumption $C_{j,t}$ and portfolio allocations across the dividend claim, the bond claim, and the derivative, subject to the constraints specified below.

Passive investors. Passive investors allocate a fraction $1-\overline{\alpha}_{p,t}$ of their wealth to the risk-free asset and a fraction $\overline{\alpha}_{p,t}$ to a *market index* fund, which holds the dividend and bond claims in proportion to their market-capitalization weights. Passive investors do not trade the derivative.

The share of wealth invested in the market index fund, $\overline{\alpha}_{p,t}$, evolves according to an Ornstein-Uhlenbeck process:

$$d\overline{\alpha}_{p,t} \;=\; \theta_p\big(\overline{\alpha}-\overline{\alpha}_{p,t}\big)dt \;+\; \sigma_{\alpha_p}\,dZ_t, \tag{10}$$

with long-run mean $\overline{\alpha} > 0$, mean-reversion $\theta_p > 0$, and loading $\sigma_{\alpha_p} \in \mathbb{R}^{1\times d}$ on the d-dimensional Brownian motion Z_t. Innovations to $\overline{\alpha}_{p,t}$ represent *portfolio flow shocks*, i.e., fluctuations in the fraction of wealth passive investors allocate to risky claims. As in Section 2, a flow shock is an asymmetric demand shock affecting only passive investors and triggering portfolio reallocation.

Portfolio flows may comove with cash-flow shocks or arise independently. We focous on the case where $\sigma = (\sigma_1, 0, 0)$ for aggregate endowment risk, $\sigma_{\alpha_p} = (\sigma_{\alpha_p,1}, \sigma_{\alpha_p,2}, 0)$ for the passive-share process, and $\sigma_s = (\sigma_{s,1}, 0, \sigma_{s,3})$ for the dividend share process. The first shock represents aggregate endowment risk, the second represents portfolio-flow shocks, and the third represents dividend-share shocks. We allow portfolio-flow shocks and dividend-share shocks to load on the aggregate endowment risk factor. This implies that aggregate endowment risk can be correlated with both portfolio flows and dividend-share shocks.

The exposure of the passive share to aggregate shocks can reflect *imperfect rebalancing* or *performance-induced flows*. Without rebalancing, a positive return on the risky asset mechanically increases the passive share when $\overline{\alpha}_{p,t} < 1$. Similarly, investors may increase risky exposure after good performance. This specification aligns with empirical evidence on passive behavior: households exhibit substantial portfolio inertia (Ameriks and Zeldes, 2004; Brunnermeier and Nagel, 2008); flows chase performance (Chevalier and Ellison, 1997; Dannhauser and Pontiff, 2024); and regulatory changes can sharply increase equity allocations via TDF adoption (Parker, Schoar, Cole, and Simester, 2022), illustrating portfolio flow shocks not tied to cash flow fundamentals.

Active investors. Investors of type $j = 1, 2$ are *active investors*. Active investors optimally rebalance their portfolios, choosing allocations across the dividend claim, the bond claim, and the derivative, $\alpha_{j,t} \equiv [\alpha_{j,D,t}, \alpha_{j,B,t}, \alpha_{j,F,t}]^\top$. Type-2 investors have low risk aversion, $\gamma_2 < \gamma_1$, and therefore have an incentive to lever up their exposure to risky assets. They borrow short-term to increase their risk exposure, subject to a leverage constraint that limits total portfolio risk:

$$\|\sigma_{j,t}\| \leq \overline{\sigma},$$

where $\sigma_{j,t} \equiv \alpha_{j,t}^\top \Sigma_t = \alpha_{j,D,t}\, \sigma_{R_D,t} + \alpha_{j,B,t}\, \sigma_{R_B,t} + \alpha_{j,F,t}\, \sigma_{R_F,t}$ is the investor's total risk exposure and $\| \cdot \|$ denotes the Euclidean norm. This constraint resembles a Value-at-Risk (VaR) limit commonly applied to banks and leveraged institutions (see, e.g., Adrian and Shin 2014 and Brunnermeier and Pedersen 2009). Type-1 investors are more risk averse and do not face a leverage constraint.

Investors' problem. An investor of type $j = 0, 1, 2$ chooses consumption $C_{j,t}$ and portfolio allocations $\alpha_{j,t}$ to solve the problem:

$$V_{j,t} = \max_{C_{j,t}, \alpha_{j,t}} \mathbb{E}_t \left[\int_t^\infty f_j(C_{j,s}, V_{j,s})\, ds \right], \qquad s.t. \tag{11}$$

$$\frac{dW_{j,t}}{W_{j,t}} = \left[r_t + \pi_t^\top \alpha_{j,t} - \frac{C_{j,t}}{W_{j,t}} \right] dt + \sigma_{j,t}\, dZ_t,$$

with $W_{j,t} \geq 0$ and $\alpha_{j,t} \in \Omega_{j,t}$, where $\sigma_{j,t} = \alpha_{j,t}^\top \Sigma_t$ is the investor's risk exposure, $W_{j,t}$ denotes the investor's wealth, and $\Omega_{j,t}$ is the investment set for type-j investor.

Portfolio allocations are restricted to lie in an *investment set* $\Omega_{j,t}$, which depends on investor type and captures differences in financial constraints across investors.[8] For passive investors, $j = 0$, holdings of the market index fund induce indirect holdings of the dividend and bond claims, so

[8]The differences in investment sets here are reminiscent of the difference in the investment universe emphasized by Koijen and Yogo (2019).

their investment set is given by:

$$\Omega_{0,t} = \{\alpha_{0,t} : \ \alpha_{0,D,t} = \overline{\alpha}_{p,t} \frac{P_{D,t}}{P_{Y,t}}, \ \alpha_{0,B,t} = \overline{\alpha}_{p,t} \frac{P_{B,t}}{P_{Y,t}}, \ \alpha_{0,F,t} = 0\},$$

Type-1 investors are unconstrained, so their investment set is given by $\Omega_{1,t} = \mathbb{R}^3$. For type-2 investors, the investment set reflects their leverage constraint:

$$\Omega_{2,t} = \{\alpha_{2,t} : \|\Sigma_t^\top \alpha_{2,t}\| \leq \overline{\sigma}\}.$$

Market clearing and equilibrium. We define equilibrium as follows.

Definition 1. *A competitive equilibrium consists of stochastic processes, all adapted to the filtration generated by Z_t. These include the aggregate endowment Y, the dividend share s, the prices P_D and P_B, the risk-free rate r, the market price of risk η_t, the weight $\overline{\alpha}_p$, and, for each investor $j \in \{0, 1, 2\}$, wealth W_j, consumption C_j, and portfolio allocations α_j. These processes satisfy the following conditions:*

(i) The aggregate endowment Y_t evolves according to (8), with $Y_0 > 0$. The dividend share s_t evolves according to (9).

(ii) Given $\{P_{D,t}, P_{B,t}, r_t, \eta_t\}$ and the passive share $\overline{\alpha}_{p,t}$, the choices $(C_{j,t}, \alpha_{j,t})$ solve investor j's problem in (11). The weight on the market index fund, $\overline{\alpha}_{p,t}$, evolves according to (10).

(iii) Markets clear for goods, the dividend and bond claims, riskless bonds, and the derivative:

$$\sum_{j=0}^{2} \omega_j C_{j,t} = Y_t, \qquad \sum_{j=0}^{2} \omega_j Q_{j,k,t} = 1, \qquad \sum_{j=0}^{2} \omega_j B_{j,t} = 0, \qquad \sum_{j=0}^{2} \omega_j F_{j,t} = 0,$$

for $k \in \{D, B\}$. Holdings are defined as follows. The number of units of the dividend and bond claims held by investor j are $Q_{j,D,t} = \alpha_{j,D,t} W_{j,t} / P_{D,t}$ and $Q_{j,B,t} = \alpha_{j,B,t} W_{j,t} / P_{B,t}$. Riskless bond holdings are given by $B_{j,t} = (1 - \alpha_{j,D,t} - \alpha_{j,B,t}) W_{j,t}$. The notional position in *the derivative is* $F_{j,t} = \alpha_{j,F,t} W_{j,t}$.[9]

3.2 Equilibrium characterization

We now characterize equilibrium in the dynamic model. We derive the optimality conditions that govern portfolio choice and consumption decisions, and show how they jointly determine prices and risk premia. We then establish a version of the *mutual fund theorem* in our economy with frictions. This result implies that investors hold the dividend and bond claims in proportion to their market-capitalization weights, so that aggregate equilibrium objects are independent of the dividend-share shock. As a result, the equilibrium admits a *block-recursive* structure, in which we first solve for the aggregate equilibrium and then recover the prices of individual claims. This representation will be central for our analytical characterization of aggregate market elasticity.

3.2.1 Equilibrium conditions

Value function and consumption-wealth ratio. With homothetic preferences, investor j's value function can be written as

$$V_{j,t} = \left(\frac{c_{j,t}}{\rho^{\psi}}\right)^{\frac{1-\gamma_j}{1-\psi}} \frac{W_{j,t}^{1-\gamma_j}}{1-\gamma_j}, \qquad \text{where} \qquad \frac{dc_{j,t}}{c_{j,t}} = \mu_{c_j,t}\, dt + \sigma_{c_j,t}\, dZ_t.$$

The function $c_{j,t} = c_j(X_t)$ denotes the consumption-wealth ratio, so that $C_{j,t}/W_{j,t} = c_{j,t}$.

From the Hamilton-Jacobi-Bellman equation, the consumption-wealth ratio satisfies

$$c_{j,t} = \psi\rho + (1-\psi)\left[r_t + \pi_t^{\top}\alpha_{j,t} - \frac{\gamma_j}{2}\|\sigma_{j,t}\|^2\right] + \xi_{j,t}, \tag{12}$$

where $\sigma_{j,t} = \alpha_{j,t}^{\top}\Sigma_t$ is the investor's risk exposure. The first term reflects time preference, while the second term captures the impact of current investment opportunities on consumption.

[9] As there is no initial cash exchange for the derivative, wealth invested in the risk-free asset equals total wealth minus the amount invested in the dividend and bond claims.

The term $\xi_{j,t}$ is given by

$$\xi_{j,t} = \mu_{c_j,t} + (1-\gamma_j)\sigma_{c_j,t}\sigma_{j,t}^\top + \frac{\psi-\gamma_j}{1-\psi}\frac{\|\sigma_{c_j,t}\|^2}{2}.$$

This term captures the effect of changes in future investment opportunities on consumption.

As in Section 2, the consumption-wealth ratio depends on both the interest rate r_t and the risk premium π_t. Importantly, it also depends on the investor's portfolio choice $\alpha_{j,t}$. Finally, $\xi_{j,t}$ acts as a state-dependent shifter, and it is equal to zero when investment opportunities are constant.

Optimal risk exposure share. The optimal risk exposure for active investors $j = 1, 2$ is given by

$$\sigma_{j,t} = \frac{1}{1+\nu_{j,t}}\left[\frac{\eta_t}{\gamma_j} + \frac{1-\gamma_j^{-1}}{\psi-1}\,\sigma_{c_j,t}\right],$$

where $\nu_{j,t}$ is the Lagrange multiplier on the leverage constraint, so $\nu_{j,t} \geq 0$ and $\nu_{1,t} = 0$. When the leverage constraint is not binding, the optimal risk exposure is the sum of two components. The first is the myopic demand, $\frac{\eta_t}{\gamma_j}$, which depends on the investor's risk tolerance and the vector of market prices of risk η_t. This term captures the compensation per unit of exposure to each risk factor: $\eta_t^\top = \Sigma_t^{-1}\pi_t$. The second is the hedging demand, $\frac{1-\gamma_j^{-1}}{\psi-1}\,\sigma_{c_j,t}$, which reflects the investor's desire to hedge fluctuations in future investment opportunities. When the constraint binds, the multiplier $\nu_{j,t} > 0$ scales down exposure uniformly across all risk factors to satisfy $\|\sigma_{j,t}\| \leq \overline{\sigma}$. This acts like an endogenous increase in effective risk aversion.

The Lagrange multiplier $\nu_{2,t}$ satisfies the complementary slackness condition:

$$\nu_{2,t}\left(\|\sigma_{2,t}\| - \overline{\sigma}\right) = 0.$$

For passive investors, risk exposure is given by

$$\sigma_{0,t} = \overline{\alpha}_{p,t}\sigma_{R_Y,t},$$

where $\sigma_{R_Y,t}$ is the risk exposure of the market index fund. Hence, $\overline{\alpha}_{p,t}$ contols the overall risk exposure of passive investors.

Given an investor's risk exposure, the corresponding portfolio shares are recovered from

$$\alpha_{j,t} = \left(\sigma_{j,t}\,\Sigma_t^{-1}\right)^\top,$$

which maps risk exposures into asset holdings.

Market clearing conditions. Market clearing implies that aggregate wealth equals the price of the aggregate claim:

$$\sum_{j=0}^{2} \omega_j W_{j,t} = P_{Y,t}.$$

Let $x_{j,t}$ denote the wealth share of type-j investors:

$$x_{j,t} \equiv \frac{\omega_j W_{j,t}}{P_{Y,t}}.$$

The market clearing conditions for goods and risky assets are given by:

$$\sum_{j=0}^{2} x_{j,t} c_{j,t} = \frac{1}{p_{Y,t}}, \qquad \sum_{j=0}^{2} x_{j,t}\sigma_{j,t} = \sigma_{R_Y,t},$$

where $p_{Y,t} \equiv P_{Y,t}/Y_t$ is the price-dividend ratio of the aggregate claim.

These conditions echo those in the simple model. The first condition states that the wealth-weighted average consumption-wealth ratio equals the dividend yield of the aggregate claim. The second condition states that the wealth-weighted average risk exposure equals the risk exposure of the economy's endowment claim.

Market price of risk. We can write the market clearing condition for risky assets as follows:

$$\underbrace{\sum_{j=1}^{2}\frac{x_{j,t}}{x_{a,t}}\sigma_{j,t}}_{\substack{\text{active demand}\\ \text{for risk}}} = \underbrace{\chi_{a,t}\sigma_{R_Y,t}}_{\substack{\text{net supply}\\ \text{of risk}}},$$

where $\chi_{a,t} \equiv \frac{1-x_{0,t}\overline{\alpha}_{p,t}}{1-x_{0,t}}$ captures the share of aggregate risk that must be absorbed by active investors, and $x_{a,t} \equiv 1 - x_{0,t}$ is the wealth share of active investors.

This is the dynamic counterpart of the risk market clearing condition in Section 2. The expression above states that active investors absorb the net supply of risk, defined as total risk minus the portion held by passive investors.

Substituting optimal risk exposures into the market clearing condition yields the market price of risk:

$$\eta_t = \gamma_{a,t}\left[\chi_{a,t}\sigma_{R_Y,t} - h_{a,t}\right], \tag{13}$$

where $h_{a,t} = \sum_{j=1}^{2}\frac{x_{j,t}}{x_{a,t}}\frac{1-\gamma_j^{-1}}{\psi-1}\frac{\sigma_{c_j,t}}{1+\nu_{j,t}}$ is the average hedging demand of active investors, adjusted for leverage constraints, and $\gamma_{a,t}$ is the average effective risk aversion:

$$\gamma_{a,t} \equiv \left[\sum_{j=1}^{2}\frac{x_{j,t}}{x_{a,t}}\frac{1}{\gamma_j(1+\nu_{j,t})}\right]^{-1}.$$

Equation (13) shows that the market price of risk is determined by three forces. First, the average effective risk aversion of active investors, $\gamma_{a,t}$. This varies over time with the wealth distribution and the extent to which leverage constraints bind. Second, the net supply of risk to active investors, captured by $\chi_{a,t}\sigma_{R_Y,t}$. As passive investors increase their holdings of the market index, the amount of risk that must be absorbed by active investors falls, reducing the compensation for risk. Third, the average hedging demand of active investors, $h_{a,t}$. Higher hedging demand implies active investors are willing to bear more risk, lowering the equilibrium compensation for risk.

Equation (13) makes clear that portfolio flows affect risk premia through their impact on the net supply of risk absorbed by active investors.

Interest rate. Combining goods market clearing with the expression for the consumption–wealth ratio yields the equilibrium interest rate:

$$r_t = \rho + \psi^{-1}\left(\mu_{P_Y,t} + \xi_t\right) + (1-\psi^{-1}) \sum_{j=0}^{2} x_{j,t} \frac{\gamma_j}{2} \|\sigma_{j,t}\|^2 - \pi_{Y,t}, \tag{14}$$

where $\xi_t \equiv \sum_{j=0}^{2} x_{j,t}\, \xi_{j,t}$.

The first term reflects time preference. The second term captures intertemporal substitution. The third term and fourth terms capture the effect of uncertainty on the interest rate.

This expression highlights that the interest rate is determined jointly by risk premia and the allocation of risk across investors, providing the key general equilibrium link between portfolio flows, risk allocation, and saving behavior.

Pricing condition. Let $p_{D,t} \equiv P_{D,t}/Y_t$ and $p_{B,t} \equiv P_{B,t}/Y_t$ denote the price normalized by the aggregate endowment of the dividend and corporate bond claims, respectively. The price-dividend ratio of the aggregate claim is $p_{Y,t} \equiv P_{Y,t}/Y_t$, which satisfies $p_{Y,t} = p_{D,t} + p_{B,t}$.

The pricing condition for each claim $k \in \{D, B\}$ is given by

$$\frac{1}{p_{k,t}} = r_t + \pi_{k,t} - \left(\mu + \mu_{p_k,t} + \sigma\, \sigma_{p_k,t}^{\top}\right), \tag{15}$$

where $\pi_{k,t} = \sigma_{R_k,t}\, \eta_t^{\top}$ is the risk premium on claim k implied by no arbitrage. The terms $\mu_{p_k,t}$ and $\sigma_{p_k,t}$ denote the drift and diffusion of the normalized prices, respectively.

State variables. The aggregate state vector consists of both endogenous and exogenous variables. The endogenous state variables are the wealth shares of active investors, $x_{1,t}$ and $x_{2,t}$, which summarize the distribution of wealth. The exogenous state variables are the passive portfolio share $\overline{\alpha}_{p,t}$ and the dividend share s_t.

Hence, the aggregate state vector is $X_t = (x_t, \overline{\alpha}_{p,t}, s_t)$, where $x_t = (x_{1,t}, x_{2,t})$. The processes $\overline{\alpha}_{p,t}$ and s_t evolve according to (10) and (9), respectively. To obtain $\mu_{X,t}$ and $\sigma_{X,t}$, we need to solve for the drift and diffusion of the endogenous state variables, $x_{1,t}$ and $x_{2,t}$.

By Itô's lemma, the law of motion for $x_{j,t}$, $j = 1, 2$, is given by

$$\frac{dx_{j,t}}{x_{j,t}} = \left[(\eta_t - \sigma_{R_Y,t})\left(\sigma_{j,t} - \sigma_{R_Y,t}\right)^\top - \left(c_{j,t} - \frac{1}{p_{Y,t}}\right) + \kappa\frac{\omega_j - x_{j,t}}{x_{j,t}}\right]dt + (\sigma_{j,t} - \sigma_{R_Y,t})\, dZ_t.$$

Endogenous volatility. Return volatility has both a cash-flow and a valuation component, $\sigma_{R_k,t} = \sigma_{k,t} + \sigma_{p_k,t}$, where $\sigma_{k,t}$ reflects cash-flow volatility and $\sigma_{p_k,t}$ arises from movements in the price-dividend ratio $p_{k,t}$ for asset $k \in \{D, B\}$. By Itô's lemma,

$$\sigma_{p_k,t} = \frac{p_{k,x}(X_t)}{p_{k,t}(X_t)}\sigma_{x,t} + \frac{p_{k,\overline{\alpha}_p}(X_t)}{p_{k,t}(X_t)}\sigma_{\overline{\alpha}_p,t} + \frac{p_{k,s}(X_t)}{p_{k,t}(X_t)}\sigma_{s,t}. \tag{16}$$

Equation (16) shows that endogenous volatility depends on the sensitivity of the price-dividend ratio to three groups of variables: (i) the wealth distribution, (ii) the passive portfolio share, and (iii) the dividend share. The first term is standard in heterogeneous-agent models (see Panageas, 2020). The third term is typical of models with multiple assets. The second term arises only in the presence of exogenous portfolio flow shocks. This second term is quantitatively important when markets are sufficiently inelastic, that is, when price-dividend ratios are highly sensitive to changes in passive demand. Thus, (16) provides a direct link between market elasticity and return volatility, as lower elasticity amplifies the sensitivity of prices to portfolio flows and therefore increases endogenous volatility.

Given the volatility of individual claims, the volatility of the aggregate claim follows from aggregation:

$$\sigma_{R_Y,t} = \sum_{k=D,B} \frac{p_{k,t}}{p_{Y,t}}\sigma_{R_k,t}.$$

It remains to determine $\sigma_{R_F,t}$. We make three assumptions about $\sigma_{R_F,t}$: (i) it does not load on the dividend-share shock, so $\sigma_{R_F,t,3} = 0$; (ii) it is orthogonal to the aggregate claim, so $\sigma_{R_F,t}\,\sigma_{R_Y,t}^\top = 0$;

(iii) it has the same norm as the aggregate claim, so $\|\sigma_{R_F,t}\| = \|\sigma_{R_Y,t}\|$. These conditions uniquely determine $\sigma_{R_F,t}$ given $\sigma_{R_Y,t}$.

Markov equilibrium. Equations (13) and (14) allow us to express the market price of risk η_t and the interest rate r_t as functions of the consumption-wealth ratios $c_j(X_t)$ and the valuation ratios $p_k(X_t)$. Using Itô's lemma, the drift and diffusion terms $(\mu_{c_j,t}, \mu_{p_k,t})$ and $(\sigma_{c_j,t}, \sigma_{p_k,t})$ can be written as functions of the state dynamics $(\mu_{X,t}, \sigma_{X,t})$.

Substituting these expressions into the HJB equation (12) and the pricing condition (15), together with the laws of motion for the state variables, yields a system of five partial differential equations.

These equations jointly determine the consumption-wealth ratios $\{c_j(X_t)\}_{j=0}^{2}$ and the normalized prices $\{p_k(X_t)\}_{k\in\{D,B\}}$ as functions of four state variables: the two active wealth shares $x_{1,t}$ and $x_{2,t}$, the passive portfolio share $\overline{\alpha}_{p,t}$, and the dividend share s_t.

3.2.2 Block recursivity and the mutual fund theorem

We show that the system of PDEs admits a block-recursive structure. First, we solve for the consumption-wealth ratios and the price-dividend ratio of the aggregate claim in a reduced set of states $\hat{X}_t = (x_{1,t}, x_{2,t}, \overline{\alpha}_{p,t})$. Second, we solve for the price-dividend ratios of the dividend and corporate bond claims in the full set of states $X_t = (x_{1,t}, x_{2,t}, \overline{\alpha}_{p,t}, s_t)$.

This provides a substantial simplification, as the equilibrium allocation and pricing kernel are identical to those of an economy with a single claim on the aggregate endowment and a derivative. At the same time, the block-recursive structure allows us to recover the pricing implications for individual claims. In particular, we can characterize how portfolio flows affect the market elasticity of the dividend claim.

An important implication of the block-recursive structure is that a version of the *mutual fund theorem* holds in our economy. Despite financial frictions, investors optimally hold risky claims in proportion to their market-capitalization weights, as in the classic mutual fund theorem.

Proposition 2 (Block recursivity and the mutual fund theorem)**.** *Define the reduced set of states as* $\hat{X}_t = (x_{1,t}, x_{2,t}, \overline{\alpha}_{p,t})$. *Then,*

(i) ***Block recursivity:*** *The consumption-wealth ratios,* $\{c_{j,t}(\hat{X}_t)\}_{j=0}^{2}$, *and the price-dividend ratio for the aggregate claim,* $p_Y(\hat{X}_t)$, *are a function of the reduced set of states* $\hat{X}_t$, *and they satisfy the following system of PDEs:*

$$c_j = \psi\rho + (1-\psi)\left[r + \eta^\top \sigma_j - \frac{\gamma_j}{2}\|\sigma_j\|^2\right] + \mu_{c_j} + (1-\gamma_j)\sigma_{c_j}\sigma_j^\top + \frac{\psi - \gamma_j}{1-\psi}\frac{\|\sigma_{c_j}\|^2}{2}$$
$$\frac{1}{p_Y} = r + \eta^\top(\sigma + \sigma_{p_Y}) - \left(\mu + \mu_{p_Y} + \sigma\,\sigma_{p_Y}^\top\right),$$

where $(r, \eta, \sigma_j, \mu_{c_j}, \sigma_{c_j}, \mu_{p_Y}, \sigma_{p_Y})$ *are all functions of* $\hat{X}_t$, (c_j, p_Y) *and their derivatives.*

(ii) ***Mutual fund theorem:*** *The portfolio shares of active investors are given by:*

$$\alpha_{j,k} = \alpha_{j,Y}\frac{p_k}{p_Y}, \quad \forall k \in \{D, B\},$$

where $\alpha_{j,Y}$ *denotes the holdings of the market index fund for investor* j, *which satisfies the condition*

$$\sigma_j = \alpha_{j,Y}\sigma_{R_Y} + \alpha_{j,F}\sigma_{R_F}.$$

Proposition 2 shows that the equilibrium with separate dividend and bond claims is equivalent to that of an economy with a single claim on the aggregate endowment and a derivative.

Two assumptions are key to this result. First, the aggregate endowment process Y_t is independent of the dividend share s_t. This assumption follows Menzly et al. (2004), but differs from the two-trees model of Cochrane, Longstaff, and Santa-Clara (2008), where the dividend share affects aggregate volatility.

Second, markets are dynamically complete. In equilibrium, investors are not exposed to the dividend-share shock, which can be perfectly diversified by holding the dividend and bond claims in proportion to their market-capitalization weights. Investors can therefore obtain any desired exposure to the fundamental and portfolio-flow shocks by trading the market index fund and the derivative.

Without the derivative, investors would need to deviate from market-capitalization weights to adjust their exposure to these risks, at the cost of bearing dividend-share risk.

Pricing the individual claims. Having solved for the aggregate objects $\{c_{j,t}(\hat{X}_t)\}_{j=0}^{2}$ and $p_Y(\hat{X}_t)$, we next recover the valuation of individual claims. The normalized prices of the dividend and bond claims are functions of the full state vector $X_t = (x_{1,t}, x_{2,t}, \overline{\alpha}_{p,t}, s_t)$, as they load on the dividend-share shock.

Given the aggregate equilibrium objects $r(\hat{X}_t)$ and $\eta(\hat{X}_t)$, and using Itô's lemma to express the drift and diffusion of valuation ratios as functions of the state dynamics and their derivatives, the pricing condition in Equation (15) yields a linear PDE for $p_{D,t}(X_t)$ and $p_{B,t}(X_t)$.

Thus, individual claim prices are obtained in a second step, conditional on the aggregate equilibrium. This separation highlights that portfolio flows affect aggregate quantities through the reduced state vector, while their impact on individual asset prices operates through valuation in the full state space.

This completes the characterization of the equilibrium with separate dividend and bond claims. We now turn to an analytical characterization of the aggregate market elasticity in this economy.

4 The Determinants of the Aggregate Market Elasticity

In this section, we analyze how the *aggregate market elasticity* governs the price impact of portfolio flows. The elasticity is determined by the coupled PDE system (**??**)–(15), which generally has no closed-form solution. To study its determinants, we extend the perturbation method of Kogan and Uppal (2001) and derive asymptotic closed-form approximations.

State-global perturbations Our perturbation approach studies economies in a neighborhood of a benchmark that admits a closed-form solution. A convenient benchmark arises when preferences are homogeneous, $\gamma_j = \gamma$, and passive investors are fully invested in the risky asset, $\overline{\alpha}_{p,t} = 1$. In this case, the economy collapses to a Lucas economy, as shown in Lemma 1. Throughout, a superscript $\star$ denotes benchmark values.

Lemma 1 (Benchmark economy)**.** *Suppose $\gamma_j = \gamma$, $\overline{\alpha}_{p,t} = 1$, and $\rho > \left(1-\psi^{-1}\right)\left(\mu - \frac{\gamma\|\sigma\|^2}{2}\right)$. Then,*

$$\pi^\star = \gamma\|\sigma\|^2, \qquad r^\star = \rho + \psi^{-1}\mu - \left(1+\psi^{-1}\right)\frac{\gamma\|\sigma\|^2}{2}, \qquad p^\star = \left[\rho - \left(1-\psi^{-1}\right)\left(\mu - \frac{\gamma\|\sigma\|^2}{2}\right)\right]^{-1},$$

$c_j^\star = (p^\star)^{-1}$, and $\alpha_j^\star = 1$ for $j = 0, \ldots, J$.

We analyze small deviations from this benchmark. Formally, consider a family of economies indexed by $\epsilon > 0$, where ϵ scales departures from the benchmark along the following dimensions.

First, investors have heterogeneous risk aversion:

$$\gamma_j = \gamma\left(1 + \hat{\gamma}_j \epsilon\right), \qquad \sum_{j=0}^{J} \omega_j \hat{\gamma}_j = 0,$$

so that γ is the population-weighted average and $\hat{\gamma}_j$ measures deviations. When $\epsilon = 0$, preferences are homogeneous; when $\epsilon = 1$, we recover the heterogeneous-investor economy.

Second, passive investors need not be fully invested in the risky asset. Their portfolio share is

$$\overline{\alpha}_{p,t} = 1 + \hat{\alpha}_{p,t}\,\epsilon,$$

where $\hat{\alpha}_{p,t}$ measures deviations from full investment benchmark and evolves as an Ornstein-Uhlenbeck process, consistent with Equation (10).

Third, the leverage constraint coefficient is perturbed according to

$$\overline{\sigma} = \|\sigma\|\left(1 + \hat{\sigma}\,\epsilon\right),$$

so that the tightness of the constraint is of order $O(\epsilon)$, matching the order of leverage demand. Finally, we assume a small mortality rate, $\kappa = \hat{\kappa}\,\epsilon$.

Equilibrium objects now depend on both the state X_t and the parameter ϵ. We expand them to second order:

$$p(X, \epsilon) = p_0(X) + p_1(X)\epsilon + p_2(X)\epsilon^2 + O(\epsilon^3),$$
$$c_j(X, \epsilon) = c_{j,0}(X) + c_{j,1}(X)\epsilon + c_{j,2}(X)\epsilon^2 + O(\epsilon^3),$$

where $p_k(X)$ and $c_{j,k}(X)$, $k \in \{0, 1, 2\}$, are functions of the state X.

Unlike standard DSGE perturbations, which linearize in both X and ϵ, our method is *state-global*: we solve directly for functions of X rather than coefficients around a steady state.[10] We refer to this approach as a *state-global perturbation*. This approach builds on the perturbation method of Kogan and Uppal (2001) and, more closely, the extension by Silva (2020). This method allows us to characterize how aggregate market elasticity varies with the state of the economy.

We proceed in two steps. First, we compute the first-order corrections $p_1(X)$ and $c_{j,1}(X)$. Second, we solve for the second-order terms $p_2(X)$ and $c_{j,2}(X)$. Zeroth-order terms are given by Lemma 1, i.e., $p_0(X) = p^\star$ and $c_{j,0}(X) = c_j^\star$.

4.1 The first-order demand system

Demand for risky asset. Investor j's demand for the risky asset is

$$Q_j(X; \epsilon) = \alpha_j(X; \epsilon)\,\frac{x_j}{\omega_j},$$

where x_j is the wealth share. Define the proportional deviation from the benchmark as

$$q_{j,t} \;\equiv\; \frac{Q_j(X_t; \epsilon) - Q_j(X_t; 0)}{Q_j(X_t; 0)} \;=\; \alpha_{j,1}(X_t)\,\epsilon + O(\epsilon^2),$$

where $\alpha_{j,1}(X)$ is the first-order correction to the portfolio weight.

[10]See Kargar, Passadore, and Silva (2023) for a comparison with standard perturbation methods.

Three results determine $\alpha_{j,1}(X)$. First, the volatility of the price-dividend ratio is second order:

$$\sigma_{p,t} = \frac{p_X(X_t)}{p(X_t)}\,\sigma_X(X_t) = O(\epsilon^2),$$

since $p_{X,0} = \sigma_{X,0} = 0$ (Lemma 1). Second, hedging demand is also $O(\epsilon^2)$ because $\sigma_{c_j}(X_t)$ is second order. Third, expanding the pricing condition (15) gives

$$\hat{\pi}_t = -\,\hat{r}_t - \frac{1}{p^\star}\hat{p}_t + O(\epsilon^2),$$

where $\hat{\pi}_t \equiv \pi_t - \pi^\star$, $\hat{r}_t \equiv r_t - r^\star$, and $\hat{p}_t \equiv \frac{p_t - p^\star}{p^\star}$, using $\mu_{p,t} = O(\epsilon^2)$.

Substituting into (**??**), the first-order demand system is

$$q_{j,t} = -\,\zeta_{j,p}\,\hat{p}_t - \zeta_{j,r}\,\hat{r}_t + f_{j,t} + O(\epsilon^2), \tag{17}$$

where, for unconstrained investors,

$$\zeta_{j,p} = \frac{1}{p^\star}\frac{1}{\gamma\|\sigma\|^2}, \qquad \zeta_{j,r} = \frac{1}{\gamma\|\sigma\|^2}, \qquad f_{j,t} = -\,\hat{\gamma}_j\,\epsilon.$$

For passive and constrained investors, own- and cross-price elasticities vanish, $\zeta_{j,p} = \zeta_{j,r} = 0$, with demand shifters $f_{j,t} = \hat{\sigma}\,\epsilon$ for constrained investors and $f_{0,t} = \hat{\alpha}_{p,t}\,\epsilon$ for passive investors.

Equation (17) shows that the first-order approximation to the dynamic heterogeneous agent model in Section 3 provides microfoundations for the demand system in Section 2. As in the simple model, unconstrained active investors exhibit nonzero own- and cross-price elasticities, here explicitly linked to structural parameters. Shocks to $\overline{\alpha}_{p,t}$ move the passive investors' demand shifter $f_{0,t}$—the dynamic analog of the comparative statics in Section 2. In addition, the model generates further demand shifts from occasionally binding leverage constraints faced by active investors.

Consumption-wealth ratio. A first-order expansion of (**??**) yields

$$c_{j,t} \;=\; \psi\rho \;+\; (1-\psi)\left[r_t + \pi_t - \frac{\gamma_j\|\sigma\|^2}{2}\right] \;+\; O(\epsilon^2).$$

Consistent with Assumption 1, the risk-free rate and the risk premium enter symmetrically at first order. Substituting the expansion $\hat{\pi}_t = -\hat{r}_t - p^{\star -1}\hat{p}_t + O(\epsilon^2)$ gives

$$\hat{c}_{j,t} \;\equiv\; c_{j,t} - c_j^\star \;=\; \chi_{j,0} \;+\; \chi_{j,p}\,\hat{p}_t \;+\; O(\epsilon^2),$$

with coefficients

$$\chi_{j,p} = (\psi - 1)\,\frac{1}{p^\star}, \qquad \chi_{j,0} = (\psi - 1)\,\frac{\gamma\|\sigma\|^2}{2}\,\hat{\gamma}_j\,\epsilon.$$

Hence, conditional on p_t, the consumption-wealth ratio is independent of r_t, as in Section 2. Its slope with respect to p_t depends on the EIS: it rises with p_t when $\psi > 1$, the standard assumption in macro-finance models. Finally, to first order, the consumption–wealth ratio is unaffected by $\overline{\alpha}_{p,t}$, a property that will be central for aggregate market elasticity.

Aggregate market elasticity. Given the demand system, equilibrium prices follow from market clearing in the risky-asset and goods markets:

$$\sum_{j=0}^{J} x_{j,t}\,q_{j,t} = 0, \qquad \sum_{j=0}^{J} x_{j,t}\,\hat{c}_{j,t} = -\frac{1}{p^\star}\,\hat{p}_t.$$

Aggregating across investors, and dropping the j subscript on coefficients that are common across types, yields the linear system

$$\begin{bmatrix} \zeta_p & \zeta_r \\ \chi_p + \frac{1}{p^\star} & 0 \end{bmatrix} \begin{bmatrix} \hat{p}_t \\ \hat{r}_t \end{bmatrix} = \begin{bmatrix} f_t \\ -\chi_{0,t} \end{bmatrix},$$

where $f_t \equiv \sum_{j=0}^{J} x_{j,t} f_{j,t}$ and $\chi_{0,t} \equiv \sum_{j=0}^{J} x_j \chi_{j,0}$.

Definition 2 (Aggregate market elasticity)**.** *The aggregate market elasticity $\epsilon_{M,t}$ is defined as the inverse of the proportional change in the price of the risky asset in response to a flow shock f_t:*

$$\epsilon_{M,t} \equiv \left[\frac{\partial \hat{p}_t}{\partial f_t} \right]^{-1}.$$

Proposition 3 (First-order impact of portfolio flows)**.** *Suppose $\rho > \left(1 - \psi^{-1}\right)\left(\mu - \frac{\gamma\|\sigma\|^2}{2}\right)$. Then:*

(i) The price impact is zero to first order: $\epsilon_{M,t}^{-1} = \frac{\partial \hat{p}_t}{\partial f_t} = 0$.

(ii) The interest rate and risk premium responses satisfy

$$\frac{\partial \hat{r}_t}{\partial f_t} = \frac{1}{\zeta_r}, \qquad \frac{\partial \hat{\pi}_t}{\partial f_t} = -\frac{1}{\zeta_r}.$$

Proposition 3 shows that the main lesson from the simple model in Section 2 extends to the dynamic economy: portfolio inflows into the risky asset reduce the risk premium, while the corresponding shift out of bonds raises the risk-free rate. To first order, these effects offset exactly, leaving the price-dividend ratio unchanged. Hence, the aggregate market is infinitely elastic, regardless of the individual investors' own-price elasticities $\zeta_{j,p}$.

Why is the market infinitely elastic? As shown in Figure 1, price impact is absent as long as the consumption-wealth ratio does not respond directly to the portfolio-flow shock. Otherwise, price changes would generate excess demand or supply in the goods market.

To see why flows leave consumption unchanged to first order, differentiate (**??**) with respect to $\alpha_{j,t}$:

$$\frac{\partial c_{j,t}}{\partial \alpha_{j,t}} = (1-\psi)\,\gamma_j\,\|\sigma_{R,t}\|^2 \left[\frac{\pi_t}{\gamma_j \|\sigma_{R,t}\|^2} + \frac{1-\gamma_j^{-1}}{\psi - 1} \frac{\sigma_{c_{j,t}}\,\sigma_{R,t}^{\top}}{\|\sigma_{R,t}\|^2} - \alpha_{j,t} \right].$$

The term in brackets is equal to zero at the privately optimal portfolio, as implied by the investor's first-order condition. By the envelope theorem, small changes in $\alpha_{j,t}$ around the optimum have negligible welfare effects on welfare and hence do not affect $c_{j,t}$.

In the first-order approximation, this derivative is evaluated at the benchmark Lucas economy, where all investors hold optimal portfolios. The derivative is thus zero, and by Proposition 3, price impact is negligible for small (first-order) deviations from the frictionless allocation: the consumption-wealth ratio does not react to flow shocks. In the simple model of Section 2, this was imposed as an *assumption*; here it emerges as a *result* of expanding around the efficient allocation.

This reasoning also clarifies when aggregate elasticity becomes finite: when the initial allocation of risk is *inefficient*. To study such economies requires larger departures from the benchmark, captured by a second-order approximation of the demand system.

4.2 Inefficient passive demand

We next turn to a second-order approximation of the demand system to study how risk allocation affects aggregate market elasticity. To focus on the mechanism most clearly, consider the special case without preference heterogeneity or leverage constraints.

The role of risk misallocation. In this setting, demand for the risky asset is still given by (17) up to second order. Without heterogeneity, the relevant demand shifter is

$$f_t \equiv x_{0,t}\left(\overline{\alpha}_{p,t} - 1\right),$$

where $x_{0,t}$ denotes the wealth share of passive investors.

The key difference relative to the benchmark arises from the consumption-wealth ratio. Aggregating (**??**) across investors, letting $c_t \equiv \sum_{j=0}^{J} x_{j,t} c_{j,t}$, and imposing market clearing $\sum_{j=0}^{J} x_{j,t}\alpha_{j,t} = 1$, yields

$$c_t = \psi\rho + (1-\psi)\left[r_t + \pi_t - \frac{\gamma\|\sigma\|^2}{2}\sum_{j=0}^{J} x_{j,t}\alpha_{j,t}^2\right] + O(\epsilon^3).$$

Under the efficient allocation $\alpha_{j,t} = 1$, so $\sum_j x_{j,t}\alpha_{j,t}^2 = 1$. Any deviation from this efficient allocation creates dispersion in portfolios, leading to $\sum_j x_{j,t}\alpha_{j,t}^2 > 1$ by Jensen's inequality, which lowers the risk-adjusted expected return $r_t + \pi_t - \frac{\gamma\|\sigma\|^2}{2}\sum_j x_{j,t}\alpha_{j,t}^2$. When $\psi > 1$, this reduction weakens incentives to save and raises the consumption-wealth ratio for a given $r_t + \pi_t$.

Movements in the passive portfolio therefore shift the aggregate consumption–wealth ratio. Its second-order expansion is

$$\hat{c}_t \;\equiv\; c_t - c^\star \;=\; \chi_{0,t} \;+\; \chi_p\,\hat{p}_t \;+\; O(\epsilon^3),$$

with coefficients

$$\chi_p = (\psi - 1)\,\frac{1}{p^\star}, \qquad \chi_{0,t} = (\psi - 1)\,\frac{\gamma\|\sigma\|^2}{2}\,\frac{f_t^2}{x_{0,t}(1 - x_{0,t})}.$$

Because $\chi_{0,t}$ is quadratic in the flow f_t, it is $O(\epsilon^2)$. For example, when passive investors are initially underexposed to the risky asset ($\overline{\alpha}_{p,t} < 1$ and $\psi > 1$), an inflow into stocks reduces the consumption-wealth ratio:

$$\frac{\partial \chi_{0,t}}{\partial f_t} \;=\; -\,(\psi - 1)\,\gamma\|\sigma\|^2\,\frac{1 - \overline{\alpha}_{p,t}}{1 - x_{0,t}}.$$

Note that $\chi_{0,t}$ is locally insensitive to portfolio flow changes when passive portfolio is efficient ($\overline{\alpha}_{p,t} = 1$), consistent with the first-order analysis.

Figure 2 illustrates the effects of portfolio flows when passive investors are initially underexposed to risk, i.e., $\overline{\alpha}_{p,t} < 1$. An inflow into the risky asset shifts both the net supply curve (left panel) and the average consumption-wealth ratio schedule (right panel) leftward. The interest rate adjusts so that active demand meets net supply at the goods market clearing price. In this case, portfolio flows have a positive price impact and the aggregate elasticity is *finite*.

Proposition 4 (Aggregate elasticity: inefficient passive demand)**.** *Without preference heterogeneity*

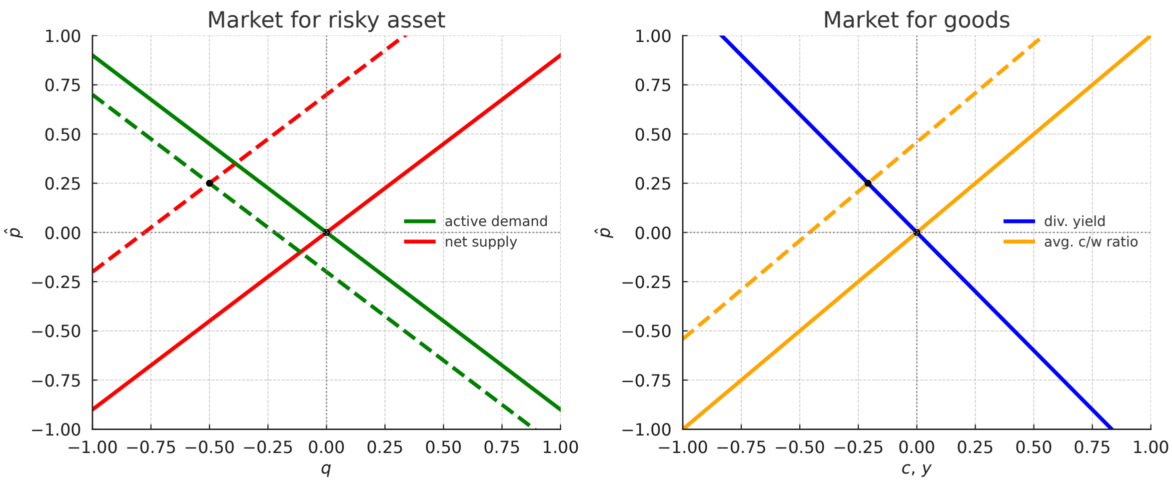


Figure 2. Equilibrium with inefficient risk allocation: risky asset (left) and goods (right) markets.

or leverage constraints, the inverse aggregate elasticity is

$$\varepsilon_{M,t}^{-1} = -(\chi_p + y^\star)^{-1} \frac{\partial \chi_{0,t}}{\partial f_t} = \left(1 - \psi^{-1}\right) \frac{\gamma \|\sigma\|^2}{y^\star} \frac{1 - \overline{\alpha}_{p,t}}{x_{a,t}} + O\left(\epsilon^2\right),$$

where $x_{a,t} = 1 - x_{0,t}$ *is the wealth share of active investors and* $y^\star = 1/p^\star$ *is the dividend-price ratio in the benchmark economy.*

Proposition 4 shows that, unlike in Gabaix and Koijen (2023), the risky-asset elasticities ζ_p and ζ_r are not the direct drivers of aggregate elasticity. Because the demand system is recursive (interest rates only enters the risky asset demand) the overall price impact is governed instead by how portfolio flows impact the consumption-wealth ratio.

The macro elasticity is finite whenever $\overline{\alpha}_{p,t} \neq 1$ and $\psi \neq 1$. In the unit EIS case, the price-dividend ratio is pinned down by the subjective discount rate, so portfolio flows generate offsetting movements in the risk premium and the risk-free rate. With efficient risk allocation ($\overline{\alpha}_{p,t} = 1$), there is again no price impact.

The price impact is positive when $\psi > 1$ and $\overline{\alpha}_{p,t} < 1$: interest rate adjustments no longer fully offset the risk premium response, so the latter dominates. This mirrors representative agent models in which uncertainty shocks affect prices only when $\psi > 1$ (e.g., Bansal and Yaron, 2004).

The state-global perturbation analysis also makes clear that elasticity is state dependent. Figure 3

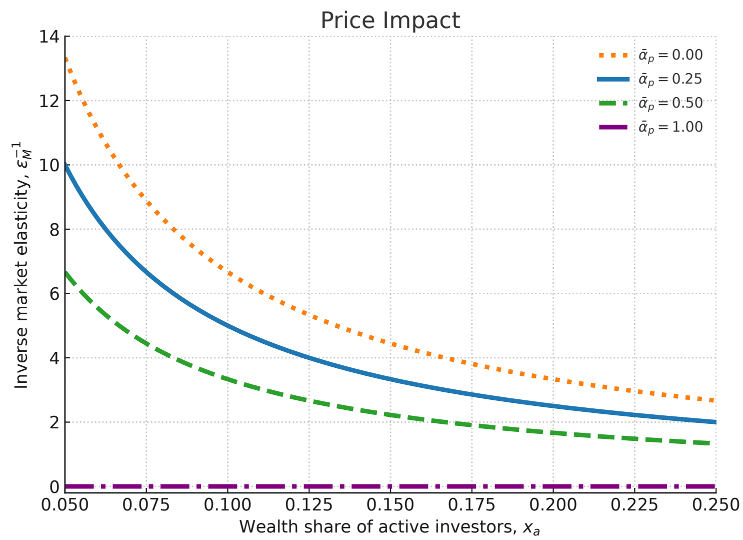


Figure 3. Price impact: inefficient passive demand.

plots the price impact as a function of active investors' wealth for different passive-share levels. Price impact rises when active investors are under capitalized, reflecting their limited risk-bearing capacity, and when the passive share deviates further from its efficient level. In the extreme case where passive investors hold no equities ($\overline{\alpha}_{p,t} = 0$), we recover a version of Basak and Cuoco (1998) in which price impact is maximized.

4.3 Preference heterogeneity and leverage constraints

We now reintroduce preference heterogeneity and leverage constraints into the model. These features shift the conditions under which passive investors' portfolios are efficient and therefore alter the aggregate market elasticity. The next proposition provides the main result.

Proposition 5 (Aggregate elasticity: heterogeneity and leverage constraints)**.** *The inverse aggregate market elasticity is*

$$\varepsilon_{M,t}^{-1} \;=\; \left(1-\psi^{-1}\right)\frac{\gamma\,\|\sigma\|^2}{y^\star}\,\frac{\overline{\alpha}_{p,t}^{\rm opt}-\overline{\alpha}_{p,t}}{x_{a,t}-x_{c,t}}\left(1-x_{c,t}\right)\;+\;O\!\left(\epsilon^2\right), \tag{18}$$

where $x_{c,t}$ is the wealth share of constrained active investors and the passive investors' optimal *portfolio share is*

$$\overline{\alpha}_{p,t}^{\text{opt}} \;=\; 1 \;-\; \frac{x_{a,t} - x_{c,t}}{1 - x_{c,t}} \, \frac{\gamma_0 - \mathbb{E}_t^u[\gamma_j]}{\gamma} \;-\; \frac{x_{c,t}}{1 - x_{c,t}} \left(\frac{\overline{\sigma}}{\|\sigma\|} - 1 \right), \tag{19}$$

with $x_{u,t} \equiv x_{a,t} - x_{c,t}$ the wealth share of unconstrained active investors and

$$\mathbb{E}_t^u[\gamma_j] \;\equiv\; \sum_{j \in \mathcal{J}^u} \frac{x_{j,t}}{x_{u,t}} \gamma_j .$$

Proposition 5 characterizes aggregate market elasticity in the full model. As in the frictionless benchmark, the market remains infinitely elastic when passive investors hold their optimal portfolio share, $\overline{\alpha}_{p,t} = \overline{\alpha}_{p,t}^{\text{opt}}$. The optimal portfolio share corresponds to the value of $\overline{\alpha}_{p,t}$ such that the passive investor would have no incentive to change their portfolio. Under homogeneous preferences ($\gamma_j = \gamma$) and slack leverage constraints ($x_{c,t} = 0$), this condition reduces to $\overline{\alpha}_{p,t} = 1$. More generally, preference heterogeneity and binding leverage constraints shift the efficient passive share away from unity, with its level determined by the wealth distribution and the relative risk aversion of different investor types.

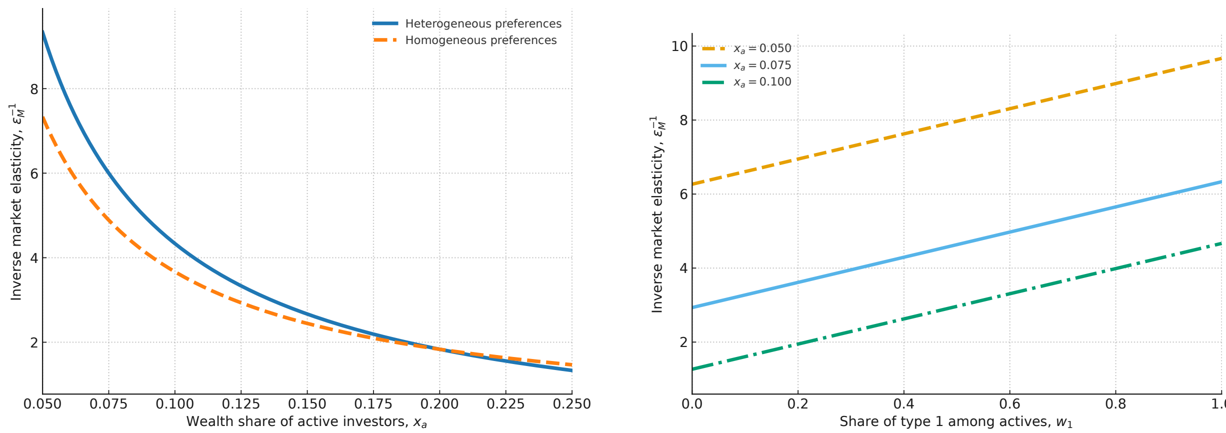
Preference heterogeneity. With slack leverage constraints ($x_{c,t} = 0$), (19) simplifies to

$$\overline{\alpha}_{p,t}^{\text{opt}} = 1 - x_{a,t} \, \frac{\gamma_0 - \mathbb{E}_t^u[\gamma_j]}{\gamma},$$

so, unlike the homogeneous preference case, the efficient passive share $\overline{\alpha}_{p,t}^{\text{opt}}$ differs from unity whenever $\gamma_0 \neq \mathbb{E}_t^u[\gamma_j]$. The inverse elasticity in (18) becomes

$$\varepsilon_{M,t}^{-1} = \left(1 - \psi^{-1}\right) \frac{\gamma \|\sigma\|^2}{y^\star} \, \frac{\overline{\alpha}_{p,t}^{\text{opt}} - \overline{\alpha}_{p,t}}{x_{a,t}} \;+\; O(\epsilon^2).$$

Optimal risk sharing implies that when passive investors are more risk averse than the average



(a) Homogeneous vs. heterogeneous preferences

(b) Role of wealth distribution of active investors

Figure 4. Price impact comparisons.
Panel (a) compares heterogeneous and homogeneous preferences. Panel (b) shows the sensitivity of the price impact to wealth distribution among active investors when $J = 2$ for different passive portfolio shares.

unconstrained active ($\gamma_0 > \mathbb{E}_t^u[\gamma_j]$),

$$\frac{\partial \overline{\alpha}_{p,t}^{\text{opt}}}{\partial x_{a,t}} = -\frac{\gamma_0 - \mathbb{E}_t^u[\gamma_j]}{\gamma} < 0,$$

so the efficient passive equity share rises as the active wealth share falls, typically after adverse aggregate shocks that erode active wealth that is more exposed to risk. If passive investors do not rebalance in such episodes, market inelasticity increases because $x_{a,t}$ declines (reducing risk-bearing capacity) while the gap $\overline{\alpha}_{p,t}^{\text{opt}} - \overline{\alpha}_{p,t}$ widens (moving the economy farther from the efficient allocation). Consequently, price impact rises as $x_{a,t}$ falls, as shown in the left panel (a) of Figure 4: the heterogeneous- and homogeneous-preference curves coincide at $x_a = 0.20$, but only the former steepens markedly as x_a declines.

Panel (b) of Figure 4 shows how the distribution of wealth across active investors shapes price impact. With two active types ($J = 2$) and $\gamma_1 > \gamma_2$, the average risk active aversion is

$$\mathbb{E}_t^u[\gamma_j] \;=\; w_{1,t}\gamma_1 + \big(1 - w_{1,t}\big)\gamma_2, \qquad w_{1,t} \equiv \frac{x_{1,t}}{x_{a,t}}.$$

When leverage constraints are slack ($x_{c,t} = 0$), the efficient passive share satisfies $\overline{\alpha}_{p,t}^{\text{opt}} = 1 -$

$x_{a,t}\,(\gamma_0 - \mathbb{E}_t^u[\gamma_j])/\gamma$, so holding $x_{a,t}$ fixed,

$$\frac{\partial\,\overline{\alpha}_{p,t}^{\text{opt}}}{\partial w_{1,t}} = \frac{x_{a,t}}{\gamma}\,(\gamma_1 - \gamma_2) \;>\; 0.$$

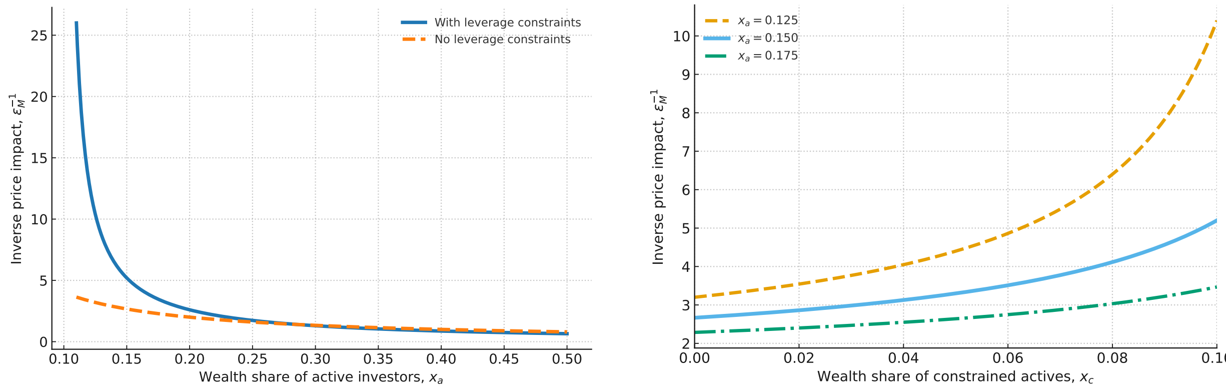
Thus, as the wealth share of the more risk averse active type rises, $\mathbb{E}_t^u[\gamma_j]$ increases and the passive sector should optimally hold more of the risky asset. If the passive portfolio is not adjusted, the deviation $\overline{\alpha}_{p,t}^{\text{opt}} - \overline{\alpha}_{p,t}$ widens, worsening risk allocation and raising price impact, consistent with the upward shift in the panel (b) of Figure 4.

Leverage constraints. When some active investors are constrained ($x_{c,t} > 0$), the marginal investors are the *unconstrained* active agents, and their risk-bearing capacity,

$$x_{u,t} \;\equiv\; x_{a,t} - x_{c,t},$$

becomes the key determinant of price impact. The panel (a) of Figure 5 plots price impact against the active wealth share $x_{a,t} = 1 - x_{0,t}$, with and without binding constraints. To isolate the role of constraints, we set $\gamma_0 = \mathbb{E}_t^u[\gamma_j]$, removing pure preference heterogeneity effects. As $x_{a,t}$ declines, e.g., after a negative aggregate shock when $\overline{\alpha}_{p,t} < 1$, price impact rises more sharply under binding constraints because $x_{u,t}$ shrinks and the risk-bearing capacity of marginal investors is depleted. Panel (b) holds $x_{a,t}$ fixed and varies the constrained share $x_{c,t}$. The price impact increases monotonically with $x_{c,t}$. This is intuitive: price impact is larger when a larger share of active investors is constrained.

Taking stock. Proposition 5 clarifies the main determinants of aggregate market elasticity. The elasticity is *finite* only when risk is misallocated: small portfolio flows around the frictionless allocation have no first-order price impact. Preference heterogeneity amplifies price impact in downturns because, as the active wealth share falls, the efficient passive share rises, widening the gap between efficient and actual portfolios. Binding leverage constraints further increase price



(a) Role of leverage constraints *(assumes $x_{c,t} = 0.10$)*

(b) Varying the wealth share of constrained actives, $x_{c,t}$

Figure 5. Price impact under leverage constraints. Panel (a) compares the cases with and without leverage constraints (holding $x_{c,t} = 0.10$). Panel (b) plots the price impact as a function of $x_{c,t}$ for different values of $x_{a,t}$.

impact by shrinking the marginal (unconstrained) risk-bearing capacity, $x_{u,t}$. Taken together, these forces imply rich state dependence of aggregate elasticity in economies with heterogeneous investors and financial frictions.

5 Quantitative Implications

We now evaluate the quantitative implications of the dynamic model of Section 3. The exercise has two goals. The first is to ask whether a calibration disciplined by U.S. data on sectoral portfolio holdings and passive equity flows can simultaneously deliver an empirically plausible price impact and the standard asset-pricing moments. The second is to quantify how much of the variation in return volatility, return predictability, and the variance of the price–dividend ratio is attributable to how portfolio flows interact with preference heterogeneity, leverage constraints, and passive-sector risk misallocation. We do the latter by comparing the baseline economy to a set of counterfactuals that turn off these frictions one at a time.

We solve the full dynamic system numerically using the neural-network method of Duarte et al. (2024), which accommodates the four state variables: passive demand, the dividend share, and the wealth shares of the constrained and unconstrained active sectors. Appendix J describes the numerical procedure in detail.

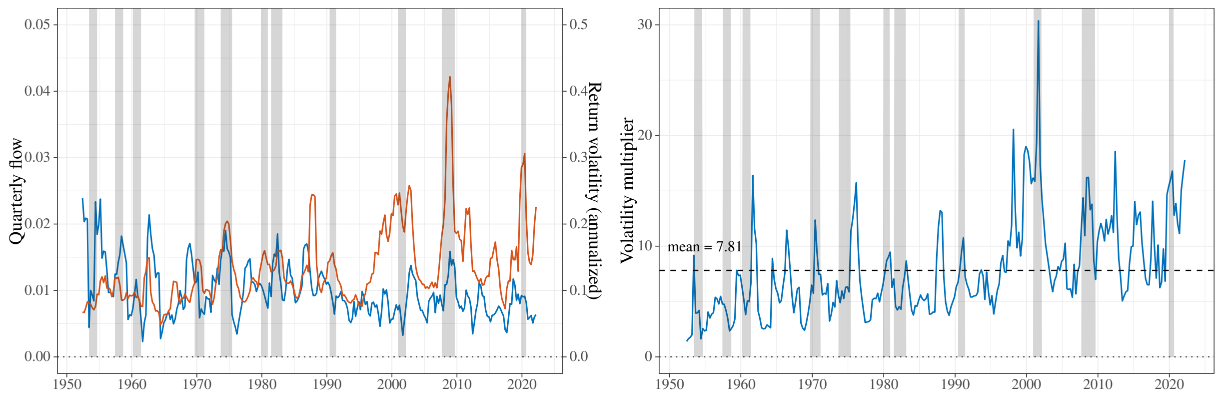


Figure 6. Quarterly flows and return volatility.
The left panel plots quarterly aggregate equity flows in blue and realized return volatility in red. Quarterly flows are scaled by lagged aggregate market capitalization, following Appendix D3 of Gabaix and Koijen (2023). The right panel plots the "volatility multiplier," defined as the ratio of return volatility to flow; this multiplier is a reduced-form analog of the price impact targeted in the calibration. Source: Flow of Funds and CRSP.

5.1 Calibration

We aggregate Flow of Funds (FoF) sectors into three ultimate sectors: a passive sector, a constrained active sector, and an unconstrained active sector.[11] The passive sector is modeled as an exogenous Ornstein-Uhlenbeck process for the portfolio share $\overline{\alpha}_{p,t}$ in Equation (10). The two active sectors differ both in risk aversion and in whether they are subject to the leverage constraint. Table 1 reports the calibrated parameter values, and the next subsection describes the moments used to discipline them.

Figure 6 motivates two of the calibration targets. The left panel plots quarterly aggregate equity flows, scaled by lagged market capitalization as in Appendix D3 of Gabaix and Koijen (2023), together with realized return volatility. The right panel reports the "volatility multiplier," defined as the ratio of realized return volatility to quarterly flows. The multiplier is a reduced-form analog of the price impact that our calibration aims to reproduce.

[11] We follow the aggregation in Koijen et al. (2024). Appendix I describes the sector mapping and the treatment of pass-through vehicles in detail.

	Parameter	Choice
	Preferences & distribution	
ψ	EIS	1.50
γ_0	Risk aversion of passive investors	10.86
γ_1	Risk aversion of unconstrained investors	25.52
γ_2	Risk aversion of constrained investors	6.15
ρ	Effective discount rate	0.01
κ	Agent turnover rate	0.10
ω_0	Share of passive agents	0.50
ω_1	Share of unconstrained agents	0.25
ω_2	Share of constrained agents	0.25
	Technology	
μ	Endowment growth rate	0.018
$\|\sigma\|$	Endowment volatility	0.033
$\|\sigma_s\|$	Dividend share volatility	0.11
$\bar{s}$	Mean dividend share	0.60
κ_s	Mean reversion of dividend share	0.09
ν_s	Corr. of dividend share and fundamental shock	0.30
	Passive demand	
$\overline{\alpha}$	Mean	0.52
θ_p	Mean reversion parameter	0.75
σ_p	Passive demand shock exposures	0.02
ν_p	Corr. of passive demand and fundamental shock	0.50
	Leverage constraints	
$\overline{\sigma}$	Tightness of the leverage constraint	0.11

Table 1. Parameter values.
This table reports the parameter values used to calibrate the baseline economy. Risk aversion parameters and sectoral masses (γ_j, ω_j) are chosen to match average sectoral wealth shares and sectoral "betas" from time-series regressions of risky-asset holdings on aggregate risky supply. Technology parameters $(\mu, \sigma, \sigma_s, \bar{s}, \kappa_s, \nu_s)$, where $\nu_s = \frac{\sigma\sigma_s^\top}{\|\sigma\|\,\|\sigma_s\|}$, are standard macro–finance targets. Passive-demand parameters $(\overline{\alpha}, \theta_p, \sigma_p, \nu_p)$, where $\nu_p = \frac{\sigma\sigma_p^\top}{\|\sigma\|\,\|\sigma_p\|}$, are disciplined by the average passive portfolio share, household portfolio inertia (Brunnermeier and Nagel, 2008), and the volatility and correlation of quarterly aggregate equity flows with market returns (Gabaix and Koijen, 2023). The leverage-constraint tightness $\overline{\sigma}$ targets the empirical asset-to-equity ratio of the constrained sector.

5.2 Moments

We discipline the calibration using three blocks of moments: technology parameters, preference and cross-sectional parameters, and the passive-demand process.

Technology. We set the endowment growth rate to $\mu = 1.8\%$ and endowment volatility to $\sigma = 3.3\%$, consistent with U.S. aggregate consumption data (Campbell and Cochrane, 1999). Following Lettau and Wachter (2011), we set the correlation between dividend-share shocks and the fundamental shock to 0.3, and the volatility of dividend-share shocks to $\sigma_s = 11\%$. We set the mean-reversion of the dividend share to $\kappa_s = 0.09$, following Santos and Veronesi (2006). Finally, we set the mean dividend share to $\bar{s} = 0.6$, which implies a debt-to-equity ratio of 0.66. This target is consistent with the average debt-to-equity ratio of the U.S. corporate sector since the 1980s.

Preferences and cross section of holdings. We set a common elasticity of intertemporal substitution $\psi = 1.5$ for all agent types, in line with the value used in Bansal and Yaron (2004). The discount rate ρ is chosen to match the average real risk-free rate, yielding $\rho = 0.01$. The sector-specific risk aversion coefficients γ_j are chosen to match sectoral *asset shares*, defined as the value of risky holdings of sector j divided by the aggregate value of risky holdings. This leads to average risk aversion of 9.7, within the range of values used in the macro-finance literature. The sector masses ω_j are chosen to match the asset-to-equity ratios of the active sectors. The agent turnover rate $\kappa = 0.10$ is calibrated so that the wealth share of the passive sector in the simulated ergodic distribution matches its empirical counterpart of about 0.45. The leverage constraint tightness $\overline{\sigma}$ is set so that the asset-to-equity ratio of the constrained sector matches its Flow of Funds counterpart.

Passive demand. The long-run mean $\overline{\alpha}$ is set to the household-weighted aggregate implied by our FoF aggregation, which yields $\overline{\alpha} = 0.52$. This value is intermediate between the one-third estimate in Chinco and Sammon (2024), inferred from index-reconstitution events, and the roughly two-thirds estimate in Koijen et al. (2024), which uses active-share measures in the spirit of Cremers and Petajisto (2009). The volatility σ_p is disciplined by the observed dispersion of quarterly aggregate equity flows shown in the left panel of Figure 6. Following Appendix D3 of Gabaix and Koijen (2023), we scale each sector's dollar equity flow by lagged aggregate market capitalization, remove mechanical revaluation, and aggregate gross flows across sectors, dividing by two to avoid double counting. The resulting series replicates the quarterly flows in their Figure D.7. The persistence θ_p

Moment	Model	Data
Panel A. Dividend-claim moments		
Risk premium (%)	6.6	6.4
Risk-free rate (%)	0.6	0.9
Return volatility (%)	22.4	18.5
Quarterly flows (%)	0.7	1.0
Price impact	6.9	4.9
Panel B. Endowment-claim moments		
Risk premium (%)	2.0	—
Return volatility (%)	4.5	—
Price impact	0.6	—
Panel C. Sector balance-sheet moments		
Assets/Equity (unconstrained)	0.5	0.5
Assets/Equity (constrained)	2.0	1.6
Asset share (passive, %)	45.4	38.8
Asset share (unconstrained, %)	22.3	20.1
Asset share (constrained, %)	32.2	29.2

Table 2. Unconditional moments.
This table reports model-implied and empirical unconditional moments for the dividend claim and the endowment claim in the baseline economy. Model moments are computed from a 1,000-year monthly simulation of the calibrated economy. The price impact is estimated as the slope coefficient $\hat{b}_1$ of the regression $\Delta \log(P/D)_t = a_0 + b_1\, \overline{\alpha}^s_{p,t}\, x_{p,t-1} + e_t$, where $\overline{\alpha}^s_{p,t}$ is the innovation to passive demand. Asset/Equity denotes the ratio of risky-asset holdings to equity for the constrained and unconstrained sectors; asset shares report the fraction of the aggregate risky market held by each sector. All values are annualized.

is informed by evidence on household portfolio inertia. Using PSID data, Brunnermeier and Nagel (2008) regress the actual change in the household risky-asset share on a counterfactual change that assumes no rebalancing, and find an inertia coefficient near 0.75, implying that portfolio shares adjust slowly to capital gains and to flow-induced changes in wealth.[12] We choose θ_p so that the half-life of the passive-demand process in the simulated model matches the half-life implied by this inertia coefficient. Finally, we estimate the loading $\nu_p = \frac{\sigma\sigma_p^\top}{\|\sigma\|\|\sigma_p\|}$ of passive-flow shocks on the aggregate shock indirectly by regressing the quarterly passive-flow series on contemporaneous market returns. This yields a model-implied correlation between flows and returns of 0.005, close to the empirical value of 0.011.

[12] Parker, Schoar, and Sun (2023) show that the adoption of target-date funds has further stabilized household portfolio shares.

5.3 Unconditional moments

Price impact regression. We simulate the calibrated economy at monthly frequency for 1,000 years and report unconditional moments in Table 2. To estimate the model-implied price impact of the dividend claim, we regress the change in the log price–dividend ratio on the innovation to passive demand, scaled by the passive sector's lagged wealth share:

$$\Delta \log(P/D)_t = a_0 + b_1 \, \overline{\alpha}^s_{p,t} \, x_{p,t-1} + e_t, \tag{20}$$

where $\overline{\alpha}^s_{p,t}$ denotes the innovation to passive demand.

Equation (20) is the model counterpart of the price-impact regression in Gabaix and Koijen (2023). Scaling by the lagged passive wealth share $x_{p,t-1}$ converts the per-unit-mass shock into an aggregate flow and mirrors the size-weighting in their granular instrumental variable (GIV) estimator.[13]

In our model, passive-demand innovations are observed and exogenous by construction, so OLS directly identifies the model-implied counterpart of the GIV estimate without requiring an instrument.

Price impact and asset-pricing moments. The model simultaneously matches a broad set of empirical facts, including asset pricing moments, the magnitude of portfolio flows, price impact, and sectoral balance sheets.

As shown in Panel A of Table 2, the model generates a large price impact, with an estimated slope $\hat{b}_1 = 6.85$, close to the empirical benchmark of about 5 (Gabaix and Koijen, 2023). For the dividend claim, the model also matches standard asset pricing moments, delivering a risk premium of 6.6%, a risk-free rate of 0.6%, and a return volatility of 22.4%. As we show below, these two successes are closely related: generating realistic price impact for the dividend claim is crucial for

[13] Gabaix and Koijen (2023), following Gabaix and Koijen (2024), decompose sector-level flows into a common component and an idiosyncratic shock, and construct the granular instrumental variable (GIV) as the difference between size-weighted and equal-weighted averages of these idiosyncratic shocks. This instrument isolates exogenous variation in aggregate flows and addresses the simultaneity bias in OLS, since prices and flows are jointly determined.

matching asset-pricing moments such as the risk premium and return volatility.

A key driver of this result is the explicit modeling of the dividend process. Panel B of Table 2 shows that the price impact for the endowment claim is only 0.57, an order of magnitude smaller than for the dividend claim. The risk premium and return volatility for the endowment claim are also much smaller than for the dividend claim. Even in a model with heterogeneity and frictions, the endowment claim remains fairly elastic.

The model also matches sectoral balance sheets, as shown in Panel C of Table 2. The constrained sector is levered (asset-to-equity ratio of 1.98 in the model versus 1.62 in the data), the unconstrained sector is conservative (0.52 versus 0.53), and the passive sector holds about 45% of risky assets.

5.4 Counterfactuals: the role of frictions

To quantify the role of each friction, we compare the baseline economy to a sequence of counterfactuals that progressively shut down individual frictions. We vary one parameter at a time, holding all others fixed at their baseline values in Table 1, so each column isolates the marginal contribution of a given friction.

The baseline economy (column 1) simultaneously delivers a large price impact, a sizable risk premium, and high return volatility, consistent with the data. Removing the leverage constraint (column 2) reduces the price impact to 4.91 and attenuates both the risk premium and return volatility, but these remain substantial. Eliminating preference heterogeneity (column 3) reduces the price impact sharply to 0.10, implying a very large market elasticity. In this case, excess volatility disappears and the model generates only a modest risk premium.

The next two counterfactuals vary passive demand. Setting $\overline{\alpha} = 1$ (column 4) raises the passive sector's exposure to risk and reduces the degree of risk misallocation, lowering the price impact to 0.25. Reducing the mass of passive investors (column 5), by setting $\omega_0 = 0.25$ and $\omega_1 = 0.75$, also lowers the price impact to 4.52, along with the risk premium and return volatility.

Finally, when passive flows are uncorrelated with fundamentals (column 6), so that $\sigma\sigma_p^\top = 0$, the model generates a small price impact for the dividend claim, with the opposite sign. This

	Baseline	No leverage constraints	No preference heterogeneity	High passive portfolio share	Low passive population share	Flows uncorrelated with fundamentals
r (%)	0.58	0.42	1.31	1.27	0.49	0.43
π_D (%)	6.66	5.89	1.21	1.52	3.62	4.51
$\sigma_{R,D}$ (%)	22.55	17.02	3.87	5.17	8.28	15.75
π_Y (%)	2.03	2.18	1.17	1.26	2.12	2.17
$\sigma_{R,Y}$ (%)	4.54	4.66	3.73	3.76	4.59	4.71
PI (Y)	0.57	0.68	0.09	0.10	1.20	0.02
PI (D)	6.85	4.91	0.10	0.25	4.52	−0.66
Cond. PI (D)	8.50	5.10	0.10	0.10	—	—
Discount-rate beta β_r	1.16	1.24	−0.02	−0.21	—	—
Γ	9.71	9.68	6.15	10.97	10.97	9.67

Table 3. Counterfactual experiments.
This table reports annualized model-implied moments for the baseline economy and five counterfactuals. *No leverage constraints* sets $\overline{\sigma} = 10{,}000$, so leverage constraints are slack. *No preference heterogeneity* sets $\gamma_c = \gamma_p = \gamma_u = 6.15$, eliminating heterogeneity in risk aversion, combined with the slack leverage constraint. *High passive portfolio share* sets $\overline{\alpha} = 1$. *Low passive population share* reduces the passive sector mass (via a lower ω_0) while increasing the mass of the unconstrained sector. *Flows uncorrelated with fundamentals* sets $\nu_p = 0$, so passive-demand shocks are orthogonal to fundamentals. All other parameters are as in Table 1. Price impacts PI (Y) and PI (D) denote the slope of the regression in equation (20) for the endowment and dividend claim, respectively. Conditional price impact restricts the regression to “bad states” in which $(P/D)_t$ is below its 25th percentile. The discount-rate coefficient β_r reports the share of the variance of the log price–dividend ratio attributable to discount-rate news at a 40-quarter horizon. The variable Γ reports the unconditional mean of the wealth-weighted harmonic mean of the risk aversion parameters.

points to the importance of imperfect rebalancing and performance-induced flows, which induce correlation between passive flows and fundamentals, in generating price impact.

These results point to a tight connection between price impact, risk premium, and return volatility. The model generates asset-pricing moments consistent with the data only when the market is sufficiently inelastic. In versions of the model with modest or no price impact, the risk premium and return volatility are also much smaller. These results highlight that matching both price impact and asset-pricing moments requires the interaction of frictions with shocks to passive demand.

Related evidence on passive behavior. These results connect closely to recent empirical evidence on the effects of passive investing. Haddad, Huebner, and Loualiche (2025b) show that the rise of passive investing has made the demand for individual stocks more inelastic. Our results extend this insight to the aggregate level, showing that the presence of passive investors plays a central role in determining aggregate market elasticity. In particular, the comparison between the baseline and the counterfactual with a low passive share illustrates that an increase in the passive sector raises price impact and reduces aggregate market elasticity.

Lochstoer and Muir (2026) provide causal evidence that the secular rise of market-index trading has roughly doubled U.S. stock market volatility since the 1950s, even as macroeconomic fundamentals have become calmer. Our counterfactuals offer a complementary structural mechanism, highlighting how an increased importance of portfolio flows amplifies both price impact and return volatility.

Parker et al. (2023) show that the introduction of target-date funds (TDFs) led to more contrarian flows, ultimately dampening volatility. In our framework, the expansion of TDFs can be interpreted as reducing the correlation between passive portfolio shares and fundamentals. Imperfect rebalancing induces a positive correlation between passive demand and return shocks, which is weakened by the rebalancing behavior of TDFs. Consistent with this evidence, our counterfactuals show that reducing this correlation lowers both price impact and return volatility.

5.5 Return predictability and the variance of the price–dividend ratio

The tight link between price impact and return volatility documented above has a natural counterpart in the literature. Since Cochrane (1992), excess volatility has been understood as reflecting variation in discount rates rather than in expected cash flows. In an inelastic market, portfolio flows move the price–dividend ratio without changing expected dividend growth, so the resulting variation must be absorbed by expected returns. The P/D ratio should therefore forecast future excess returns, with a slope that is itself informative about market inelasticity. We examine this prediction in the simulated baseline economy.

Figure 7 plots cumulative future excess returns against the current P/D ratio. Table 4 reports the corresponding predictive regression

$$\sum_{k=1}^{T} r_{t+k} - r_t = a + b\,(P/D)_t + \varepsilon_{t+T}, \qquad T = 4 \text{ quarters},$$

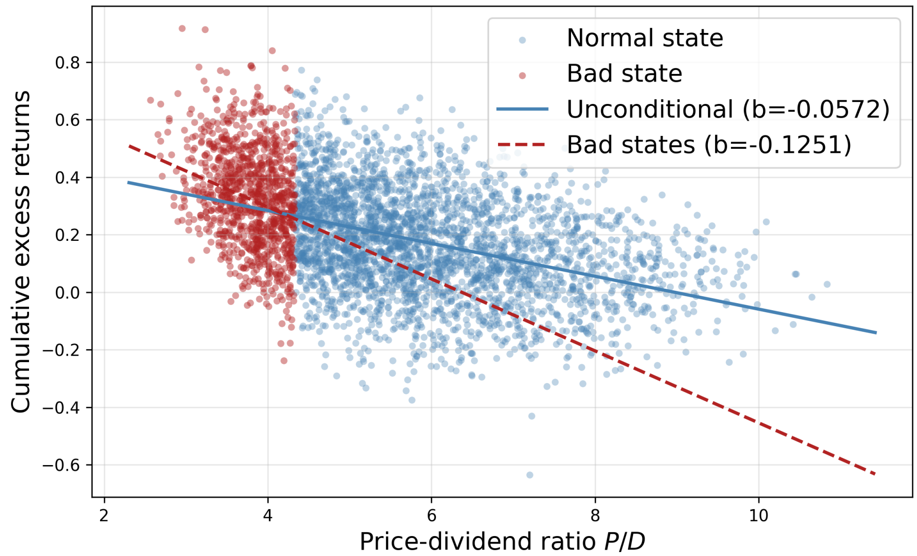


Figure 7. Return predictability.
Scatter of cumulative future excess returns ($T = 4$ quarters) against the current price–dividend ratio in the simulated baseline economy. Blue dots denote normal states; red dots denote bad states with $(P/D)_t$ below the 25th percentile of its long-run distribution. Lines show the corresponding OLS fits.

estimated on the full sample and on "bad states" in which $(P/D)_t$ falls below its 25th percentile. The slope is negative in both samples and about 2.5 times larger in magnitude in bad states. Risk premia are therefore countercyclical: when the P/D ratio is low, expected returns are high, and the P/D ratio is especially informative about them.

Following Campbell and Shiller (1988) and Cochrane (2011), we decompose the unconditional variance of $pd_t \equiv \log(P/D)_t$ as

$$\mathrm{Var}(pd_t) = \mathrm{Cov}\left(pd_t, \sum_{k=1}^{\infty} \rho^{k-1} \Delta d_{t+k}\right) - \mathrm{Cov}\left(pd_t, \sum_{k=1}^{\infty} \rho^{k-1} r_{t+k}\right).$$

Scaling by $\mathrm{Var}(pd_t)$ gives the discount-rate share

$$\beta_r = -\frac{\mathrm{Cov}\left(pd_t, \sum_{k=1}^{\infty} \rho^{k-1} r_{t+k}\right)}{\mathrm{Var}(pd_t)},$$

which we compute at a 40-quarter horizon. In the baseline economy, $\beta_r = 1.16$: nearly all variation in pd_t is driven by expected returns. The cash-flow term pushes in the opposite direction. Because the dividend share is positively correlated with fundamentals, expected dividend growth is higher

	Dependent variable: $\sum_{k=1}^{4}(r_{t+k} - r_t)$	
	(1) Full sample	(2) Bad states
Price–dividend ratio, $(P/D)_t$	−0.050	−0.120
	(0.0005)	(0.0048)
R^2	0.19	0.06
Observations	79,120	19,780
Bad-state threshold	$(P/D)_t$ < 25th percentile	

Table 4. Return predictability regression.
OLS estimates of the predictive regression $\sum_{k=1}^{T} r_{t+k} - r_t = a + b\,(P/D)_t + \varepsilon_{t+T}$ ($T = 4$ quarters), based on a 1,000-year monthly simulation of the calibrated baseline economy. Column (1): full sample. Column (2): "bad states" with $(P/D)_t$ below the 25th percentile. Standard errors in parentheses.

when pd_t is low, producing a negative cash-flow covariance that lifts β_r above one.

The counterfactuals in Table 3 trace this pattern to heterogeneity and frictions. Removing leverage constraints leaves $\beta_r = 1.24$, close to the baseline. Eliminating preference heterogeneity collapses β_r to -0.02: variation in pd_t is then driven by cash flows rather than discount rates. Setting $\overline{\alpha} = 1$, which reduces risk misallocation, yields $\beta_r = -0.21$ and a much lower price impact. Reducing the mass of passive investors or the correlation between passive demand and fundamentals has the same qualitative effect, lowering both price impact and β_r. The large baseline value of β_r is therefore a joint product of frictions and movements in passive demand, in line with the evidence in Cochrane (2011) that discount-rate variation dominates movements in U.S. equity valuation ratios.

Taken together, these results provide a demand-based explanation of the excess-volatility puzzle of Shiller (1981) and LeRoy and Porter (1981). Consistent with the inelastic market hypothesis of Gabaix and Koijen (2023), modest portfolio flows generate pronounced price fluctuations and predictable returns, even in an economy where the interest rate responds endogenously to those flows. Only economies with sufficiently inelastic markets jointly match return volatility, price impact, and return predictability, so market inelasticity is a central ingredient for matching standard macro-finance moments.

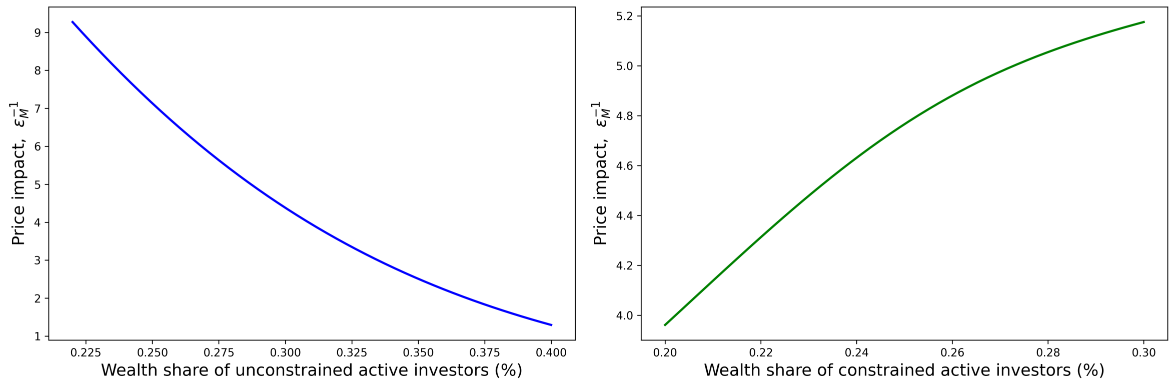


Figure 8. Conditional price impact.
Model-implied price impact of the dividend claim evaluated off the ergodic mean. In each panel, $\overline{\alpha}_{p,t}$ and s_t are fixed at their unconditional means and one wealth-share coordinate is varied. Conditional price impact is computed as the change in $\log(P/D)$ per unit of passive-flow shock, where the shock is propagated one quarter forward through the state-variable dynamics.

5.6 Conditional price impact and dynamics

Averaging across states masks the state dependence of aggregate elasticity emphasized in Section 4. To trace this dependence, we compute the model-implied *conditional* price impact off the ergodic mean, plotted in Figure 8. We fix passive demand $\overline{\alpha}_{p,t}$ and the dividend share s_t at their unconditional means, vary one wealth-share coordinate at a time, and apply a fixed one-quarter passive-flow shock. The shock is propagated through the dynamic transition of the state variables. We then recompute $\log(P/D)$ at the shocked and unshocked states, and measure the conditional price impact as the resulting change in $\log(P/D)$ per unit of passive-flow shock.

The left panel of Figure 8 shows that the price impact rises sharply as the wealth share of the unconstrained active sector $x_{u,t}$ falls. When the sector most capable of absorbing risk is small, the market is less elastic and flow shocks translate into larger price movements. The right panel shows that the price impact also rises monotonically in the wealth share of the constrained active sector $x_{c,t}$. As more of the active sector hits its leverage limit, the residual risk-bearing capacity in $x_{u,t}$ shrinks and price impact grows. These patterns mirror the analytical characterization in Proposition 5 and explain why the conditional price impact reported in Table 3 (the row labeled "Cond. price impact (D)" with a value of 8.50) is substantially larger than its unconditional counterpart.

Combined with the counterfactual results, Figure 8 makes clear that the very states in which risk-bearing capacity is depleted, with low x_u or high x_c, are the states in which discount rates are most volatile and the P/D ratio is most informative about future returns. The large unconditional β_r in Table 3 is therefore driven by these tail states.

5.7 Takeaways

Three central results emerge from the quantitative analysis. First, when calibrated to sectoral FoF holdings and passive equity flows, the model jointly matches aggregate price impact, the volatility of dividend-claim returns, and the risk premium. Second, heterogeneity and risk misallocation are essential for matching the asset-pricing moments. Removing either preference heterogeneity or risk misallocation collapses price impact, return volatility, and return predictability. Third, leverage constraints further amplify these effects by tightening risk-sharing when risk-bearing capacity is low. Finally, the model delivers a demand-based explanation of the excess-volatility puzzle and return predictability, based on how portfolio flows interact with inelastic markets.

6 Conclusion

This paper develops a general equilibrium model with heterogeneous investors, passive demand, and financial constraints to study the determinants of aggregate market elasticity. We show how general equilibrium adjustments, particularly the joint movement of the risk-free rate and the risk premium, fundamentally shape the price impact of portfolio flows.

A central insight is the role of cross-price elasticity. When demand for risky assets responds to interest rates, shifts in the risk premium are offset by changes in the risk-free rate. In this case, the market can be infinitely elastic even if individual investors are highly inelastic. Aggregate elasticity becomes finite only when risk is initially misallocated: portfolio flows then alter aggregate savings behavior, preventing interest rates from fully offsetting risk-premium movements.

Macro elasticity is both state-dependent and time-varying. Passive investors amplify price impacts, leverage constraints further tighten risk-bearing capacity, while preference heterogeneity increases elasticity by improving risk allocation. Inelasticity alone does not generate excess volatility, nor do flows in infinitely elastic markets; both elements are needed to explain observed fluctuations and the countercyclical volatility multiplier.

Our analytical results rely on a state-global perturbation method that provides closed-form characterizations of elasticity, while our quantitative analysis solves the full dynamic model numerically. The calibrated model highlights how the wealth distribution across households, intermediaries, and long-term investors shapes aggregate elasticity and volatility.

Overall, our findings suggest that market inelasticity is best understood as a symptom of inefficient risk allocation, rather than a structural feature of financial markets. Improving the allocation of risk across sectors may therefore be more effective for reducing excess volatility than targeting individual investor behavior. Future work could extend this framework to study asset pricing anomalies, monetary policy transmission, and the role of market structure in shaping aggregate elasticity.

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Appendix

A Derivations for Section 2

A.1 The bond demand view

Define the normalized bond demand $b_j \equiv \frac{R_f^{-1} B_j}{W_j} = 1 - c_j(r, \pi) - \alpha_j(1 - c_j(r, \pi))$ as the bond-to-wealth ratio:

$$b_j = (1 - \alpha_j)(1 - c_j(r, \pi)) = 1 - c_j(r, \pi) - \frac{PQ_j}{W_j}.$$

Notice that the bond demand depends not only on the interest rate, but also on the price of the risky asset, that is, the cross-elasticity will generically be non-zero.

The market clearing condition for bonds can be expressed as follows

$$\underbrace{x_a b_a}_{\textbf{active bond demand}} = \underbrace{-x_p b_p}_{\textbf{net bond supply}},$$

where $x_j \equiv \frac{\omega_j W_j}{\omega_a W_a + \omega_p W_p} = \omega_j Q_{j,-1}$ denotes the wealth share of investor j.

Notice that $\frac{PQ_j}{W_j} = \frac{P}{Y+P} \frac{\omega_j Q_j}{x_j}$. We can then write the bond market-clearing condition as follows:

$$1 - \overline{c}(\mu - p) - \frac{e^p}{1 + e^p}\left[\omega_a Q_a + \omega_p Q_p\right] = 0 \iff \underbrace{\frac{1}{1 + e^p} - \overline{c}(\mu - p)}_{\text{excess supply of goods}} - \frac{e^p}{1 + e^p}\left[\omega_a Q_a + \omega_p Q_p - 1\right] = 0$$

The left-hand side of the two expressions above gives you total bond demand. In the partial equilibrium case, the risk premium adjusts to clear the market for risky assets, so $\omega_a Q_a + \omega_p Q_p = 1$, keeping the interest rate fixed. But, due to the cross-elasticity in bond demand, a change in the price of the risky asset — keeping the interest rate fixed — would create a disequilibrium in the bond market. For instance, in the case $\chi_p \equiv -\overline{c}'(\mu - p) > 0$, a decrease in the price of the risky asset would create an insufficient demand for goods and an excess demand for bonds. The excess demand for bonds would create an incentive to reduce the interest rate,

B Derivations for Section 3

B.1 Investors' problem

The Hamilton-Jacobi-Bellman (HJB) equation for investor j can be written as

$$0 = \max_{C_j,\alpha_j} f_j(C_j, V_j) + V_{j,W}\left[rW_j + \pi^\top \alpha_j W_j - C_j\right] + V_{j,X}\mu_X + \frac{1}{2}V_{j,WW}W_j^2\|\sigma_j\|^2 + V_{j,WX}W_j\sigma_X\sigma_j^\top + \frac{1}{2}\sum_{k=1}^{3}\sigma_{X,k}^\top V_{j,XX}\sigma_{X,k},$$

subject to

$$\alpha_j = \overline{\alpha}_p[\alpha_{D,t}^m, \alpha_{B,t}^m, 0]^\top \qquad \text{for } j = 0, \qquad\qquad \sigma_j\sigma_j^\top \leq \overline{\sigma}^2 \qquad \text{for } j = 2.$$

given $\sigma_j = \alpha_j^\top\Sigma$, where $\alpha_{k,t}^m \equiv \frac{P_{k,t}}{P_{Y,t}}$ for $k \in \{D, B\}$. For ease of notation, we dropped time subscripts. Note that $V_{j,X}$ and $V_{j,WX}$ are $1 \times N$ vectors, $V_{j,XX}$ is a $N \times N$ matrix, and both $V_{j,W}$ and $V_{j,WW}$ are scalars. The drift μ_X is a $N \times 1$ vector, the diffusion σ_X is a $N \times 3$ matrix, while σ_R is a 1×3 vector. The notation $\sigma_{X,k}$ denotes the $k-$th column of σ_X, that is, the exposure to the $k-$th Brownian motion.

The optimality condition for consumption and portfolio share are given by

$$\begin{aligned} f_{j,C}(C_j, V_j) &= V_{j,W} \\ V_{j,W}\pi &= -V_{j,WW}W_j\Sigma\Sigma^\top\alpha_j - \Sigma(V_{j,WX}\sigma_X)^\top + \tilde{\nu}_{j,t}\Sigma\Sigma^\top\alpha_j. \end{aligned}$$

where the second optimality condition holds for $j = 1, 2$, and $\tilde{\nu}_{j,t}$ is the Lagrange multiplier on the leverage constraint.

The optimal consumption is given by

$$C_j = \rho^\psi((1-\gamma_j)V_j)^{\frac{1-\gamma_j\psi}{1-\gamma_j}}V_{j,W}^{-\psi}.$$

The optimal portfolio share for an active investor is given by

$$\sigma_j = -\frac{V_{j,W}}{V_{j,WW}W_j + \tilde{\nu}_{j,t}}\eta_j - \frac{V_{j,WX}}{V_{j,WW}W_j + \tilde{\nu}_{j,t}}\sigma_X.$$

using the fact that $\pi = \Sigma\eta^\top$.

Given the homotheticity of preferences, the value function for investor j can be written as

$$V_{j,t} = \left(\frac{c_{j,t}}{\rho^{\psi}}\right)^{\frac{1-\gamma_j}{1-\psi}} \frac{W_{j,t}^{1-\gamma_j}}{1-\gamma_j}. \tag{B.1}$$

This particular parametrization of the value function implies that the consumption-wealth ratio is given by

$$\frac{C_{j,t}}{W_{j,t}} = c_{j,t}.$$

The optimal risk exposure for active investors is given by

$$\sigma_{j,t} = \frac{1}{1+\nu_{j,t}}\left[\frac{\eta_t}{\gamma_j} + \frac{1-\gamma_j^{-1}}{\psi-1}\sigma_{c_{j,t}}\right],$$

where $\nu_{j,t} \equiv \tilde{\nu}_{j,t}/(V_{j,WW,t}W_{j,t})$ is the normalized Lagrange multiplier on the leverage constraint.

Plugging the consumption-wealth ratio into the HJB equation and using Equation (B.1), we obtain

$$\begin{aligned} 0 = \frac{\rho}{1-\psi^{-1}}\left[\rho^{-1}c_j - 1\right] + r + \eta\sigma_j^{\top} - c_j + \frac{1}{1-\psi}\left[\frac{\xi_{j,X}}{\xi_j}\mu_X + \frac{1}{2}\sum_{k=1}^{d}\sigma_{X,k}^{\top}\frac{\xi_{j,XX}}{\xi_j}\sigma_{X,k}\right] \\ - \frac{\gamma_j}{2}\|\sigma_j\|^2 + \frac{1-\gamma_j}{1-\psi}\frac{c_{j,X}}{c_j}\sigma_X\sigma_j^{\top} + \frac{1}{2}\frac{\psi-\gamma_j}{(1-\psi)^2}\sum_{k=1}^{d}\sigma_{X,k}^{\top}\frac{c_{j,X}^{\top}}{c_j}\frac{c_{j,X}}{c_j}\sigma_{X,k}. \end{aligned}$$

Rearranging the expression above, we obtain

$$c_{j,t} = \psi\rho + (1-\psi)\left[r_t + \eta_t\sigma_{j,t}^{\top} - \frac{\gamma_j}{2}\|\sigma_{j,t}\|^2\right] + \mu_{c_{j,t}} + (1-\gamma_j)\sigma_{c_{j,t}}\sigma_{j,t}^{\top} + \frac{\psi-\gamma_j}{1-\psi}\frac{\|\sigma_{c_{j,t}}\|^2}{2},$$

where the law of motion of $c_{j,t}$ is given by

$$\frac{dc_{j,t}}{c_{j,t}} = \mu_{c_{j,t}}dt + \sigma_{c_{j,t}}dZ_t,$$

and the drift and diffusion of $c_{j,t}$ are given by Ito's lemma:

$$\mu_{c_{j,t}} = \frac{c_{j,X}}{c_j}\mu_{X,t} + \frac{1}{2}\sum_{k=1}^{d}\sigma_{X,k,t}^{\top}\frac{c_{j,XX,t}}{c_{j,t}}\sigma_{X,k,t}, \qquad \sigma_{c_{j,t}} = \frac{c_{j,X}}{c_j}\sigma_{X,t}.$$

B.2 Pricing condition

Let $p_{D,t} = \frac{P_{D,t}}{Y_t}$ and $p_{B,t} = \frac{P_{B,t}}{Y_t}$ denote the normalized prices of the dividend and bond claims, respectively. The normalized price of the aggregate claim is $p_{Y,t} = \frac{P_{Y,t}}{Y_t}$. The pricing condition for asset $k \in \{D, B\}$ is given by

$$r_t + \sigma_{R_k,t}\eta_t^\top = \frac{1}{p_{k,t}} + \mu + \mu_{p_k,t} + \sigma\sigma_{p_k,t}^\top,$$

where $\sigma_{R,t} = \sigma + \sigma_{p_k,t}$ and $(\mu_{p_k,t}, \sigma_{p_k,t})$ are given by Ito's lemma:

$$\mu_{p_k,t} = \frac{p_{k,X,t}}{p_{k,t}}\mu_{X,t} + \frac{1}{2}\sum_{k=1}^{d}\sigma_{X,k,t}^\top \frac{p_{k,XX,t}}{p_{k,t}}\sigma_{X,k,t}, \qquad \sigma_{p_k,t} = \frac{p_{k,X,t}}{p_{k,t}}\sigma_{X,t}.$$

The expected return and risk exposure on the aggregate claim are given by $\mu_{R_Y,t} = \alpha_{D,t}^m\mu_{R_D,t} + \alpha_{B,t}^m\mu_{R_B,t}$ and $\sigma_{R_Y,t} = \alpha_{D,t}^m\sigma_{R_D,t} + \alpha_{B,t}^m\sigma_{R_B,t}$, respectively.

B.3 Aggregate state variable

Define the share of wealth of type-j investors as follows

$$x_{j,t} \equiv \frac{\omega_j W_{j,t}}{\sum_{i=0}^2 \omega_i W_{i,t}}.$$

We define the aggregate state variable as $X_t = (x_{1,t}, x_{2,t}, \overline{\alpha}_{p,t}, s_t)$.

Market clearing conditions. We can write the market clearing condition for the equity claim, corporate bond claim, and short-term bonds as follows:

$$\sum_{j=0}^{2}\omega_j W_{j,t}\alpha_{j,k,t} = P_{k,t}, \qquad \sum_{j=0}^{2}\omega_j W_{j,t}(1 - \alpha_{j,D,t} - \alpha_{j,B,t}) = 0,$$

for $k \in \{D, B\}$.

Using the fact that $P_{Y,t} = P_{D,t} + P_{B,t}$, we obtain:

$$\sum_{j=0}^{2}\omega_j W_{j,t} = P_{Y,t}.$$

The market clearing condition for goods is given by

$$\sum_{j=0}^{2} x_{j,t} c_{j,t} = \frac{1}{p_{Y,t}}.$$

We can then write the market clearing condition for the equity claim, corporate bond claim, and the derivative as follows:

$$\sum_{j=0}^{2} x_{j,t} \alpha_{j,D,t} = \alpha_{D,t}^m, \qquad \sum_{j=0}^{2} x_{j,t} \alpha_{j,B,t} = \alpha_{B,t}^m, \qquad \sum_{j=0}^{2} x_{j,t} \alpha_{j,F,t} = 0.$$

Let $\alpha_t^m = [\alpha_{D,t}^m, \alpha_{B,t}^m, 0]^\top$ denote the market weights of the dividend, bond, and derivative claims. We can then write the market clearing conditions above in a compact way:

$$\sum_{j=0}^{2} x_{j,t} \alpha_{j,t} = \alpha_t^m.$$

Multiplying both sides by Σ and using the fact that $\sigma_{R_Y,t} = \omega_{D,t}\sigma_{R_D,t} + \omega_{B,t}\sigma_{R_B,t}$, we obtain

$$\sum_{j=0}^{2} x_{j,t} \sigma_{j,t} = \sigma_{R_Y,t}.$$

Law of motion of the aggregate state variable. The law of motion of $\overline{\alpha}_{p,t}$ is given by (10). The law of motion of s_t is given by (9). To compute the law of motion of $x_{j,t}$, first, note that the law of motion of wealth for a type-j investor can be written as

$$\frac{dW_{j,t}}{W_{j,t}} = \left[r_t + \eta_t \sigma_{j,t}^\top - c_{j,t}\right] dt + \sigma_{j,t} dZ_t.$$

From Ito's lemma, the law of motion of $x_{j,t}$ is given by

$$\frac{dx_{j,t}}{x_{j,t}} = \left[r_t + \pi_t^\top \alpha_{j,t} - c_{j,t} - \mu_{P_Y,t} - (\sigma_{j,t} - \sigma_{R_Y,t})\, \sigma_{R_Y,t}^\top + \kappa \frac{\omega_j - x_{j,t}}{x_{j,t}}\right] dt + (\sigma_{j,t} - \sigma_{R_Y,t})\, dZ_t.$$

using $\mu_{P_Y,t} = \mu_{R_Y,t} - \mu_{R_D,t}\alpha_{D,t}^m - \mu_{R_B,t}\alpha_{B,t}^m$.

B.4 Block recursivity and the mutual fund theorem

To compute the equilibrium, one needs to solve a system of 5 partial differential equations (PDEs), involving the three consumption-wealth ratios $\{c_{j,t}(X_t)\}_{j=0}^2$ and the two price-dividend ratios $\{p_{k,t}(X_t)\}_{k=D,B}$. These functions depend on 4 state variables, the 2−dimensional vector x_t, the portfolio share of passive investors $\overline{\alpha}_{p,t}$, and the dividend share s_t.

The system of PDEs satisfies an important *block-recursive* structure. We can first solve for a reduced system of PDEs involving (c_j, p_Y) in the set of states $\hat{X}_t = (x_{1,t}, x_{2,t}, \overline{\alpha}_{p,t})$. Given the solution of the reduced system of PDEs, we can then solve for $p_{D,t}$ and $p_{B,t}$ in the full set of states $X_t = (x_{1,t}, x_{2,t}, \overline{\alpha}_{p,t}, s_t)$.

B.4.1 Block recursivity

We next derive the system of PDEs. We will show that one can solve for a reduced system of PDEs involving (c_j, p_Y). This is the crucial in the obtaining the mutual fund theorem in our setting.

Lemma B.1. *The consumption-wealth ratios, $\{c_{j,t}(X_t)\}_{j=0}^{2}$, and the price-dividend ratio for the aggregate claim, $p_Y(X_t)$, satisfy the following system of PDEs:*

$$c_{j,t} = \psi\rho + (1-\psi)\left[r_t + \eta_t\,\sigma_{j,t}^\top - \frac{\gamma_j}{2}\|\sigma_{j,t}\|^2\right] + \mu_{c_j,t} + (1-\gamma_j)\sigma_{c_j,t}\sigma_{j,t}^\top + \frac{\psi-\gamma_j}{1-\psi}\frac{\|\sigma_{c_j,t}\|^2}{2}$$

$$\frac{1}{p_{Y,t}} = r_t + \eta_t\,(\sigma + \sigma_{p_Y,t})^\top - \left(\mu + \mu_{p_Y,t} + \sigma\,\sigma_{p_Y,t}^\top\right),$$

where $(r_t, \eta_t, \sigma_{j,t}, \mu_{c_j,t}, \sigma_{c_j,t}, \mu_{p_Y,t}, \sigma_{p_Y,t})$ are all functions of X_t, (c_j, p_Y) and their derivatives.

Proof. We proceed in four steps. First, we show how to solve for the diffusion of the endogenous state variables, $\sigma_{x_1,t}$ and $\sigma_{x_2,t}$, in terms of X_t, (c_j, p_Y) and their derivatives. Second, we show how to solve for the drift of the endogenous state variables, $\mu_{x_1,t}$ and $\mu_{x_2,t}$, again in terms of X_t, (c_j, p_Y) and their derivatives. Third, we show how to solve for the interest rate, r_t. Finally, we show how to obtain the PDEs for (c_j, p_Y).

In the derivations below, we drop the explicit dependence on X for brevity.

Step 1: Risk exposure of endogenous state variables. Consider the expression for the risk exposure of the aggregate claim and the consumption-wealth ratio:

$$\sigma_{R_Y} = \sigma + \sum_{k=1}^{2}\frac{p_{Y,x_k}}{p_Y}\sigma_{x_k} + \frac{p_{Y,\overline{\alpha}_p}}{p_Y}\sigma_{\overline{\alpha}_p} + \frac{p_{Y,s}}{p_Y}\sigma_s,$$

$$\sigma_{c_j} = \sum_{k=1}^{2}\frac{c_{j,x_k}}{c_j}\sigma_{x_k} + \frac{c_{j,\overline{\alpha}_p}}{c_j}\sigma_{\overline{\alpha}_p} + \frac{c_{j,s}}{c_j}\sigma_s,$$

Notice that we can write σ_{R_Y} as follows:

$$\sigma_{R_Y} = \chi_{\sigma_{R_Y},0} + \chi_{\sigma_{R_Y},1}\sigma_{x_1} + \chi_{\sigma_{R_Y},2}\sigma_{x_2},$$

where the coefficients $\chi_{\sigma_{R_Y},k}$, $k = 0, 1, 2$, are functions of X given by

$$\chi_{\sigma_{R_Y},0} = \sigma + \frac{p_{Y,\overline{\alpha}_p}}{p_Y}\sigma_{\overline{\alpha}_p} + \frac{p_{Y,s}}{p_Y}\sigma_s,$$
$$\chi_{\sigma_{R_Y},k} = \frac{p_{Y,x_k}}{p_Y}, k = 1, 2.$$

Similarly, we can write σ_{c_j} as follows:

$$\sigma_{c_j} = \chi_{\sigma_{c_j},0} + \chi_{\sigma_{c_j},1}\sigma_{x_1} + \chi_{\sigma_{c_j},2}\sigma_{x_2},$$

where the coefficients $\chi_{\sigma_{c_j},k}$, $k = 0, 1, 2$, are functions of X given by

$$\chi_{\sigma_{c_j},0} = \frac{c_{j,\overline{\alpha}_p}}{c_j}\sigma_{\overline{\alpha}_p} + \frac{c_{j,s}}{c_j}\sigma_s$$
$$\chi_{\sigma_{c_j},k} = \frac{c_{j,x_k}}{c_j}, k = 1, 2.$$

The market price of risk

$$\eta = \gamma_a \left[\chi_a\sigma_{R_Y} - \sum_{j=0}^{2} x_j \frac{1-\gamma_j^{-1}}{\psi - 1}\frac{\sigma_{c_j}}{1+\nu_j}\right],$$

where $\chi_a = \frac{1-(1-x_1-x_2)\overline{\alpha}_p}{x_1+x_2}$ is a known function of the aggregate state variables X.

We can then write η as follows:

$$\eta = \chi_{\eta,0} + \chi_{\eta,1}\sigma_{x_1} + \chi_{\eta,2}\sigma_{x_2},$$

where the coefficients $\chi_{\eta,k}$, $k = 0, 1, 2$, are functions of X given by

$$\chi_{\eta,k} = \gamma_a \left[\chi_a\chi_{\sigma_{R_Y},k} - \sum_{j=0}^{2} x_j \frac{1-\gamma_j^{-1}}{\psi - 1}\frac{\chi_{\sigma_{c_j},k}}{1+\nu_j}\right].$$

The risk exposure of investor $j = 0, 1, 2$ is given by

$$\sigma_0 = \overline{\alpha}_p\sigma_{R_Y}$$
$$\sigma_j = \frac{1}{1+\nu_j}\left[\frac{\eta}{\gamma_j} + \frac{1-\gamma_j^{-1}}{\psi - 1}\sigma_{c_j}\right], \qquad j = 1, 2.$$

We can then solve for σ_j in terms of the risk exposure of the endogenous state variables:

$$\sigma_j = \chi_{\sigma_j,0} + \chi_{\sigma_j,1}\sigma_{x_1} + \chi_{\sigma_j,2}\sigma_{x_2},$$

where the coefficients $\chi_{\sigma_j,k}$, $k = 0, 1, 2$, are functions of X given by

$$\begin{aligned}
\chi_{\sigma_0,k} &= \overline{\alpha}_p \chi_{\sigma_{R_Y},k} \\
\chi_{\sigma_j,k} &= \frac{1}{1+\nu_j}\left[\frac{\chi_{\eta,k}}{\gamma_j} + \frac{1-\gamma_j^{-1}}{\psi - 1}\chi_{\sigma_{c_j},k}\right], \qquad j = 1, 2.
\end{aligned}$$

The risk exposure of the endogenous state variables is given by

$$\sigma_{x_j} = x_j\left[\sigma_j - \sum_{k=0}^{2} x_k \sigma_k\right],$$

using the fact that $\sigma_{R_Y} = \sum_{k=0}^{2} x_k \sigma_k$.

This gives us a system of equations determining $\sigma_{x_j,t}$:

$$\begin{bmatrix} 1 - x_1 \sum_{k=0}^{2}\left(\chi_{\sigma_1,1} - \chi_{\sigma_k,1}\right) x_k & -x_1 \sum_{k=0}^{2}\left(\chi_{\sigma_1,2} - \chi_{\sigma_k,2}\right) x_k \\ -x_2 \sum_{k=0}^{2}\left(\chi_{\sigma_2,1} - \chi_{\sigma_k,1}\right) x_k & 1 - x_2 \sum_{k=0}^{2}\left(\chi_{\sigma_2,2} - \chi_{\sigma_k,2}\right) x_k \end{bmatrix} \begin{bmatrix} \sigma_{x_1} \\ \sigma_{x_2} \end{bmatrix} = \begin{bmatrix} x_1 \sum_{k=0}^{2}\left(\chi_{\sigma_1,0} - \chi_{\sigma_k,0}\right) x_k \\ x_2 \sum_{k=0}^{2}\left(\chi_{\sigma_2,0} - \chi_{\sigma_k,0}\right) x_k \end{bmatrix}.$$

Solving the system above, we obtain σ_{x_j} as a function of X and the functions c_j and p_Y and their derivatives. Given σ_{x_j}, we can solve for σ_j, η, σ_{c_j}, and σ_{R_Y} using the expressions above. The risk premium on the aggregate claim is given by $\pi_Y = \sigma_{R_Y}\eta^\top$.

Step 2: Drift of endogenous state variables. Given $(\sigma_j, \eta, \sigma_{R_Y})$, we can solve for the drift of the endogenous state variables:

$$\mu_{x_j} = x_j\left[(\eta - \sigma_{R_Y})\left(\sigma_j - \sigma_{R_Y}\right)^\top - \left(c_j - \frac{1}{p_Y}\right)\right] + \kappa(\omega_j - x_j),$$

This gives us μ_X and σ_X in terms of X, (c_j, p_Y) and their derivatives. We can then solve for the drift of c_j and p_Y using Ito's lemma:

$$\begin{aligned}
\mu_{c_j} &= \frac{c_{j,X}}{c_j}\mu_X + \frac{1}{2}\sum_{k=1}^{3}\sigma_{X,k}^\top \frac{c_{j,XX}}{c_j}\sigma_{X,k}, \\
\mu_{p_Y} &= \frac{p_{Y,X}}{p_Y}\mu_X + \frac{1}{2}\sum_{k=1}^{3}\sigma_{X,k}^\top \frac{p_{Y,XX}}{p_Y}\sigma_{X,k}.
\end{aligned}$$

Step 3: Interest rate. Given the expressions derived in the previous steps, we can solve for the shifter of the consumption-wealth ratio $\xi_{j,t}$:

$$\xi_j = \mu_{c_j} + (1-\gamma_j)\sigma_{c_j}\sigma_j^\top + \frac{\psi - \gamma_j}{1-\psi}\frac{\|\sigma_{c_j}\|^2}{2}.$$

We can then solve for $\xi = \sum_{j=0}^{2} x_j \xi_j$ and the interest rate:

$$r = \rho + \psi^{-1}\left(\mu_{p_Y} + \mu + \sigma\sigma_{p_Y}^\top + \xi\right) + (1-\psi^{-1})\sum_{j=0}^{2} x_j \frac{\gamma_j}{2}\|\sigma_j\|^2 - \pi_Y.$$

Step 4: PDEs for the consumption-wealth ratio and price-dividend ratio. From the expression for the consumption-wealth ratio and the pricing condition for the aggregate claim, we can write the PDE for c_j and p_Y as follows:

$$\begin{aligned}
c_j &= \psi\rho + (1-\psi)\left[r + \eta\,\sigma_j^\top - \frac{\gamma_j}{2}\|\sigma_j\|^2\right] + \mu_{c_j} + (1-\gamma_j)\sigma_{c_j}\sigma_j^\top + \frac{\psi-\gamma_j}{1-\psi}\frac{\|\sigma_{c_j}\|^2}{2} \\
\frac{1}{p_Y} &= r + \eta\,\sigma_{R_Y}^\top - (\mu + \mu_{p_Y} + \sigma\,\sigma_{p_Y}^\top),
\end{aligned}$$

Notice that $(r, \eta, \sigma_j, \mu_{c_j}, \sigma_{c_j}, \mu_{p_Y}, \sigma_{p_Y})$ are all functions of X, (c_j, p_Y) and their derivatives. Hence, these equations form a system of PDEs for (c_j, p_Y). □

A corollary of the above proposition is that the consumption-wealth ratio and price-dividend ratio are independent of the dividend share s.

Corollary 1. *The consumption-wealth ratio c_j and price-dividend ratio p_Y are a function of the reduced state vector $\hat{X} = (x_1, x_2, \overline{\alpha}_p)$.*

Proof. The dividend share s does not appear in the expressions for PDE system for (c_j, p_Y). They only appear indirectly through μ_s and σ_s. If (c_j, p_Y) are a function of $\hat{X}$, then their derivatives with respect to s are zero, and s completely drops out of the PDE system for (c_j, p_Y). □

B.4.2 The mutual fund theorem

Lemma B.1 and its corollary show that the consumption-wealth ratio and price-dividend ratio of the aggregate claim are independent of the dividend share s. This implies that investor's consumption or wealth choices are not exposed to idiosyncratic dividend-share shocks:

$$\sigma_{j,t,3} = \sigma_{R_Y,t,3} = \sigma_{R_F,t,3} = 0,$$

where the third component of the risk exposure vector is the dividend-share shock.

We can implement the risk exposure $\sigma_{j,t}$ by holding the market index fund and a derivative. Let $\hat{\sigma}_{j,t} = [\sigma_{j,t,1}, \sigma_{j,t,2}]$, $\hat{\Sigma}_t = \begin{bmatrix} \hat{\sigma}_{R_Y,t} \\ \hat{\sigma}_{R_F,t} \end{bmatrix}$, and $\hat{\alpha}_{j,t} = [\alpha_{j,Y,t}, \alpha_{j,F,t}]^\top$, then we can define the portfolio shares of the market index fund and the derivative as:

$$\hat{\sigma}_{j,t} = \hat{\alpha}_{j,t}^\top \hat{\Sigma}_t \Rightarrow \hat{\alpha}_{j,t}^\top = \hat{\sigma}_{j,t} \hat{\Sigma}_t^{-1}.$$

We can then write the risk exposure $\sigma_{j,t}$ as:

$$\sigma_{j,t} = \alpha_{j,Y,t}\sigma_{R_Y,t} + \alpha_{j,F,t}\sigma_{R_F,t}.$$

Using the fact that $\sigma_{R_Y,t} = \frac{P_{D,t}}{P_{Y,t}}\sigma_{R_D,t} + \frac{P_{B,t}}{P_{Y,t}}\sigma_{R_B,t}$, we obtain:

$$\sigma_{j,t} = \alpha_{j,Y,t}\frac{P_{D,t}}{P_{Y,t}}\sigma_{R_D,t} + \alpha_{j,Y,t}\frac{P_{B,t}}{P_{Y,t}}\sigma_{R_B,t} + \alpha_{j,F,t}\left(\frac{P_{B,t}}{P_{Y,t}}\sigma_{R_Y,t} + \frac{P_{D,t}}{P_{Y,t}}\sigma_{R_D,t}\right).$$

Hence, this implies that the portfolio shares of dividend and bond claims are given by:

$$\alpha_{j,D,t} = \alpha_{j,Y,t}\frac{P_{D,t}}{P_{Y,t}}, \quad \alpha_{j,B,t} = \alpha_{j,Y,t}\frac{P_{B,t}}{P_{Y,t}},$$

where $\alpha_{j,Y,t}$ is the portfolio share of the market index fund defined.

Pricing the individual claims. The normalized price of the dividend claim satisfies the pricing condition:

$$\frac{s_t}{p_{D,t}} = r_t + \sigma_{R_D,t}\eta_t^\top - \left(\mu + \mu_{p_D,t} + \sigma\, \sigma_{p_D,t}^\top\right),$$

where

$$\mu_{p_{D,t}} = \frac{p_{D,X,t}}{p_{D,t}}\mu_{X,t} + \frac{1}{2}\sum_{k=1}^{3}\sigma_{X,k,t}^\top \frac{p_{D,XX,t}}{p_{D,t}}\sigma_{X,k,t}, \qquad \sigma_{p_{D,t}} = \frac{p_{D,X,t}}{p_{D,t}}\sigma_{X,t}.$$

C Derivations for Section 4

In this section, we discuss the derivations of the perturbation solution in detail. Consistent with the block-recursivity property in Proposition 2, the consumption-wealth ratios and the price-dividend ratio of the aggregate claim are independent of the dividend share s.

For these variables, we can focus on the reduced set of states $\hat{X} = (x, \overline{\alpha}_p)$, where $x = (x_1, x_2)$ is the vector of wealth shares of the active investors. To price the individual claims, we need to consider the full set of states $X = (x, \overline{\alpha}_p, s)$.

The perturbation is defined by the following parameter restrictions:

$$\gamma_j = \gamma(1 + \hat{\gamma}_j \epsilon), \qquad \overline{\sigma} = \|\sigma\|(1 + \hat{\sigma}\epsilon), \qquad \kappa = \hat{\kappa}\epsilon,$$

where $\sum_{j=0}^{2} \omega_j \hat{\gamma}_j = 0$.

We also make assumptions about the dynamics of the exogenous state variables. First, the portfolio share of passive investors is written as $\overline{\alpha}_p = 1 + \hat{\alpha}_p \epsilon$, where $\hat{\alpha}_p$ follows the process

$$d\overline{\alpha}_p = \kappa_p(\overline{\alpha} - \overline{\alpha}_p)dt + \sigma_p \epsilon dZ_t,$$

where $\overline{\alpha} = 1 + \hat{\alpha}\epsilon$. Equivalently, $d\hat{\alpha}_p = \kappa_p(\hat{\alpha} - \hat{\alpha}_p)dt + \sigma_p dZ_t$.

Second, the dividend share is written as $s = \overline{s} + \hat{s}\epsilon$, where $\hat{s}$ follows the process induced by

$$ds = \kappa_s(\overline{s} - s)dt + s(1 - s)\sigma_s \epsilon dZ_t.$$

The case $\epsilon = 0$ corresponds to the benchmark economy, in which there is no preference heterogeneity and the exogenous state variables are constant. In particular, passive investors are fully invested in the risky asset and the dividend share is fixed at $s = \overline{s}$.

We expand the consumption-wealth ratio of investor $j \in \{0, 1, 2\}$ and the price-dividend ratio of the aggregate claim as follows:

$$c_j(x, \overline{\alpha}_p; \epsilon) = c_{j,0}(x, \hat{\alpha}_p) + c_{j,1}(x, \hat{\alpha}_p)\epsilon + c_{j,2}(x, \hat{\alpha}_p)\epsilon^2 + O(\epsilon^3),$$
$$p_Y(x, \overline{\alpha}_p; \epsilon) = p_{Y,0}(x, \hat{\alpha}_p) + p_{Y,1}(x, \hat{\alpha}_p)\epsilon + p_{Y,2}(x, \hat{\alpha}_p)\epsilon^2 + O(\epsilon^3).$$

For each individual claim $k \in \{D, B\}$, we use the expansion

$$p_k(x, \overline{\alpha}_p, s; \epsilon) = p_{k,0}(x, \hat{\alpha}_p, \hat{s}) + p_{k,1}(x, \hat{\alpha}_p, \hat{s})\epsilon + p_{k,2}(x, \hat{\alpha}_p, \hat{s})\epsilon^2 + O(\epsilon^3).$$

The following lemma summarizes some basic properties of the perturbation solution that will be useful for the derivations below.

Lemma C.1 (Basic properties of the perturbation solution)**.** *The perturbation solution satisfies the following properties:*

1. *The zeroth-order terms are constant:*

$$c_{j,0}(x, \hat{\alpha}_p) = c_j^*, \qquad p_{Y,0}(x, \hat{\alpha}_p) = p_Y^*, \qquad p_{k,0}(x, \hat{\alpha}_p, \hat{s}) = p_k^*, \qquad k \in \{D, B\},$$

for some constants c_j^*, p_Y^*, p_k^* independent of $(x, \hat{\alpha}_p, \hat{s})$.

2. *The drift and diffusion of the state variables are first-order in ϵ:*

$$\mu_X = \mathcal{O}(\epsilon), \qquad \sigma_X = \mathcal{O}(\epsilon).$$

3. *The first-order and second-order corrections to the consumption-wealth ratio and the price-dividend ratio for the aggregate claim can be written as:*

$$c_{j,1}(x, \hat{\alpha}_p) = \psi_{j,0}(x) + \psi_{j,\hat{\alpha}_p}(x)\hat{\alpha}_p, \qquad c_{j,2}(x, \hat{\alpha}_p) = \Psi_{j,0}(x) + \Psi_{j,\hat{\alpha}_p}(x)\hat{\alpha}_p + \Psi_{j,\hat{\alpha}_p^2}(x)\hat{\alpha}_p^2,$$
$$p_{Y,1}(x, \hat{\alpha}_p) = \psi_{Y,0}(x) + \psi_{Y,\hat{\alpha}_p}(x)\hat{\alpha}_p, \qquad p_{Y,2}(x, \hat{\alpha}_p) = \Psi_{Y,0}(x) + \Psi_{Y,\hat{\alpha}_p}(x)\hat{\alpha}_p + \Psi_{Y,\hat{\alpha}_p^2}(x)\hat{\alpha}_p^2,$$

where $\psi_{j,l}(x)$ and $\Psi_{j,l}(x)$, for $j \in \{0, 1, 2, Y\}$ and $l \in \{0, \hat{\alpha}_p, \hat{\alpha}_p^2\}$, are functions of x.

4. *The first-order and second-order corrections to the price-dividend ratio for the individual claim can be written as:*

$$p_{k,1}(x, \hat{\alpha}_p, \hat{s}) = \psi_{k,0}(x) + \psi_{k,\hat{\alpha}_p}(x)\hat{\alpha}_p + \psi_{k,\hat{s}}(x)\hat{s},$$
$$p_{k,2}(x, \hat{\alpha}_p, \hat{s}) = \Psi_{k,0}(x) + \Psi_{k,\hat{\alpha}_p}(x)\hat{\alpha}_p + \Psi_{k,\hat{s}}(x)\hat{s} + \Psi_{k,\hat{\alpha}_p^2}(x)\hat{\alpha}_p^2 + \Psi_{k,\hat{\alpha}_p\hat{s}}(x)\hat{\alpha}_p\hat{s} + \Psi_{k,\hat{s}^2}(x)\hat{s}^2,$$

where $\psi_{k,l}(x)$ and $\Psi_{k,l}(x)$, for $k \in \{D, B\}$ and $l \in \{0, \hat{\alpha}_p, \hat{\alpha}_p^2, \hat{s}, \hat{\alpha}_p\hat{s}, \hat{s}^2\}$, are functions of x.

These properties follow from the definition of the perturbation solution. When $\epsilon = 0$, the exogenous state variables are constant and investors have identical preferences. Hence, the zeroth-order terms are independent of $(x, \hat{\alpha}_p, \hat{s})$. The fact that the drift and diffusion of the state variables are first-order in ϵ comes from the assumptions on the dynamics of the exogenous state variables, and the fact that the zeroth-order terms are constant. Properties 3 and 4 describe a solution that is global in the endogenous wealth shares and local in the exogenous state variables.

Our goal is to solve for the coefficients in this perturbation expansion. We proceed in steps. We first characterize the benchmark economy. We then derive the first- and second-order corrections to the consumption-wealth ratios and the price-dividend ratio of the aggregate claim. Finally, we derive the corresponding corrections to the price-dividend ratios of the individual claims.

C.1 Proof of Lemma 1

Proof. The assumption $\epsilon = 0$ implies that there is no preference heterogeneity and that passive investors are fully invested in the aggregate risky claim. We guess and verify that, in this benchmark economy, expected returns are constant and the leverage constraint is slack. In particular, the wealth distribution plays no role in equilibrium. This implies that $\mu_{c_j,0}(X) = \mu_{p_Y,0}(X) = 0$ and $\sigma_{c_j,0}(X) = \sigma_{p_Y,0}(X) = 0_{1\times 3}$, where $0_{1\times 3}$ is a 1×3 vector of zeros.

Risk exposure and market price of risk. The risk exposure vector for the aggregate claim is given by

$$\sigma_{R_Y,0}(X) = \sigma.$$

The market price of risk is therefore

$$\eta_0(X) = \gamma\sigma,$$

using the fact that $h_{a,t} = 0$ and $\chi_{a,t} = 1$.

The risk exposure of investor j is therefore

$$\sigma_{j,0}(X) = \sigma.$$

Because $\overline{\sigma} = \|\sigma\|$, the leverage constraint is satisfied in the benchmark economy.

Risk premium and interest rate. The risk premium on the aggregate claim is given by

$$\pi_{Y,0}(X) = \eta_0(X)\sigma_{R_Y,0}(X)^\top = \gamma\|\sigma\|^2,$$

where $\|\sigma\|^2 = \sigma\sigma^\top$.

The interest rate is given by

$$r_0(X) = \rho + \psi^{-1}\mu - (1+\psi^{-1})\frac{\gamma}{2}\|\sigma\|^2,$$

using the fact that $\xi_t = 0$, $\sigma_{j,t} = \sigma$, $\mu_{p_Y,t} = 0$, and $\sigma_{p_Y,t} = 0_{1\times 3}$ in the benchmark economy.

Consumption-wealth ratio and price-dividend ratio. The consumption-wealth ratio $c_{j,0}(X)$ is given by

$$c_{j,0}(X) = \psi\rho + (1-\psi)\left[r_0(X) + \eta_0(X)\,\sigma^\top - \frac{\gamma}{2}\|\sigma\|^2\right].$$

Plugging in the expression for $r_0(X)$ and $\eta_0(X)$, we obtain

$$c_{j,0}(X) = \rho - \left(1-\psi^{-1}\right)\left(\mu - \frac{\gamma\|\sigma\|^2}{2}\right),$$

We assume $\rho > \left(1-\psi^{-1}\right)\left(\mu - \frac{\gamma\|\sigma\|^2}{2}\right)$, so the benchmark consumption-wealth ratio is positive.

The goods-market-clearing condition then implies

$$\frac{1}{p_{Y,0}(X)} = \rho - \left(1-\psi^{-1}\right)\left(\mu - \frac{\gamma\|\sigma\|^2}{2}\right).$$

Drift and diffusion of wealth shares. The drift and diffusion of the wealth shares are given by

$$\mu_{x_j,0}(X) = 0, \qquad \sigma_{x_j,0}(X) = 0_{1\times 3}, \qquad \forall j \in \{1,2\}.$$

using the fact that $\kappa = 0$, $\sigma_{j,0}(X) = \sigma_{R_Y,0}(X)$, and $c_{j,0}(X) = \frac{1}{p_{Y,0}(X)}$.

Derivative claim. The risk exposure of the derivative claim is given by

$$\sigma_{R_F,0} = [\sigma^1_{R_F,0}, \sigma^2_{R_F,0}, \sigma^3_{R_F,0}].$$

These three quantities are defined by the normalization conditions

$$\sigma_{R_Y,0}\sigma^\top_{R_F,0} = 0, \qquad \sigma_{R_F,0}\sigma^\top_{R_F,0} = \sigma_{R_Y,0}\sigma^\top_{R_Y,0}, \qquad \sigma^3_{R_F,0} = 0.$$

Using $\sigma_{R_Y,0} = [\sigma^1, 0, 0]$, we obtain

$$\sigma^1_{R_F,0} = 0, \qquad \sigma^2_{R_F,0} = \sigma^1, \qquad \sigma^3_{R_F,0} = 0.$$

Portfolio shares. The portfolio shares on the aggregate claim and the derivative satisfy

$$\sigma_{j,0} = \alpha_{j,Y}\sigma_{R_Y,0} + \alpha_{j,F}\sigma_{R_F,0}.$$

Using $\sigma_{j,0} = \sigma_{R_Y,0}$, we obtain $\alpha_{j,Y} = 1$ and $\alpha_{j,F} = 0$ for $j = 0, 1, 2$.

□

C.2 Closed-form solution to individual claims

I consider next the characterization of the price of the individual claims when the interest rate and the price of risk are constant. In contrast to the benchmark economy, we do not assume that s is close to $\bar{s}$, instead we show that one can obtain closed-form solutions, analogous to the ones in Menzly et al. (2004), in the special case $\gamma = 1$. The closed-form solutions provide a useful reference to compare with the perturbation solution derived below.

Pricing the individual claims. The price of the dividend claim satisfies the pricing condition:

$$P_{D,t} = \mathbb{E}_t \left[\int_t^\infty \frac{\Lambda_\tau}{\Lambda_t} s_\tau Y_\tau d\tau \right],$$

The SDF and the aggregate endowment follow the processes:

$$\frac{d\Lambda_t}{\Lambda_t} = -r^* dt - \eta^* dZ_t, \qquad \frac{dY_t}{Y_t} = \mu dt + \sigma dZ_t.$$

We can solve for the price of the dividend claim in closed form in the case $\gamma = 1$, so $\eta^* = \sigma$. In this case, we can then write the SDF and the aggregate endowment as:

$$\Lambda_t = \exp\left(-\int_0^t \left(r^* + \frac{\sigma\sigma^\top}{2} \right) ds - \sigma Z_t \right) \Lambda_0, \qquad Y_t = \exp\left(\int_0^t \left(\mu - \frac{\sigma\sigma^\top}{2} \right) ds + \sigma Z_t \right) Y_0.$$

Combining the expressions for Λ_t and Y_t, we obtain:

$$P_{D,t} = \int_t^\infty e^{-\delta(\tau - t)} \mathbb{E}_t \left[s_\tau \right] d\tau Y_t,$$

where $\delta = r^* + \sigma\sigma^\top - \mu$, using the fact that Z_t cancels out in the product $\Lambda_t Y_t$.

The conditional expectation of the dividend share is given by

$$\mathbb{E}_t \left[s_\tau \right] = \bar{s} + (s_t - \bar{s}) e^{-\kappa_s(\tau - t)}.$$

The price of the dividend claim is given by

$$P_{D,t} = \left[\frac{\bar{s}}{\delta} + \frac{s_t - \bar{s}}{\delta + \kappa_s} \right] Y_t$$

where $Y_t = \exp\left(\int_0^t \left(\mu - \frac{\sigma\sigma^\top}{2} \right) ds + \sigma Z_t \right) Y_0$.

Hence, the normalized price of the dividend and corporate bond claims are given by

$$p_{D,0}(X) = \frac{1}{\delta} \left[\frac{\delta}{\delta + \kappa_s} s_t + \frac{\kappa_s}{\delta + \kappa_s} \bar{s} \right], \qquad p_{B,0}(X) = \frac{1}{\delta} \left[\frac{\delta}{\delta + \kappa_s} (1 - s_t) + \frac{\kappa_s}{\delta + \kappa_s} (1 - \bar{s}) \right].$$

Portfolio share of individual claims. The portfolio share of investor j on the dividend and bond claims are given by

$$\alpha_{j,D,t} = \frac{p_{D,t}}{p_{Y,t}}, \qquad \alpha_{j,B,t} = \frac{p_{B,t}}{p_{Y,t}}.$$

In the benchmark economy with $\gamma = 1$, we have that

$$\alpha_{j,D,0}(X) = \frac{\delta}{\delta + \kappa_s} s + \frac{\kappa_s}{\delta + \kappa_s} \overline{s}, \qquad \alpha_{j,B,0}(X) = \frac{\delta}{\delta + \kappa_s}(1 - s) + \frac{\kappa_s}{\delta + \kappa_s}(1 - \overline{s}),$$

using the fact that $\delta = 1/p_{Y,0}(X)$ when $\gamma = 1$.

C.3 Proof of Proposition 3

Proof. Consistent with Proposition 2, the consumption-wealth ratios and the price-dividend ratio of the aggregate claim are independent of the dividend-share shock s. We can then focus on the reduced set of states $\hat{X} = (x, \overline{\alpha}_p)$, where $x = (x_1, x_2)$ is the vector of wealth shares of the active investors.

We write the portfolio share of passive investors as $\overline{\alpha}_p = 1 + \hat{\alpha}_p \epsilon$, where $\hat{\alpha}_p$ follows a process independent of ϵ:

$$d\hat{\alpha}_p = \kappa_p(\hat{\alpha} - \hat{\alpha}_p)dt + \sigma_p dZ_t.$$

This implies $\overline{\alpha} = 1 + \hat{\alpha}\epsilon$ and that the diffusion of $\overline{\alpha}_p$ is of order $O(\epsilon)$.

We consider the following expansion for the consumption-wealth ratio of investor j and the price-dividend ratio of the aggregate claim:

$$\begin{aligned} c_j(x, \overline{\alpha}_p; \epsilon) &= c_{j,0}(x, \hat{\alpha}_p) + c_{j,1}(x, \hat{\alpha}_p)\epsilon + O(\epsilon^2), \\ p_Y(x, \overline{\alpha}_p; \epsilon) &= p_{Y,0}(x, \hat{\alpha}_p) + p_{Y,1}(x, \hat{\alpha}_p)\epsilon + O(\epsilon^2). \end{aligned}$$

The terms $c_{j,0}(x, \hat{\alpha}_p)$ and $p_{Y,0}(x, \hat{\alpha}_p)$ are independent of $(x, \hat{\alpha}_p)$ and are characterized in Lemma 1. We focus on the first-order correction terms $c_{j,1}(x, \hat{\alpha}_p)$ and $p_{Y,1}(x, \hat{\alpha}_p)$.

Diffusion for consumption-wealth ratios and price-dividend ratio. Given the specification of $\overline{\alpha}_p$, we have $\mu_{\overline{\alpha}_p} = O(\epsilon)$ and $\sigma_{\overline{\alpha}_p} = O(\epsilon)$. Similarly, $\mu_{x_j} = O(\epsilon)$ and $\sigma_{x_j} = O(\epsilon)$ for $j = 1, 2$. Hence, the drift and diffusion of the state variables are first-order in ϵ, that is, $\mu_{\hat{X}} = O(\epsilon)$ and $\sigma_{\hat{X}} = O(\epsilon)$.

We characterize next the derivatives of the consumption-wealth ratios and the price-dividend ratio with respect to $\hat{X}$. Given the consumption-wealth ratios and the price-dividend ratio for the aggregate claim are independent of $(x, \hat{\alpha}_p)$ at $\epsilon = 0$, we have that $c_{j,x_j} = O(\epsilon)$ and $p_{Y,x_j} = O(\epsilon)$, as these derivatives are zero at order zero. Moreover, we guess-and-verify that the derivatives of the consumption-wealth ratios and the price-dividend ratio with respect to $\overline{\alpha}_p$ are first-order in ϵ:

$$c_{j,\overline{\alpha}_p} = O(\epsilon); \qquad p_{Y,\overline{\alpha}_p} = O(\epsilon).$$

From the expansion for the consumption-wealth ratios, we can express the derivatives with respect to $\overline{\alpha}_p$ in terms of the derivatives of the first-order correction terms with respect to $\hat{\alpha}_p$:

$$c_{j,\overline{\alpha}_p} = \frac{\partial c_{j,1}}{\partial \hat{\alpha}_p} + O(\epsilon),$$

using the fact that $c_{j,0}$ is independent of $\hat{\alpha}_p$.

Hence, $c_{j,\overline{\alpha}_p} = O(\epsilon)$ implies $\frac{\partial c_{j,1}}{\partial \hat{\alpha}_p} = 0$, so $c_{j,1}$ is independent of $\hat{\alpha}_p$. A similar argument shows that $p_{Y,\overline{\alpha}_p} = O(\epsilon)$ implies that $p_{Y,1}$ is independent of $\hat{\alpha}_p$.

We can then write the diffusion terms for the consumption-wealth ratios and the price-dividend ratio are second-order terms in ϵ:

$$\sigma_{c_j} = \underbrace{\frac{c_{j,\hat{X}}}{c_j}}_{O(\epsilon)} \times \underbrace{\sigma_{\hat{X}}}_{O(\epsilon)} = O(\epsilon^2), \qquad \sigma_{p_Y} = \underbrace{\frac{p_{Y,\hat{X}}}{p_Y}}_{O(\epsilon)} \times \underbrace{\sigma_{\hat{X}}}_{O(\epsilon)} = O(\epsilon^2),$$

This implies that $\sigma_{R_Y,1} = 0_{1\times 3}$.

Risk exposure and market price of risk. The risk exposure for the passive investor is given by

$$\sigma_{0,1} = \hat{\alpha}_p \sigma.$$

The first-order correction to the risk exposure for an active investor is given by

$$\sigma_{j,1} = \frac{\eta_1}{\gamma} - \frac{\eta_0}{\gamma}(\hat{\gamma}_j + \hat{v}_j),$$

where $\hat{v}_j$ is the Lagrange multiplier on the leverage constraint for investor j.

Expanding the market clearing condition for risky assets, we obtain

$$\sum_{j=0}^{2} x_j \sigma_{j,1} = 0$$

using the fact that $\sigma_{R_Y,1} = 0$.

The first-order correction to the market price of risk is given by

$$\eta_1 = \gamma\sigma \left[\sum_{j=1}^{2} \frac{x_j}{x_a}(\hat{\gamma}_j + \hat{v}_j) - \frac{x_0}{x_a}\hat{\alpha}_p \right],$$

where $x_a = 1 - x_0$.

The risk exposure for the first active investor is

$$\sigma_{1,1} = -\left[\frac{x_2}{x_a}(\hat{\gamma}_1 - \hat{\gamma}_2 - \hat{\nu}_2) + \frac{x_0}{x_a}\hat{\alpha}_p\right]\sigma.$$

and the risk exposure for the second active investor is

$$\sigma_{2,1} = \left[\frac{x_1}{x_a}(\hat{\gamma}_1 - \hat{\gamma}_2 - \hat{\nu}_2) - \frac{x_0}{x_a}\hat{\alpha}_p\right]\sigma.$$

Lagrange multiplier on the leverage constraint. The leverage constraint for investor j is

$$\|\sigma_j\| \leq \overline{\sigma}.$$

Using $\sigma_j = \sigma + \sigma_{j,1}\epsilon + O(\epsilon^2)$ and $\overline{\sigma} = \|\sigma\|(1 + \hat{\sigma}\epsilon)$, the first-order expansion of the constraint is

$$\sqrt{(\sigma + \sigma_{j,1}\epsilon)(\sigma + \sigma_{j,1}\epsilon)^\top} = \|\sigma\|\left(1 + \frac{\sigma_{j,1}\sigma^\top}{\sigma\sigma^\top}\epsilon\right) + O(\epsilon^2).$$

Therefore, the first-order constraint is

$$\frac{\sigma_{j,1}\sigma^\top}{\sigma\sigma^\top} \leq \hat{\sigma}.$$

We assume that investor 1 is unconstrained and investor 2 may be constrained, so $\hat{\nu}_1 = 0$ and $\hat{\nu}_2 \geq 0$. For investor 2, using the expression for $\sigma_{2,1}$ above, the first-order leverage constraint is

$$\frac{x_1}{x_a}(\hat{\gamma}_1 - \hat{\gamma}_2 - \hat{\nu}_2) - \frac{x_0}{x_a}\hat{\alpha}_{p,1} \leq \hat{\sigma}.$$

If the constraint binds, this condition holds with equality, which implies

$$\hat{\nu}_2 = \hat{\gamma}_1 - \hat{\gamma}_2 - \frac{x_0}{x_1}\hat{\alpha}_{p,1} - \frac{x_a}{x_1}\hat{\sigma}.$$

If the constraint is slack, then $\hat{\nu}_2 = 0$ and

$$\frac{x_1}{x_a}(\hat{\gamma}_1 - \hat{\gamma}_2) - \frac{x_0}{x_a}\hat{\alpha}_{p,1} < \hat{\sigma},$$

or, equivalently,

$$\hat{\gamma}_1 - \hat{\gamma}_2 - \frac{x_0}{x_1}\hat{\alpha}_{p,1} - \frac{x_a}{x_1}\hat{\sigma} < 0.$$

Hence, the first-order Lagrange multiplier is

$$\hat{\nu}_2 = \max\left\{\hat{\gamma}_1 - \hat{\gamma}_2 - \frac{x_0}{x_1}\hat{\alpha}_{p,1} - \frac{x_a}{x_1}\hat{\sigma}, 0\right\}.$$

When the constraint binds, substituting the expression for $\hat{\nu}_2$ into the market price of risk yields

$$\eta_1 = \gamma\sigma\left[\hat{\gamma}_1 - \frac{x_2}{x_1}\hat{\sigma} - \frac{x_0}{x_1}\hat{\alpha}_{p,1}\right].$$

When the constraint is slack, $\hat{\nu}_2 = 0$, and the market price of risk is

$$\eta_1 = \gamma\sigma\left[\sum_{j=1}^{2}\frac{x_j}{x_a}\hat{\gamma}_j - \frac{x_0}{x_a}\hat{\alpha}_{p,1}\right].$$

Equivalently, both cases can be summarized as

$$\eta_1 = \gamma\sigma\left[\sum_{j=1}^{2}\frac{x_j}{x_a}(\hat{\gamma}_j + \hat{\nu}_j) - \frac{x_0}{x_a}\hat{\alpha}_{p,1}\right],$$

with $\hat{\nu}_1 = 0$ and $\hat{\nu}_2$ given above.

Risk premium. The first-order risk premium is

$$\pi_{Y,1} = \eta_1\sigma^\top = \gamma\|\sigma\|^2\left[\sum_{j=1}^{2}\frac{x_j}{x_a}(\hat{\gamma}_j + \hat{\nu}_j) - \frac{x_0}{x_a}\hat{\alpha}_{p,1}\right].$$

It increases with the average risk aversion of active investors, with tighter (binding) leverage constraints, and when active investors bear a larger share of aggregate risk.

Drift for consumption-wealth ratios and price-dividend ratio. Itô's lemma implies that the drifts are

$$\mu_{c_j} = \frac{c_{j,\hat{X}}}{c_j}\mu_{\hat{X}} + \frac{1}{2}\sum_{k=1}^{3}\sigma_{\hat{X},k}^\top\frac{c_{j,\hat{X}\hat{X}}}{c_j}\sigma_{\hat{X},k},$$

$$\mu_{p_Y} = \frac{p_{Y,\hat{X}}}{p_Y}\mu_{\hat{X}} + \frac{1}{2}\sum_{k=1}^{3}\sigma_{\hat{X},k}^\top\frac{p_{Y,\hat{X}\hat{X}}}{p_Y}\sigma_{\hat{X},k},$$

where $\sigma_{\hat{X},k}$ is the k-th column of the matrix $\sigma_{\hat{X}}$.

Hence, $\mu_{c_j} = O(\epsilon^2)$ and $\mu_{p_Y} = O(\epsilon^2)$, since the gradients are $O(\epsilon)$ and both $\mu_{\hat{X}}$ and $\sigma_{\hat{X}}$ are $O(\epsilon)$.

Interest rate. The first-order correction to the interest rate is

$$r_1 = (1-\psi^{-1})\frac{\gamma\|\sigma\|^2}{2}\sum_{j=0}^{2} x_j\hat{\gamma}_j + (1-\psi^{-1})\gamma\|\sigma\|^2\sum_{j=0}^{2} x_j\frac{\sigma_{j,1}\sigma^\top}{\|\sigma\|^2} - \pi_{Y,1},$$

where we used the fact that $\xi_j = O(\epsilon^2)$, as $\mu_{c_j} = O(\epsilon^2)$ and $\sigma_{c_j} = O(\epsilon^2)$.

Using $\sum_{j=0}^{2} x_j\sigma_{j,1} = 0_{1\times 3}$, we obtain

$$r_1 = -\gamma\|\sigma\|^2\left[\sum_{j=1}^{2}\frac{x_j}{x_a}(\hat{\gamma}_j+\hat{\nu}_j) - \sum_{j=0}^{2} x_j\hat{\gamma}_j - \frac{x_0}{x_a}\hat{\alpha}_{p,1} + \frac{1+\psi^{-1}}{2}\sum_{j=0}^{2} x_j\hat{\gamma}_j\right].$$

Notice that we can write the expression above as

$$r_1 = -\gamma\|\sigma\|^2\frac{1+\psi^{-1}}{2}\sum_{j=0}^{2} x_j\hat{\gamma}_j - \left[\pi_{Y,1} - \pi_{Y,1}^{fb}\right],$$

where

$$\pi_{Y,1}^{fb} = \gamma\|\sigma\|^2\sum_{j=0}^{2} x_j\hat{\gamma}_j.$$

The first term captures the effect of heterogeneity in risk aversion in the frictionless benchmark (no passive investors or leverage constraints). The second term captures the impact of frictions through the wedge between the equilibrium risk premium and its frictionless counterpart.

Consumption-wealth ratio. The consumption-wealth ratio for investor j is given by

$$c_{j,1} = (1-\psi)\left[r_1 + \eta_1\,\sigma^\top + \eta_0\sigma_{j,1}^\top - \gamma\sigma\sigma_{j,1}^\top - \frac{\gamma}{2}\|\sigma\|^2\hat{\gamma}_j\right].$$

Substituting for r_1 and $\pi_{Y,1}$, the first-order correction to the consumption-wealth ratio is

$$c_{j,1} = \frac{\gamma\|\sigma\|^2}{2}(1-\psi)\left[\left(1-\psi^{-1}\right)\sum_{k=0}^{2} x_k\hat{\gamma}_k - \hat{\gamma}_j\right],$$

which is independent of $\hat{\alpha}_p$ as conjectured.

Price-dividend ratio of aggregate claim. The pricing condition implies

$$-\frac{1}{(p_Y^*)^2} p_{Y,1} = r_1 + \pi_{Y,1}.$$

Hence, the first-order correction to the price-dividend ratio is

$$\frac{p_{Y,1}}{p_Y^*} = -p_Y^*(1-\psi^{-1})\frac{\gamma\|\sigma\|^2}{2}\sum_{j=0}^{2} x_j \hat{\gamma}_j,$$

which is independent of $\hat{\alpha}_p$ as conjectured. If $\psi > 1$, an increase in average risk aversion reduces the price-dividend ratio of the aggregate claim.

State dynamics. The diffusion of the wealth share of investor j, $j = 1, 2$, is given by

$$\begin{aligned}\sigma_{x_j,1} &= x_j(\sigma_{j,1} - \sigma_{R,1}) \\ &= x_j \left[\sum_{k=1}^{2} \frac{x_k}{x_a}(\hat{\gamma}_k + \hat{\nu}_k) - \frac{x_0}{x_a}\hat{\alpha}_{p,1} - (\hat{\gamma}_j + \hat{\nu}_j)\right]\sigma.\end{aligned}$$

The drift of the wealth share of investor j, $j = 1, 2$, is given by

$$\mu_{x_j,1} = x_j \left[(\gamma - 1)\sigma\sigma_{j,1}^{\top} + \frac{\gamma\|\sigma\|^2}{2}(\psi - 1)\left(\sum_{k=0}^{2} x_k\hat{\gamma}_k - \hat{\gamma}_j\right)\right] + \hat{\kappa}(\omega_j - x_j),$$

Finally, the law of motion of $\hat{\alpha}_p$ is

$$d\hat{\alpha}_p = \kappa_p(\hat{\alpha} - \hat{\alpha}_p)dt + \sigma_p dZ_t,$$

so the drift and diffusion of $\overline{\alpha}_p$ are $\mu_{\overline{\alpha}_p} = \kappa_p(\hat{\alpha} - \hat{\alpha}_p)\epsilon$ and $\sigma_{\overline{\alpha}_p} = \sigma_p\epsilon$.

□

C.4 First-order correction of individual claims

We next price the dividend and bond claims, whose payoffs are $D = sY$ and $B = (1-s)Y$. It is convenient to work with prices normalized by aggregate output,

$$p_k \equiv \frac{P_k}{Y}, \qquad k \in \{D, B\}.$$

We consider the expansion

$$p_k(x, \overline{\alpha}_p, s; \epsilon) = p_{k,0} + p_{k,1}(x, \hat{\alpha}_p, \hat{s})\epsilon + O(\epsilon^2),$$

where $p_{k,0}$ is constant. Let $s_D = s$ and $s_B = 1 - s$, so that

$$s_k = \overline{s}_k + \hat{s}_k \epsilon,$$

with

$$\overline{s}_D = \overline{s}, \quad \overline{s}_B = 1 - \overline{s}, \qquad \hat{s}_D = \hat{s}, \quad \hat{s}_B = -\hat{s}.$$

The pricing condition for claim k is

$$r + \sigma_{R_k} \eta^\top = \frac{s_k}{p_k} + \mu + \mu_{p_k} + \sigma \sigma_{p_k}^\top.$$

Zeroth-order term. At $\epsilon = 0$, we have

$$p_{k,0} = \frac{\overline{s}_k}{\delta}, \qquad \delta \equiv r^* + \gamma \|\sigma\|^2 - \mu.$$

First-order correction. The first-order pricing equation implies

$$\delta p_{k,1} + (r_1 + \pi_{Y,1}) p_{k,0} = \hat{s}_k - \kappa_s \hat{s} \frac{\partial p_{k,1}}{\partial \hat{s}} + (1 - \gamma) \sigma \frac{\partial p_{k,1}}{\partial \hat{s}} \overline{s}(1 - \overline{s}) \sigma_s^\top.$$

Dividend claim. We conjecture

$$p_{D,1} = a_D(x) + b_D(x) \hat{s}.$$

Matching coefficients gives

$$b_D = \frac{1}{\delta + \kappa_s},$$

and

$$a_D = -(r_1 + \pi_{Y,1}) \frac{\overline{s}}{\delta^2} - \frac{\gamma - 1}{\delta(\delta + \kappa_s)} \overline{s}(1 - \overline{s}) \sigma \sigma_s^\top.$$

Hence

$$p_{D,1} = \frac{\hat{s}}{\delta + \kappa_s} - (r_1 + \pi_{Y,1}) \frac{\overline{s}}{\delta^2} - \frac{\gamma - 1}{\delta(\delta + \kappa_s)} \overline{s}(1 - \overline{s}) \sigma \sigma_s^\top.$$

Bond claim. Similarly, the normalized price of the bond claim is given by

$$p_{B,1} = -\frac{\hat{s}}{\delta + \kappa_s} - (r_1 + \pi_{Y,1}) \frac{1 - \overline{s}}{\delta^2} + \frac{\gamma - 1}{\delta(\delta + \kappa_s)} \overline{s}(1 - \overline{s}) \sigma \sigma_s^\top.$$

Consistency. Adding both claims yields

$$p_{D,1} + p_{B,1} = -\frac{r_1 + \pi_{Y,1}}{\delta^2},$$

which is consistent with the first-order correction of the aggregate claim.

C.5 Second-order correction: no dynamics in passive portfolio share

We next derive the second-order correction. To keep the algebra transparent, we proceed in two steps. First, we consider the case in which the portfolio share of passive investors is fixed, so

$$\kappa_p = 0, \qquad \sigma_p = 0_{1\times 3}.$$

Equivalently, $\hat{\alpha}_p$ is a fixed parameter rather than a stochastic state variable. This removes all drift and diffusion terms associated with $\overline{\alpha}_p$ and leaves the endogenous wealth shares $x = (x_1, x_2)$ as the only state variables that contribute to the second-order drift and diffusion of equilibrium objects. After deriving this benchmark second-order system, we then allow for non-zero drift and diffusion in the passive investors' portfolio share.

Step 1: Diffusion for c_j and y_Y.

The expansion of $\sigma_{c_j,t}$ in ϵ is given by

$$\begin{aligned}
\sigma_{c_j}(\hat{X}, \epsilon) &= \frac{c_{j,\hat{X}}(\hat{X}, \epsilon)}{c_j(\hat{X}, \epsilon)} \sigma_{\hat{X}}(\hat{X}, \epsilon) \\
&= \frac{c_{j,1,x}(x, \hat{\alpha}_p)}{c_{j,0}(x, \hat{\alpha}_p)} \sigma_{x,1}(x, \hat{\alpha}_p)\epsilon^2 + O(\epsilon^3) \\
&= -(\psi - 1)\frac{\left(1 - \psi^{-1}\right)\gamma\|\sigma\|^2}{2c_{j,0}(x, \hat{\alpha}_p)} \sum_{k=1}^{2} (\hat{\gamma}_k - \hat{\gamma}_0)\, x_k \sigma_{k,1}(x, \hat{\alpha}_p)\epsilon^2 + O(\epsilon^3),
\end{aligned}$$

This follows from $c_{j,x} = O(\epsilon)$ and $\sigma_x = O(\epsilon)$, so their product is second order.

At second order, it is convenient to work with the dividend yield of the aggregate claim,

$$y_Y \equiv \frac{1}{p_Y}.$$

This is only a change of variables. After solving for the second-order correction to y_Y, we recover the correction to p_Y from

$$p_Y = \frac{1}{y_Y}.$$

If

$$y_Y = y_{Y,0} + y_{Y,1}\epsilon + y_{Y,2}\epsilon^2 + O(\epsilon^3),$$

then

$$p_Y = p_{Y,0} - p_{Y,0}^2 y_{Y,1}\epsilon + \left(p_{Y,0}^3 y_{Y,1}^2 - p_{Y,0}^2 y_{Y,2}\right)\epsilon^2 + O(\epsilon^3).$$

We now derive the dynamics of y_Y directly:

$$\begin{aligned}
\sigma_{y_Y}(\hat{X},\epsilon) &= \frac{y_{Y,\hat{X}}(\hat{X},\epsilon)}{y_Y(\hat{X},\epsilon)}\sigma_{\hat{X}}(\hat{X},\epsilon) \\
&= \left(1-\psi^{-1}\right)\frac{\gamma\|\sigma\|^2}{2y_{Y,0}(x,\hat{\alpha}_p)}\sum_{k=1}^{2}(\hat{\gamma}_k - \hat{\gamma}_0)\,x_k\sigma_{k,1}(x,\hat{\alpha}_p)\epsilon^2 + O(\epsilon^3).
\end{aligned}$$

The same scaling applies to y_Y, implying that its diffusion starts at order ϵ^2.

The drift of $c_{j,t}$ is given by

$$\begin{aligned}
\mu_{c_j}(\hat{X},\epsilon) &= \frac{c_{j,\hat{X}}(\hat{X},\epsilon)}{c_j(\hat{X},\epsilon)}\mu_{\hat{X}}(\hat{X},\epsilon) + \sum_{k=1}^{3}\frac{1}{2}\sigma_{\hat{X},k}^{\top}(\hat{X},\epsilon)\frac{c_{j,\hat{X}\hat{X}}(\hat{X},\epsilon)}{c_j(\hat{X},\epsilon)}\sigma_{\hat{X},k}(\hat{X},\epsilon) \\
&= \frac{c_{j,1,x}(x,\hat{\alpha}_p)}{c_{j,0}(x,\hat{\alpha}_p)}\mu_{x,1}(x,\hat{\alpha}_p)\epsilon^2 + O(\epsilon^3) \\
&= -(\psi-1)\frac{\left(1-\psi^{-1}\right)\gamma\|\sigma\|^2}{2c_{j,0}(x,\hat{\alpha}_p)}\sum_{k=1}^{2}(\hat{\gamma}_k - \hat{\gamma}_0)\,x_k\mu_{x_k,1}(x,\hat{\alpha}_p)\epsilon^2 + O(\epsilon^3).
\end{aligned}$$

The drift of y_Y, t is given by

$$\begin{aligned}
\mu_{y_Y}(\hat{X},\epsilon) &= \frac{y_{Y,\hat{X}}(\hat{X},\epsilon)}{y_Y(\hat{X},\epsilon)}\mu_{\hat{X}}(\hat{X},\epsilon) + \sum_{k=1}^{3}\frac{1}{2}\sigma_{\hat{X},k}^{\top}(\hat{X},\epsilon)\frac{y_{Y,\hat{X}\hat{X}}(\hat{X},\epsilon)}{y_Y(\hat{X},\epsilon)}\sigma_{\hat{X},k}(\hat{X},\epsilon) \\
&= \frac{y_{Y,1,x}(x,\hat{\alpha}_p)}{y_{Y,0}(x,\hat{\alpha}_p)}\mu_{x,1}(x,\hat{\alpha}_p)\epsilon^2 + O(\epsilon^3) \\
&= \left(1-\psi^{-1}\right)\frac{\gamma\|\sigma\|^2}{2y_{Y,0}(x,\hat{\alpha}_p)}\sum_{k=1}^{2}(\hat{\gamma}_k - \hat{\gamma}_0)\,x_k\mu_{x_k,1}(x,\hat{\alpha}_p)\epsilon^2 + O(\epsilon^3).
\end{aligned}$$

Step 2: Risk exposures of investors.

The risk exposure of an active investor is

$$\sigma_{j,t} = \frac{1}{1+\nu_{j,t}}\left[\frac{\eta_t}{\gamma_j} + \frac{1-\gamma_j^{-1}}{\psi-1}\,\sigma_{c_j,t}\right].$$

It is useful to first define the desired exposure before imposing the leverage multiplier:

$$q_{j,t} \equiv \frac{\eta_t}{\gamma_j} + \frac{1-\gamma_j^{-1}}{\psi-1}\,\sigma_{c_j,t}.$$

We write

$$\begin{aligned}
\eta_t &= \eta_0 + \eta_1\epsilon + \eta_2\epsilon^2 + O(\epsilon^3),\\
\nu_{j,t} &= \hat{\nu}_{j,1}\epsilon + \hat{\nu}_{j,2}\epsilon^2 + O(\epsilon^3),\\
\sigma_{c_j,t} &= \sigma_{c_j,2}\epsilon^2 + O(\epsilon^3),
\end{aligned}$$

where $\hat{\nu}_{j,1} = \hat{\nu}_j$ is the first-order multiplier characterized above. Since $\gamma_j = \gamma(1+\hat{\gamma}_j\epsilon)$, we have

$$\gamma_j^{-1} = \gamma^{-1}\left(1 - \hat{\gamma}_j\epsilon + \hat{\gamma}_j^2\epsilon^2\right) + O(\epsilon^3).$$

Therefore,

$$q_{j,t} = q_{j,0} + q_{j,1}\epsilon + q_{j,2}\epsilon^2 + O(\epsilon^3),$$

with

$$\begin{aligned}
q_{j,0} &= \frac{\eta_0}{\gamma} = \sigma,\\
q_{j,1} &= \frac{\eta_1}{\gamma} - \sigma\hat{\gamma}_j,\\
q_{j,2} &= \frac{\eta_2}{\gamma} - \frac{\eta_1}{\gamma}\hat{\gamma}_j + \sigma\hat{\gamma}_j^2 + \frac{1-\gamma^{-1}}{\psi-1}\sigma_{c_j,2}.
\end{aligned}$$

Expanding $\sigma_{j,t} = q_{j,t}/(1+\nu_{j,t})$ gives

$$\begin{aligned}
\sigma_{j,1} &= q_{j,1} - \hat{\nu}_{j,1}\sigma,\\
\sigma_{j,2} &= q_{j,2} - \hat{\nu}_{j,1}q_{j,1} + \left(\hat{\nu}_{j,1}^2 - \hat{\nu}_{j,2}\right)\sigma.
\end{aligned}$$

Equivalently,

$$\sigma_{j,2} = \frac{\eta_2}{\gamma} - \frac{\eta_1}{\gamma}\hat{\gamma}_j + \sigma\hat{\gamma}_j^2 + \frac{1-\gamma^{-1}}{\psi-1}\sigma_{c_j,2} - \hat{\nu}_{j,1}\left(\frac{\eta_1}{\gamma} - \sigma\hat{\gamma}_j\right) + \left(\hat{\nu}_{j,1}^2 - \hat{\nu}_{j,2}\right)\sigma. \tag{C.1}$$

For an unconstrained investor, $\hat{\nu}_{j,1} = \hat{\nu}_{j,2} = 0$, and this reduces to

$$\sigma_{j,2} = \frac{\eta_2}{\gamma} - \frac{\eta_1}{\gamma}\hat{\gamma}_j + \sigma\hat{\gamma}_j^2 + \frac{1-\gamma^{-1}}{\psi-1}\sigma_{c_j,2}. \tag{C.2}$$

For a constrained investor, the multiplier is determined by the second-order expansion of the leverage constraint. The leverage constraint is

$$\|\sigma_j\| \leq \|\sigma\|(1+\hat{\sigma}\epsilon).$$

Using

$$\sigma_j = \sigma + \sigma_{j,1}\epsilon + \sigma_{j,2}\epsilon^2 + O(\epsilon^3),$$

we obtain

$$\|\sigma_j\| = \|\sigma\| \left[1 + \frac{\sigma_{j,1}\sigma^\top}{\sigma\sigma^\top}\epsilon + \left(\frac{\sigma_{j,2}\sigma^\top}{\sigma\sigma^\top} + \frac{1}{2}\left(\frac{\sigma_{j,1}\sigma_{j,1}^\top}{\sigma\sigma^\top} - \left(\frac{\sigma_{j,1}\sigma^\top}{\sigma\sigma^\top}\right)^2\right)\right)\epsilon^2\right] + O(\epsilon^3).$$

At first order, $\sigma_{j,1}$ is proportional to σ. Therefore, the curvature term in the second line is zero. If the constraint binds through second order and $\overline{\sigma} = \|\sigma\|(1 + \hat{\sigma}\epsilon)$ has no second-order correction, then

$$\frac{\sigma_{j,2}\sigma^\top}{\sigma\sigma^\top} = 0. \tag{C.3}$$

Combining (C.1) and (C.3) yields

$$\hat{\nu}_{j,2} = \frac{q_{j,2}\sigma^\top}{\sigma\sigma^\top} - \hat{\nu}_{j,1}\frac{q_{j,1}\sigma^\top}{\sigma\sigma^\top} + \hat{\nu}_{j,1}^2.$$

Substituting for $q_{j,1}$ and $q_{j,2}$ gives

$$\hat{\nu}_{j,2} = \frac{\eta_2\sigma^\top}{\gamma\sigma\sigma^\top} - \hat{\gamma}_j\frac{\eta_1\sigma^\top}{\gamma\sigma\sigma^\top} + \hat{\gamma}_j^2 + \frac{1-\gamma^{-1}}{\psi - 1}\frac{\sigma_{c_j,2}\sigma^\top}{\sigma\sigma^\top} - \hat{\nu}_{j,1}\left(\frac{\eta_1\sigma^\top}{\gamma\sigma\sigma^\top} - \hat{\gamma}_j\right) + \hat{\nu}_{j,1}^2. \tag{C.4}$$

In the maintained two-active-investor case, investor 1 is unconstrained and investor 2 may be constrained. Thus $\hat{\nu}_{1,1} = \hat{\nu}_{1,2} = 0$. If investor 2 is constrained, then $\hat{\nu}_{2,1} = \hat{\nu}_2$ and $\hat{\nu}_{2,2}$ is given by (C.4) with $j = 2$. If investor 2 is slack, then $\hat{\nu}_{2,1} = \hat{\nu}_{2,2} = 0$ and (C.2) applies to both active investors.

Step 3: Market price of risk.

The market price of risk is given by

$$\eta_t = \gamma_{a,t}\left[\chi_{a,t}\sigma_{R_Y,t} - h_{a,t}\right],$$

where

$$\chi_{a,t} \equiv \frac{1 - x_0\overline{\alpha}_p}{x_a}$$

is the aggregate risky exposure that must be absorbed by active investors per unit of active wealth,

$$h_{a,t} \equiv \sum_{j=1}^{2}\frac{x_j}{x_a}\frac{1-\gamma_j^{-1}}{\psi - 1}\frac{\sigma_{c_j,t}}{1 + \nu_{j,t}}$$

is the average hedging demand of active investors, adjusted for leverage constraints, and

$$\gamma_{a,t} \equiv \left[\sum_{j=1}^{2} \frac{x_j}{x_a} \frac{1}{\gamma_j (1 + \nu_{j,t})} \right]^{-1}$$

is the average effective risk aversion of active investors.

We now expand each term to second order. Let

$$a_j \equiv \hat{\gamma}_j + \hat{\nu}_{j,1},$$

and define active-wealth-weighted averages by

$$\mathbb{E}_a[z_j] \equiv \sum_{j=1}^{2} \frac{x_j}{x_a} z_j.$$

Since

$$\frac{1}{\gamma_j (1 + \nu_{j,t})} = \frac{1}{\gamma} \left[1 - a_j \epsilon + \left(a_j^2 - \hat{\nu}_{j,2} - \hat{\gamma}_j \hat{\nu}_{j,1} \right) \epsilon^2 \right] + O(\epsilon^3),$$

we have

$$\gamma_{a,t} = \gamma \left[1 + \Gamma_{a,1} \epsilon + \Gamma_{a,2} \epsilon^2 \right] + O(\epsilon^3),$$

where

$$\begin{aligned}
\Gamma_{a,1} &= \mathbb{E}_a[a_j], \\
\Gamma_{a,2} &= \mathbb{E}_a[\hat{\nu}_{j,2} + \hat{\gamma}_j \hat{\nu}_{j,1}] - \mathrm{Var}_a(a_j).
\end{aligned}$$

Here

$$\mathrm{Var}_a(a_j) \equiv \mathbb{E}_a[a_j^2] - \mathbb{E}_a[a_j]^2.$$

Because $\overline{\alpha}_p = 1 + \hat{\alpha}_p \epsilon$ and $\kappa_p = \sigma_p = 0$ in this step,

$$\chi_{a,t} = 1 - \frac{x_0}{x_a} \hat{\alpha}_p \epsilon.$$

Moreover, since $y_Y = 1/p_Y$, the risk exposure of the aggregate claim is

$$\sigma_{R_Y,t} = \sigma - \sigma_{y_Y,2} \epsilon^2 + O(\epsilon^3).$$

Finally, since $\sigma_{c_j,t} = \sigma_{c_j,2}\epsilon^2 + O(\epsilon^3)$, hedging demand starts at second order:

$$h_{a,t} = h_{a,2}\epsilon^2 + O(\epsilon^3),$$

with

$$h_{a,2} = \frac{1-\gamma^{-1}}{\psi - 1}\mathbb{E}_a[\sigma_{c_j,2}].$$

Therefore,

$$\chi_{a,t}\sigma_{R_Y,t} - h_{a,t} = \sigma - \frac{x_0}{x_a}\hat{\alpha}_p\sigma\epsilon - \left(\sigma_{y_Y,2} + h_{a,2}\right)\epsilon^2 + O(\epsilon^3).$$

Combining these expansions gives

$$\begin{aligned}
\eta_0 &= \gamma\sigma, \\
\eta_1 &= \gamma\sigma\left[\mathbb{E}_a[a_j] - \frac{x_0}{x_a}\hat{\alpha}_p\right], \\
\eta_2 &= \gamma\left[\Gamma_{a,2}\sigma - \frac{x_0}{x_a}\hat{\alpha}_p\Gamma_{a,1}\sigma - \sigma_{y_Y,2} - h_{a,2}\right].
\end{aligned}$$

The first-order term coincides with the expression derived above because $a_j = \hat{\gamma}_j + \hat{v}_{j,1}$. The second-order term depends on $\hat{v}_{j,2}$ through $\Gamma_{a,2}$. Thus, when a leverage constraint binds at second order, (C.4) and the expression for η_2 form a linear system for η_2 and the second-order multiplier.

Step 4: Risk premium.

The risk premium on the aggregate claim is given by

$$\pi_{Y,t} = \sigma_{R_Y,t}\eta_t^\top.$$

Using that $\sigma_{R_Y,1} = 0$, expanding the risk premium to second order gives

$$\pi_{Y,2} = \sigma_{R_Y,2}\eta_0^\top + \sigma\eta_2^\top = -\sigma_{y_Y,2}\eta_0^\top + \sigma\eta_2^\top.$$

where we used $\sigma_{R_Y,t} = \sigma - \sigma_{y_Y,2}\epsilon^2 + O(\epsilon^3)$.

Step 5: Interest rate.

$$r_t = \rho + \psi^{-1}\left(\mu_{P_Y,t} + \xi_t\right) + (1-\psi^{-1})\sum_{j=0}^{2} x_{j,t}\frac{\gamma_j}{2}\|\sigma_{j,t}\|^2 - \pi_{Y,t},$$

where $\mu_{P_Y,t}$ is the drift of the aggregate claim value.

Since $P_Y = p_Y Y$, it satisfies

$$\mu_{P_Y,t} = \mu + \mu_{p_Y,t} + \sigma\sigma_{p_Y,t}^{\top}.$$

At second order, working with the dividend yield implies

$$\mu_{P_Y,2} = -\mu_{y_Y,2} - \sigma\sigma_{y_Y,2}^{\top},$$

up to terms of order higher than ϵ^2, since $\sigma_{y_Y,t} = O(\epsilon^2)$ and $\sigma_{p_Y,2} = -\sigma_{y_Y,2}$. Moreover, the intertemporal-hedging term satisfies

$$\xi_2 = \sum_{j=0}^{2} x_j \left[\mu_{c_j,2} + (1-\gamma)\sigma_{c_j,2}\sigma^{\top}\right],$$

where terms involving $\|\sigma_{c_j,t}\|^2$ are fourth order and therefore do not enter at order ϵ^2.

Expanding the risk-compensation term gives

$$\left[\sum_{j=0}^{2} x_j \frac{\gamma_j}{2}\|\sigma_j\|^2\right]_2 = \gamma \sum_{j=0}^{2} x_j \left[\sigma_{j,2}\sigma^{\top} + \frac{1}{2}\sigma_{j,1}\sigma_{j,1}^{\top} + \hat{\gamma}_j \sigma_{j,1}\sigma^{\top}\right].$$

Therefore, the second-order correction to the interest rate is

$$\begin{aligned} r_2 = {} & -\psi^{-1}\left(\mu_{y_Y,2} + \sigma\sigma_{y_Y,2}^{\top}\right) + \psi^{-1}\sum_{j=0}^{2} x_j \left[\mu_{c_j,2} + (1-\gamma)\sigma_{c_j,2}\sigma^{\top}\right] \\ & + (1-\psi^{-1})\gamma \sum_{j=0}^{2} x_j \left[\sigma_{j,2}\sigma^{\top} + \frac{1}{2}\sigma_{j,1}\sigma_{j,1}^{\top} + \hat{\gamma}_j \sigma_{j,1}\sigma^{\top}\right] - \pi_{Y,2}. \end{aligned} \tag{C.5}$$

Using market clearing for the risky claim,

$$\sum_{j=0}^{2} x_j \sigma_{j,2} = \sigma_{R_Y,2} = -\sigma_{y_Y,2},$$

we can also write

$$r_2 = -\psi^{-1}\left(\mu_{y_Y,2} + \sigma\sigma_{y_Y,2}^\top\right) + \psi^{-1}\sum_{j=0}^{2} x_j\left[\mu_{c_j,2} + (1-\gamma)\sigma_{c_j,2}\sigma^\top\right]$$
$$+ (1-\psi^{-1})\gamma\left[-\sigma_{y_Y,2}\sigma^\top + \sum_{j=0}^{2} x_j\left(\frac{1}{2}\sigma_{j,1}\sigma_{j,1}^\top + \hat{\gamma}_j\sigma_{j,1}\sigma^\top\right)\right] - \pi_{Y,2}. \tag{C.6}$$

This expression separates the second-order interest-rate correction into the aggregate-claim drift term, the intertemporal-hedging term, the second-order change in average risk compensation, and the second-order risk premium.

Step 6: Dividend yield.

The dividend yield is given by

$$y_Y = r + \eta(\sigma + \sigma_{p_Y})^\top - \left(\mu + \mu_{p_Y} + \sigma\sigma_{p_Y}^\top\right),$$

where σ_{p_Y} and μ_{p_Y} denote the diffusion and drift of the price-dividend ratio. Since $y_Y = 1/p_Y$, we have

$$\sigma_{p_Y,2} = -\sigma_{y_Y,2}, \qquad \mu_{p_Y,2} = -\mu_{y_Y,2},$$

up to terms of order higher than ϵ^2. Expanding the dividend yield to second order gives

$$\begin{aligned} y_{Y,2} &= r_2 + \sigma\eta_2^\top - \eta_0\sigma_{y_Y,2}^\top + \mu_{y_Y,2} + \sigma\sigma_{y_Y,2}^\top \\ &= r_2 + \pi_{Y,2} + \mu_{y_Y,2} + \sigma\sigma_{y_Y,2}^\top, \end{aligned}$$

where the second equality uses the expression for $\pi_{Y,2}$ from Step 4. Substituting the expression for r_2 from (C.6), we obtain

$$y_{Y,2} = (1-\psi^{-1})\mu_{y_Y,2} + (1-\psi^{-1})\sigma\sigma_{y_Y,2}^\top + \psi^{-1}\sum_{j=0}^{2} x_j\left[\mu_{c_j,2} + (1-\gamma)\sigma_{c_j,2}\sigma^\top\right]$$
$$+ (1-\psi^{-1})\gamma\left[-\sigma_{y_Y,2}\sigma^\top + \sum_{j=0}^{2} x_j\left(\frac{1}{2}\sigma_{j,1}\sigma_{j,1}^\top + \hat{\gamma}_j\sigma_{j,1}\sigma^\top\right)\right].$$

From Step 1, the first-order corrections imply

$$\mu_{c_j,2} = (1-\psi)\mu_{y_Y,2}, \qquad \sigma_{c_j,2} = (1-\psi)\sigma_{y_Y,2}.$$

Using $\sum_{j=0}^{2} x_j = 1$, the terms involving $\mu_{y_Y,2}$ cancel. The terms involving $\sigma_{y_Y,2}\sigma^\top$ also cancel.

Therefore,

$$y_{Y,2} = (1 - \psi^{-1})\gamma \sum_{j=0}^{2} x_j \left(\frac{1}{2}\sigma_{j,1}\sigma_{j,1}^{\top} + \hat{\gamma}_j \sigma_{j,1}\sigma^{\top} \right). \tag{C.7}$$

Equivalently,

$$y_{Y,2} = (1 - \psi^{-1})\gamma \|\sigma\|^2 \sum_{j=0}^{2} x_j \left(\frac{1}{2} m_{j,1}^2 + \hat{\gamma}_j m_{j,1} \right), \tag{C.8}$$

where $\sigma_{j,1} = m_{j,1}\sigma$. This expression shows that the second-order correction to the dividend yield depends only on first-order risk reallocations and preference heterogeneity.

The first-order exposure coefficients are affine in $\hat{\alpha}_p$. Write

$$m_{j,1} = \bar{m}_j - \ell_j \hat{\alpha}_p.$$

For passive investors,

$$\bar{m}_0 = 0, \qquad \ell_0 = -1,$$

so that $m_{0,1} = \hat{\alpha}_p$. For active investors,

$$\begin{aligned} \bar{m}_1 &= \frac{x_2}{x_a}(\hat{\gamma}_2 - \hat{\gamma}_1 + \hat{\nu}_2), & \qquad \ell_1 &= \frac{x_0}{x_a}, \\ \bar{m}_2 &= \frac{x_1}{x_a}(\hat{\gamma}_1 - \hat{\gamma}_2 - \hat{\nu}_2), & \qquad \ell_2 &= \frac{x_0}{x_a}. \end{aligned}$$

These definitions satisfy the first-order market-clearing condition

$$\sum_{j=0}^{2} x_j m_{j,1} = 0.$$

Substituting $m_{j,1} = \bar{m}_j - \ell_j \hat{\alpha}_p$ into (C.8), the solution for $y_{Y,2}$ is quadratic in $\hat{\alpha}_p$:

$$y_{Y,2}(x, \hat{\alpha}_p) = a_Y^0(x) + b_Y^0(x)\hat{\alpha}_p + d_Y^0(x)\hat{\alpha}_p^2,$$

where

$$a_Y^0(x) = (1-\psi^{-1})\gamma\|\sigma\|^2 \sum_{j=0}^{2} x_j \left(\frac{1}{2}\bar{m}_j^2 + \hat{\gamma}_j \bar{m}_j\right),$$

$$b_Y^0(x) = -(1-\psi^{-1})\gamma\|\sigma\|^2 \sum_{j=0}^{2} x_j \ell_j \left(\bar{m}_j + \hat{\gamma}_j\right),$$

$$d_Y^0(x) = \frac{1}{2}(1-\psi^{-1})\gamma\|\sigma\|^2 \sum_{j=0}^{2} x_j \ell_j^2.$$

Using the values of ℓ_j, the quadratic coefficient can be simplified to

$$d_Y^0(x) = \frac{1}{2}(1-\psi^{-1})\gamma\|\sigma\|^2 \left[x_0 + \frac{x_0^2}{x_a}\right] = \frac{1}{2}(1-\psi^{-1})\gamma\|\sigma\|^2 \frac{x_0}{x_a}.$$

The second-order price-dividend-ratio correction follows from the change of variables above:

$$p_{Y,2} = p_{Y,0}^3 y_{Y,1}^2 - p_{Y,0}^2 y_{Y,2}.$$

Step 7: Consumption-wealth ratio.
The consumption-wealth ratio is given by

$$c_{j,t} = \psi\rho + (1-\psi)\left[r_t + \eta_t\,\sigma_{j,t}^\top - \frac{\gamma_j}{2}\|\sigma_{j,t}\|^2\right] + \mu_{c_j,t} + (1-\gamma_j)\sigma_{c_j,t}\sigma_{j,t}^\top + \frac{\psi-\gamma_j}{1-\psi}\frac{\|\sigma_{c_j,t}\|^2}{2}.$$

We expand each term to second order. Using

$$r_t = r_0 + r_1\epsilon + r_2\epsilon^2 + O(\epsilon^3),$$

$$\eta_t = \eta_0 + \eta_1\epsilon + \eta_2\epsilon^2 + O(\epsilon^3),$$

$$\sigma_{j,t} = \sigma + \sigma_{j,1}\epsilon + \sigma_{j,2}\epsilon^2 + O(\epsilon^3),$$

and

$$\gamma_j = \gamma(1 + \hat{\gamma}_j\epsilon),$$

$$\sigma_{c_j,t} = \sigma_{c_j,2}\epsilon^2 + O(\epsilon^3),$$

$$\mu_{c_j,t} = \mu_{c_j,2}\epsilon^2 + O(\epsilon^3),$$

we collect second-order terms.

The first-order term is

$$c_{j,1} = (1-\psi)\left[r_1 + \eta_1\sigma^\top - \frac{\gamma}{2}\|\sigma\|^2\hat{\gamma}_j\right],$$

which matches the previous result.

Before simplifying, the second-order term is

$$\begin{aligned}c_{j,2} = (1-\psi)&\left[r_2 + \eta_2\sigma^\top + \eta_1\sigma_{j,1}^\top + \eta_0\sigma_{j,2}^\top - \gamma\sigma\sigma_{j,2}^\top - \frac{\gamma}{2}\sigma_{j,1}\sigma_{j,1}^\top - \gamma\hat{\gamma}_j\sigma\sigma_{j,1}^\top\right]\\ &+ \mu_{c_j,2} + (1-\gamma)\sigma_{c_j,2}\sigma^\top.\end{aligned}$$

Using $\eta_0 = \gamma\sigma$, the terms involving $\sigma_{j,2}$ cancel. Therefore,

$$\begin{aligned}c_{j,2} = (1-\psi)&\left[r_2 + \eta_2\sigma^\top + \eta_1\sigma_{j,1}^\top - \frac{\gamma}{2}\sigma_{j,1}\sigma_{j,1}^\top - \gamma\hat{\gamma}_j\sigma\sigma_{j,1}^\top\right]\\ &+ \mu_{c_j,2} + (1-\gamma)\sigma_{c_j,2}\sigma^\top.\end{aligned}$$

Substituting the expressions for r_2, η_2, and $\sigma_{j,2}$ derived in Steps 2–5 yields a closed-form expression for $c_{j,2}$ in terms of first-order reallocations and heterogeneity.

Step 8: The dividend claim.

The normalized price of the dividend claim, $p_D = P_D/Y$, satisfies

$$r_t + (\sigma + \sigma_{p_D,t})\eta_t^\top = \frac{s_t}{p_{D,t}} + \mu + \mu_{p_D,t} + \sigma\sigma_{p_D,t}^\top.$$

Equivalently, after multiplying by $p_{D,t}$,

$$\left[r_t + \sigma\eta_t^\top - \mu\right]p_{D,t} = s_t + p_{D,x,t}\mu_{x,t} + p_{D,s,t}\mu_{s,t} + \frac{1}{2}\sum_{k=1}^{3}\Sigma_{D,k,t}^\top p_{D,XX,t}\Sigma_{D,k,t} + (\sigma - \eta_t)\left(p_{D,x,t}\sigma_{x,t} + p_{D,s,t}\sigma_{s,t}\right)^\top,$$

where $X = (x, s)$, $p_{D,XX}$ is the Hessian with respect to (x, s), and $\Sigma_{D,k,t}$ is the k-th column of the diffusion matrix of (x, s).

We now introduce the local coordinate

$$s = \bar{s} + \epsilon\hat{s}.$$

The function p_D is a function of the physical state (x, s), but its perturbation coefficients are functions of $(x, \hat{s})$:

$$p_D(x, s; \epsilon) = p_{D,0} + p_{D,1}(x, \hat{s})\epsilon + p_{D,2}(x, \hat{s})\epsilon^2 + O(\epsilon^3).$$

Because $\hat{s} = (s - \bar{s})/\epsilon$, derivatives with respect to the physical state s satisfy

$$
\begin{aligned}
p_{D,s} &= p_{D,1,\hat{s}} + \epsilon p_{D,2,\hat{s}} + O(\epsilon^2), \\
p_{D,ss} &= \epsilon^{-1} p_{D,1,\hat{s}\hat{s}} + p_{D,2,\hat{s}\hat{s}} + O(\epsilon).
\end{aligned}
$$

In the first-order solution below, $p_{D,1}$ is linear in $\hat{s}$, so $p_{D,1,\hat{s}\hat{s}} = 0$ and the second derivative is well behaved:

$$
p_{D,ss} = p_{D,2,\hat{s}\hat{s}} + O(\epsilon).
$$

Similarly,

$$
p_{D,x} = \epsilon p_{D,1,x} + \epsilon^2 p_{D,2,x} + O(\epsilon^3), \qquad p_{D,xs} = p_{D,1,x\hat{s}} + \epsilon p_{D,2,x\hat{s}} + O(\epsilon^2).
$$

The dividend-share process implies

$$
\begin{aligned}
\mu_{s,t} &= -\kappa_s (s_t - \bar{s}) = -\kappa_s \hat{s} \epsilon, \\
\sigma_{s,t} &= s_t(1 - s_t)\sigma_s \epsilon = \bar{s}(1 - \bar{s})\sigma_s \epsilon + (1 - 2\bar{s})\hat{s}\sigma_s \epsilon^2 + O(\epsilon^3).
\end{aligned}
$$

The wealth-share dynamics satisfy

$$
\mu_{x,t} = \mu_{x,1}\epsilon + O(\epsilon^2), \qquad \sigma_{x,t} = \sigma_{x,1}\epsilon + O(\epsilon^2).
$$

Define

$$
A_t \equiv r_t + \sigma \eta_t^\top - \mu = \delta + A_1 \epsilon + A_2 \epsilon^2 + O(\epsilon^3),
$$

where

$$
\delta = r_0 + \sigma \eta_0^\top - \mu, \qquad A_1 = r_1 + \sigma \eta_1^\top, \qquad A_2 = r_2 + \sigma \eta_2^\top.
$$

The payoff satisfies $s_t = \bar{s} + \hat{s}\epsilon$.

The zeroth-order condition is

$$
\delta p_{D,0} = \bar{s}, \qquad \text{so} \qquad p_{D,0} = \frac{\bar{s}}{\delta}.
$$

At first order, the only state-derivative terms come from the drift of s and from the order-ϵ diffusion of s evaluated at $\bar{s}$. The first-order correction satisfies

$$
\delta p_{D,1} + A_1 p_{D,0} = \hat{s} - \kappa_s \hat{s} p_{D,1,\hat{s}} + (\sigma - \eta_0)\left[p_{D,1,\hat{s}}\bar{s}(1 - \bar{s})\sigma_s\right]^\top.
$$

This equation is the same first-order pricing equation derived above. Its solution is

$$p_{D,1} = \frac{\hat{s}}{\delta + \kappa_s} - (r_1 + \pi_{Y,1})\frac{\bar{s}}{\delta^2} - \frac{\gamma - 1}{\delta(\delta + \kappa_s)}\bar{s}(1 - \bar{s})\sigma\sigma_s^\top.$$

At second order, using $\mu_{s,t} = -\kappa_s \hat{s}\epsilon$ and $p_{D,s} = p_{D,1,\hat{s}} + \epsilon p_{D,2,\hat{s}} + O(\epsilon^2)$, the correction satisfies

$$\begin{aligned}
\delta p_{D,2} + A_1 p_{D,1} + A_2 p_{D,0} = -\kappa_s \hat{s} p_{D,2,\hat{s}} &+ (\sigma - \eta_0)\left[p_{D,2,\hat{s}}\bar{s}(1-\bar{s})\sigma_s\right]^\top \\
&+ p_{D,1,x}\mu_{x,1} + (\sigma - \eta_0)\left[p_{D,1,x}\sigma_{x,1}\right]^\top - \eta_1\left[p_{D,1,\hat{s}}\bar{s}(1-\bar{s})\sigma_s\right]^\top \\
&+ (\sigma - \eta_0)\left[p_{D,1,\hat{s}}(1 - 2\bar{s})\hat{s}\sigma_s\right]^\top + \frac{1}{2}p_{D,2,\hat{s}\hat{s}}\bar{s}^2(1-\bar{s})^2\sigma_s\sigma_s^\top. \quad \text{(C.9)}
\end{aligned}$$

The potential second-order Hessian terms involving $p_{D,1,x\hat{s}}$ and $p_{D,1,\hat{s}\hat{s}}$ vanish because $p_{D,1}$ is additively separable in $(x, \hat{s})$ and linear in $\hat{s}$. Thus, after these simplifications, the right-hand side of (C.9) is at most affine in $\hat{s}$, apart from the operator applied to $p_{D,2}$. We can solve (C.9) by matching coefficients in $\hat{s}$. Let

$$B_s \equiv \bar{s}(1 - \bar{s}), \qquad \Lambda_s \equiv (\sigma - \eta_0)\sigma_s^\top = (1 - \gamma)\sigma\sigma_s^\top.$$

Write the first-order solution as

$$p_{D,1}(x, \hat{s}) = a_1(x, \hat{\alpha}_p) + b_1\hat{s},$$

where

$$b_1 = \frac{1}{\delta + \kappa_s},$$

and

$$a_1(x, \hat{\alpha}_p) = -(r_1 + \pi_{Y,1})\frac{\bar{s}}{\delta^2} - \frac{\gamma - 1}{\delta(\delta + \kappa_s)}B_s\sigma\sigma_s^\top.$$

Equivalently, using $A_1 = r_1 + \sigma\eta_1^\top = r_1 + \pi_{Y,1}$,

$$a_1(x, \hat{\alpha}_p) = -\frac{A_1 p_{D,0}}{\delta} + \frac{\Lambda_s B_s}{\delta(\delta + \kappa_s)}.$$

Conjecture that the second-order correction is quadratic in $\hat{s}$:

$$p_{D,2}(x, \hat{s}) = a_2(x, \hat{\alpha}_p) + b_2(x, \hat{\alpha}_p)\hat{s} + c_2(x, \hat{\alpha}_p)\hat{s}^2.$$

Matching the coefficient on $\hat{s}^2$ gives

$$(\delta + \kappa_s)c_2 = 0,$$

so

$$c_2 = 0.$$

Thus, in this benchmark case, the second-order correction is affine in $\hat{s}$.

Matching the coefficient on $\hat{s}$ gives

$$(\delta + \kappa_s)b_2 = -A_1 b_1 + \Lambda_s(1 - 2\bar{s})b_1,$$

so

$$b_2(x, \hat{\alpha}_p) = \frac{\Lambda_s(1 - 2\bar{s}) - A_1}{(\delta + \kappa_s)^2}.$$

Finally, matching the constant term gives

$$\delta a_2 = -A_1 a_1 - A_2 p_{D,0} + \Lambda_s B_s b_2 + \mathcal{M}_x - \eta_1 \left(b_1 B_s \sigma_s\right)^\top,$$

where

$$\mathcal{M}_x \equiv a_{1,x}\mu_{x,1} + (\sigma - \eta_0)\left[a_{1,x}\sigma_{x,1}\right]^\top.$$

Therefore,

$$a_2(x, \hat{\alpha}_p) = \frac{1}{\delta}\left[-A_1 a_1 - A_2 p_{D,0} + \Lambda_s B_s b_2 + \mathcal{M}_x - \eta_1 \left(b_1 B_s \sigma_s\right)^\top\right].$$

Combining the coefficient restrictions, the second-order correction is

$$p_{D,2}(x, \hat{s}) = a_2(x, \hat{\alpha}_p) + b_2(x, \hat{\alpha}_p)\hat{s}.$$

The term proportional to $\eta_1(b_1 B_s \sigma_s)^\top$ captures the interaction between first-order changes in the market price of risk and the first-order dividend-share exposure of the dividend claim.

Step 9: Pricing impact for the dividend claim.

We now compute the derivative of the second-order dividend-claim price with respect to the passive investors' portfolio share $\hat{\alpha}_p$. The coefficient representation from Step 8 gives

$$p_{D,2}(x, \hat{s}) = a_2(x, \hat{\alpha}_p) + b_2(x, \hat{\alpha}_p)\hat{s},$$

where

$$b_2(x, \hat{\alpha}_p) = \frac{\Lambda_s(1 - 2\bar{s}) - A_1}{(\delta + \kappa_s)^2}$$

and

$$a_2(x, \hat{\alpha}_p) = \frac{1}{\delta}\left[-A_1 a_1 - A_2 p_{D,0} + \Lambda_s B_s b_2 + \mathcal{M}_x - \eta_1 \left(b_1 B_s \sigma_s\right)^\top\right].$$

Recall that

$$A_1 = r_1 + \sigma\eta_1^\top = r_1 + \pi_{Y,1}.$$

The first-order aggregate-price term is independent of $\hat{\alpha}_p$, so

$$\frac{\partial A_1}{\partial \hat{\alpha}_p} = 0.$$

It follows that a_1 and b_2 are also independent of $\hat{\alpha}_p$, except through the wealth-share dynamics term $\mathcal{M}_x$ in a_2. Therefore,

$$\frac{\partial p_{D,2}}{\partial \hat{\alpha}_p} = \frac{\partial a_2}{\partial \hat{\alpha}_p},$$

and

$$\frac{\partial p_{D,2}}{\partial \hat{\alpha}_p} = \frac{1}{\delta}\left[-p_{D,0}\frac{\partial A_2}{\partial \hat{\alpha}_p} + \frac{\partial \mathcal{M}_x}{\partial \hat{\alpha}_p} - \frac{\partial \eta_1}{\partial \hat{\alpha}_p}\left(b_1 B_s \sigma_s\right)^\top\right].$$

Using $p_{D,0} = \bar{s}/\delta$ and $b_1 = 1/(\delta + \kappa_s)$, this can be written as

$$\frac{\partial p_{D,2}}{\partial \hat{\alpha}_p} = -\frac{\bar{s}}{\delta^2}\frac{\partial A_2}{\partial \hat{\alpha}_p} - \frac{B_s}{\delta(\delta+\kappa_s)}\frac{\partial \eta_1}{\partial \hat{\alpha}_p}\sigma_s^\top + \frac{1}{\delta}\frac{\partial \mathcal{M}_x}{\partial \hat{\alpha}_p}.$$

The first term is the aggregate-price-impact channel. The second term is specific to the dividend claim: passive demand changes the first-order market price of risk, and the dividend claim loads on dividend-share risk through $b_1 B_s \sigma_s$. The last term captures the indirect effect operating through first-order wealth-share dynamics. Since

$$\mathcal{M}_x = a_{1,x}\mu_{x,1} + (\sigma - \eta_0)\left[a_{1,x}\sigma_{x,1}\right]^\top,$$

and a_1 is independent of $\hat{\alpha}_p$, this indirect term is

$$\frac{\partial \mathcal{M}_x}{\partial \hat{\alpha}_p} = a_{1,x}\frac{\partial \mu_{x,1}}{\partial \hat{\alpha}_p} + (\sigma - \eta_0)\left[a_{1,x}\frac{\partial \sigma_{x,1}}{\partial \hat{\alpha}_p}\right]^\top.$$

This term is equal to zero. To see this, recall that the first-order diffusion of wealth shares is

$$\sigma_{x_j,1} = x_j \sigma_{j,1},$$

and the first-order drift of wealth shares is

$$\mu_{x_j,1} = x_j\left[(\eta_0 - \sigma)\sigma_{j,1}^\top - c_{j,1} + y_{Y,1}\right].$$

Since $c_{j,1}$ and $y_{Y,1}$ are independent of $\hat{\alpha}_p$, differentiating with respect to $\hat{\alpha}_p$ gives

$$\frac{\partial \mu_{x_j,1}}{\partial \hat{\alpha}_p} = x_j(\eta_0 - \sigma)\frac{\partial \sigma_{j,1}^\top}{\partial \hat{\alpha}_p}, \qquad \frac{\partial \sigma_{x_j,1}}{\partial \hat{\alpha}_p} = x_j \frac{\partial \sigma_{j,1}}{\partial \hat{\alpha}_p}.$$

Therefore,

$$\begin{aligned}\frac{\partial \mathcal{M}_x}{\partial \hat{\alpha}_p} &= \sum_{j=1}^{2} a_{1,x_j} x_j (\eta_0 - \sigma)\frac{\partial \sigma_{j,1}^\top}{\partial \hat{\alpha}_p} + (\sigma - \eta_0)\sum_{j=1}^{2} a_{1,x_j} x_j \frac{\partial \sigma_{j,1}^\top}{\partial \hat{\alpha}_p} \\ &= 0.\end{aligned}$$

Thus the indirect wealth-share-dynamics channel drops out of the dividend-claim price impact. Finally, since

$$p_D(x,s;\epsilon) = p_{D,0} + p_{D,1}(x,\hat{s})\epsilon + p_{D,2}(x,\hat{s})\epsilon^2 + O(\epsilon^3)$$

and $p_{D,1}$ is independent of $\hat{\alpha}_p$, the price impact of passive flows is

$$\frac{\partial p_D(x,s;\epsilon)}{\partial F(x,s;\epsilon)} = \frac{\epsilon}{x_0}\frac{\partial p_{D,2}}{\partial \hat{\alpha}_p} + O(\epsilon^2).$$

Thus, the dividend-claim price impact is the sum of an aggregate-price-impact channel and a direct dividend-risk-exposure channel.

C.6 Second-order correction: time-varying passive portfolio share

We next derive the second-order correction in the case the passive portfolio share is time-varying. Now, $\overline{\alpha}_p$ becomes a state variable. The dynamics of $\overline{\alpha}_p = 1 + \hat{\alpha}_p\epsilon$ is given by

$$\begin{aligned}\mu_{\overline{\alpha}_p,t} &= \kappa_p(\hat{\alpha} - \hat{\alpha}_p)\epsilon, \\ \sigma_{\overline{\alpha}_p,t} &= \sigma_p\epsilon.\end{aligned}$$

We consider the following expansion of $c_j(x,\overline{\alpha}_p;\epsilon)$:

$$c_j(x,\overline{\alpha}_p;\epsilon) = c_{j,0} + c_{j,1}(x,\hat{\alpha}_p)\epsilon + c_{j,2}(x,\hat{\alpha}_p)\epsilon^2 + O(\epsilon^3).$$

Because $\hat{\alpha}_p = (\overline{\alpha}_p - 1)/\epsilon$, derivatives with respect to the physical state $\overline{\alpha}_p$ satisfy the same chain-rule structure as for s. For a generic object $f(x,\overline{\alpha}_p;\epsilon) = f_0 + f_1(x,\hat{\alpha}_p)\epsilon + f_2(x,\hat{\alpha}_p)\epsilon^2 + O(\epsilon^3)$, we have

$$f_{\overline{\alpha}_p} = f_{1,\hat{\alpha}_p} + \epsilon f_{2,\hat{\alpha}_p} + O(\epsilon^2),$$
$$f_{\overline{\alpha}_p\overline{\alpha}_p} = \epsilon^{-1} f_{1,\hat{\alpha}_p\hat{\alpha}_p} + f_{2,\hat{\alpha}_p\hat{\alpha}_p} + O(\epsilon).$$

In our setting, $c_{j,1}$ and $y_{Y,1}$ are independent of $\hat{\alpha}_p$, so

$$f_{1,\hat{\alpha}_p} = 0, \qquad f_{1,\hat{\alpha}_p\hat{\alpha}_p} = 0,$$

and therefore

$$f_{\overline{\alpha}_p} = \epsilon f_{2,\hat{\alpha}_p} + O(\epsilon^2), \qquad f_{\overline{\alpha}_p\overline{\alpha}_p} = f_{2,\hat{\alpha}_p\hat{\alpha}_p} + O(\epsilon).$$

Similarly, cross-derivatives satisfy

$$f_{x\overline{\alpha}_p} = f_{1,x\hat{\alpha}_p} + \epsilon f_{2,x\hat{\alpha}_p} + O(\epsilon^2) = \epsilon f_{2,x\hat{\alpha}_p} + O(\epsilon^2),$$

since f_1 does not depend on $\hat{\alpha}_p$.

Step 1: Drift and diffusion for c_j and y_Y.

We avoid repeating the derivation for the case in which $\hat{\alpha}_p$ is fixed. For any second-order object z_2, write

$$z_2 = z_2^0 + \Delta z_2,$$

where z_2^0 denotes the second-order correction derived in the fixed-$\hat{\alpha}_p$ economy, and Δz_2 collects only the additional terms generated by the drift and diffusion of $\overline{\alpha}_p$. By construction, Δz_2 vanishes when $\kappa_p = 0$ and $\sigma_p = 0_{1\times 3}$.

Because $c_{j,1}$ and $y_{Y,1}$ are independent of $\hat{\alpha}_p$, the diffusion terms coming from the time variation in $\overline{\alpha}_p$ are

$$\Delta\sigma_{c_j,2} = \frac{c_{j,2,\hat{\alpha}_p}}{c_{j,0}}\sigma_p,$$
$$\Delta\sigma_{y_Y,2} = \frac{y_{Y,2,\hat{\alpha}_p}}{y_{Y,0}}\sigma_p.$$

Therefore,

$$\sigma_{c_j,2} = \sigma^0_{c_j,2} + \frac{c_{j,2,\hat{\alpha}_p}}{c_{j,0}}\sigma_p,$$
$$\sigma_{y_Y,2} = \sigma^0_{y_Y,2} + \frac{y_{Y,2,\hat{\alpha}_p}}{y_{Y,0}}\sigma_p.$$

Similarly, the new drift terms generated by the time variation in $\overline{\alpha}_p$ are

$$\Delta\mu_{c_j,2} = \frac{c_{j,2,\hat{\alpha}_p}}{c_{j,0}}\kappa_p(\hat{\alpha} - \hat{\alpha}_p) + \frac{1}{2}\frac{c_{j,2,\hat{\alpha}_p\hat{\alpha}_p}}{c_{j,0}}\sigma_p\sigma_p^\top,$$
$$\Delta\mu_{y_Y,2} = \frac{y_{Y,2,\hat{\alpha}_p}}{y_{Y,0}}\kappa_p(\hat{\alpha} - \hat{\alpha}_p) + \frac{1}{2}\frac{y_{Y,2,\hat{\alpha}_p\hat{\alpha}_p}}{y_{Y,0}}\sigma_p\sigma_p^\top.$$

Thus the only new second-order terms are proportional to κ_p or σ_p.

Step 2 (time-varying α_p): Risk exposures of investors.

Write

$$\sigma_{j,2} = \sigma_{j,2}^0 + \Delta\sigma_{j,2},$$

where $\sigma_{j,2}^0$ is the second-order exposure from the fixed-$\hat{\alpha}_p$ economy and $\Delta\sigma_{j,2}$ collects only the new terms generated by the time variation in $\overline{\alpha}_p$.

The desired exposure remains

$$q_{j,t} \equiv \frac{\eta_t}{\gamma_j} + \frac{1-\gamma_j^{-1}}{\psi - 1}\sigma_{c_j,t}.$$

Relative to the fixed-$\hat{\alpha}_p$ economy, the second-order desired exposure changes for two reasons: the market price of risk changes by $\Delta\eta_2$, and the consumption-wealth ratio has an additional diffusion induced by σ_p. Thus,

$$\Delta q_{j,2} = \frac{\Delta\eta_2}{\gamma} + \frac{1-\gamma^{-1}}{\psi - 1}\Delta\sigma_{c_j,2}.$$

Using Steps 1 and 2,

$$\Delta q_{j,2} = \frac{\Delta\eta_2}{\gamma} + \frac{1-\gamma^{-1}}{\psi - 1}\frac{c_{j,2,\hat{\alpha}_p}}{c_{j,0}}\sigma_p.$$

For an unconstrained investor, the multiplier is zero, so the incremental second-order exposure is simply

$$\Delta\sigma_{j,2} = \Delta q_{j,2}.$$

Therefore,

$$\Delta\sigma_{j,2} = \frac{\Delta\eta_2}{\gamma} + \frac{1-\gamma^{-1}}{\psi - 1}\frac{c_{j,2,\hat{\alpha}_p}}{c_{j,0}}\sigma_p.$$

For a constrained investor, the second-order multiplier adjusts to keep the constraint binding. Writing

$$\hat{\nu}_{j,2} = \hat{\nu}_{j,2}^0 + \Delta\hat{\nu}_{j,2},$$

the incremental exposure is

$$\Delta\sigma_{j,2} = \Delta q_{j,2} - \Delta\hat{\nu}_{j,2}\sigma.$$

The second-order constraint condition requires the component of $\Delta\sigma_{j,2}$ in the direction of σ to vanish:

$$\Delta\sigma_{j,2}\sigma^{\top} = 0.$$

Hence

$$\Delta\hat{v}_{j,2} = \frac{\Delta q_{j,2}\sigma^{\top}}{\sigma\sigma^{\top}},$$

and therefore

$$\Delta\sigma_{j,2} = \Delta q_{j,2} - \frac{\Delta q_{j,2}\sigma^{\top}}{\sigma\sigma^{\top}}\sigma.$$

Thus, for constrained investors, only the component of the new desired exposure orthogonal to σ affects actual risk exposure; the component parallel to σ is absorbed by the second-order multiplier.

Combining the results, the time-varying passive portfolio share affects second-order risk exposures through the new market-price-of-risk term $\Delta\eta_2$ and the new consumption-wealth diffusion term proportional to σ_p.

Step 3 (time-varying α_p): Market price of risk.

Write

$$\eta_2 = \eta_2^0 + \Delta\eta_2,$$

where η_2^0 is the second-order market price of risk from the fixed-$\hat{\alpha}_p$ economy and $\Delta\eta_2$ collects only the new terms generated by the time variation in $\overline{\alpha}_p$.

Recall that

$$\eta_t = \gamma_{a,t}\left[\chi_{a,t}\sigma_{R_Y,t} - h_{a,t}\right].$$

Relative to the fixed-$\hat{\alpha}_p$ economy, there are three new second-order effects. First, the aggregate claim has an additional diffusion induced by σ_p. Second, active investors have additional hedging demand because their consumption-wealth ratios load on σ_p. Third, if the leverage constraint binds for investor 2, the second-order multiplier changes, which changes effective aggregate risk aversion.

From Step 1,

$$\Delta\sigma_{y_Y,2} = \frac{y_{Y,2,\hat{\alpha}_p}}{y_{Y,0}}\sigma_p,$$

and the additional hedging demand is

$$\Delta h_{a,2} = \frac{1-\gamma^{-1}}{\psi-1}\mathbb{E}_a[\Delta\sigma_{c_j,2}] = \frac{1-\gamma^{-1}}{\psi-1}\mathbb{E}_a\left[\frac{c_{j,2,\hat{\alpha}_p}}{c_{j,0}}\right]\sigma_p.$$

The new contribution to effective aggregate risk aversion comes from the second-order multiplier. Using the expansion for $\gamma_{a,t}$ from the fixed-$\hat{\alpha}_p$ case,

$$\Delta\Gamma_{a,2} = \mathbb{E}_a[\Delta\hat{v}_{j,2}].$$

Therefore, the incremental market price of risk is

$$\Delta\eta_2 = \gamma \left[\mathbb{E}_a[\Delta\hat{v}_{j,2}]\sigma - \Delta\sigma_{y_Y,2} - \Delta h_{a,2}\right]. \tag{C.10}$$

If no leverage constraint binds, then $\Delta\hat{v}_{j,2} = 0$ for all active investors and (C.10) reduces to

$$\Delta\eta_2 = -\gamma \left[\frac{y_{Y,2,\hat{\alpha}_p}}{y_{Y,0}} + \frac{1-\gamma^{-1}}{\psi - 1}\mathbb{E}_a\left[\frac{c_{j,2,\hat{\alpha}_p}}{c_{j,0}}\right]\right]\sigma_p.$$

If the leverage constraint binds for investor 2, then $\Delta\hat{v}_{1,2} = 0$ and $\Delta\hat{v}_{2,2}$ is pinned down by the incremental second-order constraint. Let

$$H_j \equiv \frac{1-\gamma^{-1}}{\psi - 1}\frac{c_{j,2,\hat{\alpha}_p}}{c_{j,0}}.$$

The incremental desired exposure of investor 2 is

$$\Delta q_{2,2} = \frac{\Delta\eta_2}{\gamma} + H_2\sigma_p.$$

The binding constraint requires the component of $\Delta\sigma_{2,2}$ in the direction of σ to be zero. Thus,

$$\Delta\hat{v}_{2,2} = \frac{\Delta q_{2,2}\sigma^\top}{\sigma\sigma^\top} = \frac{\Delta\eta_2\sigma^\top}{\gamma\sigma\sigma^\top} + H_2\frac{\sigma_p\sigma^\top}{\sigma\sigma^\top}. \tag{C.11}$$

Substituting (C.11) into (C.10) gives the linear system

$$\begin{aligned}\Delta\eta_2 &= \gamma\left[\frac{x_2}{x_a}\Delta\hat{v}_{2,2}\sigma - \frac{y_{Y,2,\hat{\alpha}_p}}{y_{Y,0}}\sigma_p - \frac{1-\gamma^{-1}}{\psi-1}\mathbb{E}_a\left[\frac{c_{j,2,\hat{\alpha}_p}}{c_{j,0}}\right]\sigma_p\right], \\ \Delta\hat{v}_{2,2} &= \frac{\Delta\eta_2\sigma^\top}{\gamma\sigma\sigma^\top} + H_2\frac{\sigma_p\sigma^\top}{\sigma\sigma^\top}.\end{aligned} \tag{C.12}$$

Equations (C.12) jointly determine the new market-price-of-risk component and the new second-order multiplier when investor 2 is constrained. The key difference relative to the slack case is that part of the new risk is absorbed by the multiplier, which changes effective aggregate risk aversion.
Step 4 (time-varying α_p): Risk premium.

The risk premium on the aggregate claim is given by

$$\pi_{Y,t} = \sigma_{R_Y,t}\eta_t^\top.$$

We write

$$\pi_{Y,2} = \pi_{Y,2}^0 + \Delta\pi_{Y,2},$$

where $\pi_{Y,2}^0$ is the second-order risk premium in the fixed-$\hat{\alpha}_p$ economy.

Since

$$\sigma_{R_Y,t} = \sigma - \sigma_{y_Y,2}\epsilon^2 + O(\epsilon^3),$$

the second-order risk premium is

$$\pi_{Y,2} = -\sigma_{y_Y,2}\eta_0^\top + \sigma\eta_2^\top.$$

Therefore, the incremental risk premium induced by time variation in $\overline{\alpha}_p$ is

$$\Delta\pi_{Y,2} = -\Delta\sigma_{y_Y,2}\eta_0^\top + \sigma\Delta\eta_2^\top. \tag{C.13}$$

Using $\eta_0 = \gamma\sigma$ and Step 1,

$$\Delta\pi_{Y,2} = -\gamma\frac{y_{Y,2,\hat{\alpha}_p}}{y_{Y,0}}\sigma_p\sigma^\top + \sigma\Delta\eta_2^\top.$$

If no leverage constraint binds, substituting the expression for $\Delta\eta_2$ from Step 3 gives

$$\Delta\pi_{Y,2} = -\gamma\left[\frac{y_{Y,2,\hat{\alpha}_p}}{y_{Y,0}}\sigma_p\sigma^\top + \left(\frac{y_{Y,2,\hat{\alpha}_p}}{y_{Y,0}} + \frac{1-\gamma^{-1}}{\psi-1}\mathbb{E}_a\left[\frac{c_{j,2,\hat{\alpha}_p}}{c_{j,0}}\right]\right)\sigma\sigma_p^\top\right].$$

When the leverage constraint binds for investor 2, the same formula (C.13) applies, but $\Delta\eta_2$ is jointly determined with $\Delta\hat{v}_{2,2}$ by (C.12).

Step 5 (time-varying α_p): Interest rate.

We write the second-order interest rate as

$$r_2 = r_2^0 + \Delta r_2,$$

where r_2^0 is the fixed-$\hat{\alpha}_p$ expression in (C.5), and Δr_2 collects only the new terms generated by the dynamics of $\overline{\alpha}_p$.

From (C.5), the incremental correction is

$$\begin{aligned}\Delta r_2 = &-\psi^{-1}\left(\Delta\mu_{y_Y,2} + \sigma\Delta\sigma_{y_Y,2}^{\top}\right)\\&+\psi^{-1}\sum_{j=0}^{2} x_j\left[\Delta\mu_{c_j,2} + (1-\gamma)\Delta\sigma_{c_j,2}\sigma^{\top}\right]\\&+(1-\psi^{-1})\gamma\sum_{j=0}^{2} x_j\Delta\sigma_{j,2}\sigma^{\top} - \Delta\pi_{Y,2}.\end{aligned} \tag{C.14}$$

All first-order terms in $\sigma_{j,1}$ and $\hat{\gamma}_j$ are already included in r_2^0 and therefore do not appear in Δr_2.

Using Step 1, the new drift and diffusion terms are

$$\begin{aligned}\Delta\sigma_{c_j,2} &= \frac{c_{j,2,\hat{\alpha}_p}}{c_{j,0}}\sigma_p,\\ \Delta\sigma_{y_Y,2} &= \frac{y_{Y,2,\hat{\alpha}_p}}{y_{Y,0}}\sigma_p,\\ \Delta\mu_{c_j,2} &= \frac{c_{j,2,\hat{\alpha}_p}}{c_{j,0}}\kappa_p(\hat{\alpha}-\hat{\alpha}_p) + \frac{1}{2}\frac{c_{j,2,\hat{\alpha}_p\hat{\alpha}_p}}{c_{j,0}}\sigma_p\sigma_p^{\top},\\ \Delta\mu_{y_Y,2} &= \frac{y_{Y,2,\hat{\alpha}_p}}{y_{Y,0}}\kappa_p(\hat{\alpha}-\hat{\alpha}_p) + \frac{1}{2}\frac{y_{Y,2,\hat{\alpha}_p\hat{\alpha}_p}}{y_{Y,0}}\sigma_p\sigma_p^{\top}.\end{aligned}$$

Moreover, from Step 4,

$$\Delta\pi_{Y,2} = -\Delta\sigma_{y_Y,2}\eta_0^{\top} + \sigma\Delta\eta_2^{\top}.$$

Substituting this expression into (C.14) gives

$$\begin{aligned}\Delta r_2 = &-\psi^{-1}\Delta\mu_{y_Y,2} - \psi^{-1}\sigma\Delta\sigma_{y_Y,2}^{\top}\\&+\psi^{-1}\sum_{j=0}^{2} x_j\left[\Delta\mu_{c_j,2} + (1-\gamma)\Delta\sigma_{c_j,2}\sigma^{\top}\right]\\&+(1-\psi^{-1})\gamma\sum_{j=0}^{2} x_j\Delta\sigma_{j,2}\sigma^{\top} + \Delta\sigma_{y_Y,2}\eta_0^{\top} - \sigma\Delta\eta_2^{\top}.\end{aligned}$$

Since $\eta_0 = \gamma\sigma$, the terms involving $\Delta\sigma_{y_Y,2}$ can also be combined as

$$\left(\gamma - \psi^{-1}\right)\sigma\Delta\sigma_{y_Y,2}^{\top}.$$

Thus, equivalently,

$$\Delta r_2 = -\psi^{-1}\Delta\mu_{y_Y,2} + \psi^{-1}\sum_{j=0}^{2} x_j \Delta\mu_{c_j,2}$$
$$+ \psi^{-1}(1-\gamma)\sum_{j=0}^{2} x_j \Delta\sigma_{c_j,2}\sigma^\top + (1-\psi^{-1})\gamma\sum_{j=0}^{2} x_j \Delta\sigma_{j,2}\sigma^\top$$
$$+ \left(\gamma - \psi^{-1}\right)\sigma\Delta\sigma_{y_Y,2}^\top - \sigma\Delta\eta_2^\top. \tag{C.15}$$

This expression simplifies substantially because the baseline economy is homogeneous. In particular, $c_{j,0} = y_{Y,0}$ for all j, and the aggregation identity $\Delta y_{Y,2} = \sum_{j=0}^{2} x_j \Delta c_{j,2}$ implies

$$\sum_{j=0}^{2} x_j \Delta\mu_{c_j,2} = \Delta\mu_{y_Y,2}, \qquad \sum_{j=0}^{2} x_j \Delta\sigma_{c_j,2} = \Delta\sigma_{y_Y,2}.$$

The drift terms in (C.15) therefore cancel. Moreover, the consumption-diffusion terms combine with the aggregate-claim diffusion term as

$$\psi^{-1}(1-\gamma)\Delta\sigma_{y_Y,2}\sigma^\top + \left(\gamma-\psi^{-1}\right)\sigma\Delta\sigma_{y_Y,2}^\top = (1-\psi^{-1})\gamma\Delta\sigma_{y_Y,2}\sigma^\top.$$

Using market clearing for the risky claim,

$$\sum_{j=0}^{2} x_j \Delta\sigma_{j,2} = -\Delta\sigma_{y_Y,2},$$

the remaining exposure terms cancel as well. Hence the incremental interest-rate correction reduces to

$$\Delta r_2 = -\sigma\Delta\eta_2^\top. \tag{C.16}$$

Thus, after aggregation and market clearing, the only effect of time-varying passive demand on the second-order interest rate operates through the induced change in the market price of risk. When no leverage constraint binds, $\Delta\eta_2$ is given directly by Step 3. When the leverage constraint binds for investor 2, $\Delta\eta_2$ and $\Delta\hat{\nu}_{2,2}$ are jointly determined by (C.12).

Step 6 (time-varying α_p): Dividend yield.

We write the second-order dividend yield as

$$y_{Y,2} = y_{Y,2}^0 + \Delta y_{Y,2},$$

where $y_{Y,2}^0$ is the fixed-$\hat{\alpha}_p$ correction in (C.7), and $\Delta y_{Y,2}$ collects only the new terms generated by the dynamics of $\overline{\alpha}_p$.

From the pricing identity,

$$y_{Y,2} = r_2 + \pi_{Y,2} + \mu_{y_Y,2} + \sigma\sigma_{y_Y,2}^\top,$$

so the incremental correction satisfies

$$\Delta y_{Y,2} = \Delta r_2 + \Delta\pi_{Y,2} + \Delta\mu_{y_Y,2} + \sigma\Delta\sigma_{y_Y,2}^\top.$$

Using (C.16) and (C.13), we obtain

$$\begin{aligned}\Delta y_{Y,2} &= -\sigma\Delta\eta_2^\top + \left(-\Delta\sigma_{y_Y,2}\eta_0^\top + \sigma\Delta\eta_2^\top\right) + \Delta\mu_{y_Y,2} + \sigma\Delta\sigma_{y_Y,2}^\top \\ &= \Delta\mu_{y_Y,2} + (1-\gamma)\Delta\sigma_{y_Y,2}\sigma^\top,\end{aligned}$$

where the terms involving $\Delta\eta_2$ cancel and we use $\eta_0 = \gamma\sigma$.

Using Step 1, this becomes

$$\Delta y_{Y,2} = \frac{y_{Y,2,\hat{\alpha}_p}}{y_{Y,0}}\kappa_p(\hat{\alpha} - \hat{\alpha}_p) + \frac{1}{2}\frac{y_{Y,2,\hat{\alpha}_p\hat{\alpha}_p}}{y_{Y,0}}\sigma_p\sigma_p^\top + (1-\gamma)\frac{y_{Y,2,\hat{\alpha}_p}}{y_{Y,0}}\sigma_p\sigma^\top.$$

Equivalently, since $y_{Y,2} = y_{Y,2}^0 + \Delta y_{Y,2}$, the incremental dividend-yield correction solves

$$\Delta y_{Y,2} = \frac{(y_{Y,2}^0 + \Delta y_{Y,2})_{\hat{\alpha}_p}}{y_{Y,0}}\left[\kappa_p(\hat{\alpha} - \hat{\alpha}_p) + (1-\gamma)\sigma_p\sigma^\top\right] + \frac{1}{2}\frac{(y_{Y,2}^0 + \Delta y_{Y,2})_{\hat{\alpha}_p\hat{\alpha}_p}}{y_{Y,0}}\sigma_p\sigma_p^\top. \quad \text{(C.17)}$$

This is the time-varying-α_p counterpart of (C.7): the new correction is the drift and risk-adjusted diffusion of the total second-order dividend yield induced by the state variable $\overline{\alpha}_p$.

The second-order price-dividend-ratio correction follows from

$$p_{Y,2} = p_{Y,0}^3 y_{Y,1}^2 - p_{Y,0}^2 y_{Y,2},$$

so the incremental term is

$$\Delta p_{Y,2} = -p_{Y,0}^2\Delta y_{Y,2}.$$

Step 7 (time-varying α_p): Solving for coefficients of dividend yield.
We can solve the coefficients of $\Delta y_{Y,2}$ directly from (C.17). Write the fixed-$\hat{\alpha}_p$ correction as

$$y_{Y,2}^0(x, \hat{\alpha}_p) = a_Y^0(x) + b_Y^0(x)\hat{\alpha}_p + d_Y^0(x)\hat{\alpha}_p^2,$$

and conjecture

$$\Delta y_{Y,2}(x,\hat{\alpha}_p) = a_Y(x) + b_Y(x)\hat{\alpha}_p + d_Y(x)\hat{\alpha}_p^2.$$

Let

$$b_Y^T(x) \equiv b_Y^0(x) + b_Y(x), \qquad d_Y^T(x) \equiv d_Y^0(x) + d_Y(x)$$

denote the linear and quadratic coefficients of the total second-order dividend-yield correction $y_{Y,2}^0 + \Delta y_{Y,2}$. Then

$$(y_{Y,2}^0 + \Delta y_{Y,2})_{\hat{\alpha}_p} = b_Y^T + 2d_Y^T\hat{\alpha}_p, \qquad (y_{Y,2}^0 + \Delta y_{Y,2})_{\hat{\alpha}_p\hat{\alpha}_p} = 2d_Y^T.$$

For compactness, define

$$M_p \equiv \kappa_p\hat{\alpha} + (1-\gamma)\sigma_p\sigma^\top, \qquad V_p \equiv \sigma_p\sigma_p^\top.$$

Then (C.17) can be written as

$$\Delta y_{Y,2} = \frac{1}{y_{Y,0}}(b_Y^T + 2d_Y^T\hat{\alpha}_p)\left(M_p - \kappa_p\hat{\alpha}_p\right) + \frac{d_Y^T}{y_{Y,0}}V_p. \tag{C.18}$$

We first match the coefficient on $\hat{\alpha}_p^2$. The only term in (C.18) that generates a quadratic coefficient is the product of $2d_Y^T\hat{\alpha}_p$ with $-\kappa_p\hat{\alpha}_p$. Therefore,

$$d_Y = -\frac{2\kappa_p}{y_{Y,0}}d_Y^T = -\frac{2\kappa_p}{y_{Y,0}}(d_Y^0 + d_Y).$$

Solving for d_Y yields

$$d_Y = -\frac{\frac{2\kappa_p}{y_{Y,0}}}{1+\frac{2\kappa_p}{y_{Y,0}}}\,d_Y^0. \tag{C.19}$$

Equivalently, the quadratic coefficient of the total second-order correction is

$$d_Y^T = \frac{1}{1+\frac{2\kappa_p}{y_{Y,0}}}\,d_Y^0. \tag{C.20}$$

Thus, mean reversion in passive demand attenuates the quadratic dependence of the dividend yield on $\hat{\alpha}_p$. When $\kappa_p = 0$, we recover $d_Y = 0$ and $d_Y^T = d_Y^0$, as in the fixed-$\hat{\alpha}_p$ economy.

We can similarly solve for the coefficient on $\hat{\alpha}_p$. Matching the linear coefficient in (C.18) gives

$$b_Y = \frac{1}{y_{Y,0}}\left(2d_Y^T M_p - \kappa_p b_Y^T\right) = \frac{1}{y_{Y,0}}\left(2d_Y^T M_p - \kappa_p(b_Y^0 + b_Y)\right).$$

Solving for b_Y yields

$$b_Y = \frac{2d_Y^T M_p - \kappa_p b_Y^0}{y_{Y,0} + \kappa_p}. \tag{C.21}$$

Substituting (C.20), this can also be written as

$$b_Y = \frac{\frac{2M_p}{1+\frac{2\kappa_p}{y_{Y,0}}} d_Y^0 - \kappa_p b_Y^0}{y_{Y,0} + \kappa_p}.$$

Equivalently, the linear coefficient of the total second-order correction is

$$b_Y^T = b_Y^0 + b_Y = \frac{y_{Y,0} b_Y^0 + 2d_Y^T M_p}{y_{Y,0} + \kappa_p}. \tag{C.22}$$

Finally, we solve for the constant coefficient. Matching the constant term in (C.18) gives

$$a_Y = \frac{1}{y_{Y,0}} \left(b_Y^T M_p + d_Y^T V_p \right).$$

Using (C.22) and (C.20), this becomes

$$a_Y = \frac{1}{y_{Y,0}} \left[\frac{y_{Y,0} b_Y^0 + 2d_Y^T M_p}{y_{Y,0} + \kappa_p} M_p + d_Y^T V_p \right], \tag{C.23}$$

where

$$d_Y^T = \frac{1}{1 + \frac{2\kappa_p}{y_{Y,0}}} d_Y^0.$$

Equivalently, substituting out d_Y^T,

$$a_Y = \frac{1}{y_{Y,0}} \left[\frac{y_{Y,0} b_Y^0 + \frac{2M_p}{1+\frac{2\kappa_p}{y_{Y,0}}} d_Y^0}{y_{Y,0} + \kappa_p} M_p + \frac{1}{1 + \frac{2\kappa_p}{y_{Y,0}}} d_Y^0 V_p \right].$$

Therefore, the incremental dividend-yield correction is

$$\Delta y_{Y,2}(x, \hat{\alpha}_p) = a_Y(x) + b_Y(x)\hat{\alpha}_p + d_Y(x)\hat{\alpha}_p^2,$$

with d_Y, b_Y, and a_Y given by (C.19), (C.21), and (C.23).

Step 8 (time-varying α_p): Aggregate market elasticity (no heterogeneity or leverage). Consider the case without preference heterogeneity or leverage constraints. Then $\hat{\gamma}_j = 0$ for

$j = 0, 1, 2$ and $\hat{v}_2 = 0$. In the fixed-$\hat{\alpha}_p$ economy, Step 6 gives

$$y^0_{Y,2}(x, \hat{\alpha}_p) = d^0_Y \hat{\alpha}^2_p, \qquad d^0_Y = \frac{1}{2}(1 - \psi^{-1})\gamma\|\sigma\|^2 \frac{x_0}{x_a},$$

so $a^0_Y = b^0_Y = 0$.

With time-varying passive demand, Step 7 implies that the total second-order dividend-yield correction is

$$y^T_{Y,2}(x, \hat{\alpha}_p) \equiv y^0_{Y,2}(x, \hat{\alpha}_p) + \Delta y_{Y,2}(x, \hat{\alpha}_p) = a^T_Y + b^T_Y \hat{\alpha}_p + d^T_Y \hat{\alpha}^2_p,$$

where, using $b^0_Y = 0$,

$$\begin{aligned}
d^T_Y &= \frac{1}{1 + \frac{2\kappa_p}{y_{Y,0}}} d^0_Y, \\
b^T_Y &= \frac{2 d^T_Y M_p}{y_{Y,0} + \kappa_p}, \\
a^T_Y &= a_Y = \frac{1}{y_{Y,0}} \left(b^T_Y M_p + d^T_Y V_p \right),
\end{aligned}$$

with

$$M_p \equiv \kappa_p \hat{\alpha} + (1 - \gamma)\sigma_p \sigma^\top, \qquad V_p \equiv \sigma_p \sigma_p^\top.$$

The constant term a^T_Y does not affect the elasticity. Differentiating with respect to $\hat{\alpha}_p$ gives

$$\frac{\partial y^T_{Y,2}}{\partial \hat{\alpha}_p} = b^T_Y + 2 d^T_Y \hat{\alpha}_p = 2 d^T_Y \left(\hat{\alpha}_p + \frac{M_p}{y_{Y,0} + \kappa_p} \right). \tag{C.24}$$

Using the price-dividend-ratio expansion,

$$\frac{1}{p(X, \epsilon)} \frac{\partial p(X, \epsilon)}{\partial F(X, \epsilon)} = -\frac{1}{y_{Y,0}} \frac{\partial y^T_{Y,2}}{\partial \hat{\alpha}_p} \frac{\epsilon}{x_0} + O(\epsilon^2).$$

Substituting (C.24) and the expression for d^0_Y yields

$$\begin{aligned}
\frac{1}{p(X, \epsilon)} \frac{\partial p(X, \epsilon)}{\partial F(X, \epsilon)} &= -\frac{1}{y_{Y,0}} \frac{2 d^0_Y}{1 + \frac{2\kappa_p}{y_{Y,0}}} \left(\hat{\alpha}_p + \frac{M_p}{y_{Y,0} + \kappa_p} \right) \frac{\epsilon}{x_0} + O(\epsilon^2) \\
&= -\left(1 - \psi^{-1}\right) \frac{\gamma\|\sigma\|^2}{y_{Y,0}} \frac{1}{x_a} \frac{1}{1 + \frac{2\kappa_p}{y_{Y,0}}} \left(\hat{\alpha}_p + \frac{M_p}{y_{Y,0} + \kappa_p} \right) \epsilon + O(\epsilon^2).
\end{aligned}$$

Equivalently, using $\overline{\alpha}_p = 1 + \hat{\alpha}_p \epsilon$, this can be written as

$$\frac{1}{p(X,\epsilon)} \frac{\partial p(X,\epsilon)}{\partial F(X,\epsilon)} = \left(1 - \psi^{-1}\right) \frac{\gamma \|\sigma\|^2}{y_{Y,0}} \frac{1}{x_a} \frac{1}{1 + \frac{2\kappa_p}{y_{Y,0}}} \left[1 - \overline{\alpha}_p - \frac{\epsilon M_p}{y_{Y,0} + \kappa_p}\right] + O(\epsilon^2). \tag{C.25}$$

The first term is the fixed-passive-share elasticity attenuated by mean reversion in passive demand. The additional term involving M_p captures the fact that predictable rebalancing and passive-demand risk affect prices even when the current passive share equals its benchmark value. When $\kappa_p = 0$ and $\sigma_p = 0$, $M_p = 0$ and (C.25) collapses to the fixed-$\hat{\alpha}_p$ result.

Step 9 (time-varying α_p): Aggregate market elasticity (preference heterogeneity).
Consider next the case with preference heterogeneity but no leverage constraints. Then $\hat{\nu}_2 = 0$, and the first-order exposure coefficients are

$$m_{0,1} = \hat{\alpha}_p, \qquad m_{j,1} = \bar{\gamma}_a - \frac{x_0}{x_a}\hat{\alpha}_p - \hat{\gamma}_j, \qquad j = 1, 2,$$

where

$$\bar{\gamma}_a \equiv \sum_{j=1}^{2} \frac{x_j}{x_a} \hat{\gamma}_j, \qquad x_a \equiv 1 - x_0.$$

In the fixed-$\hat{\alpha}_p$ economy, the second-order dividend-yield correction is

$$y^0_{Y,2}(x, \hat{\alpha}_p) = a^0_Y(x) + b^0_Y(x)\hat{\alpha}_p + d^0_Y(x)\hat{\alpha}_p^2,$$

with

$$b^0_Y(x) = (1 - \psi^{-1})\gamma\|\sigma\|^2 x_0(\hat{\gamma}_0 - \bar{\gamma}_a), \qquad d^0_Y(x) = \frac{1}{2}(1 - \psi^{-1})\gamma\|\sigma\|^2 \frac{x_0}{x_a}.$$

The constant term a^0_Y does not affect the elasticity.

With time-varying passive demand, Step 7 implies that the total second-order dividend-yield correction satisfies

$$y^T_{Y,2}(x, \hat{\alpha}_p) \equiv y^0_{Y,2}(x, \hat{\alpha}_p) + \Delta y_{Y,2}(x, \hat{\alpha}_p) = a^T_Y + b^T_Y \hat{\alpha}_p + d^T_Y \hat{\alpha}_p^2,$$

where

$$d^T_Y = \frac{1}{1 + \frac{2\kappa_p}{y_{Y,0}}} d^0_Y, \qquad b^T_Y = \frac{y_{Y,0} b^0_Y + 2 d^T_Y M_p}{y_{Y,0} + \kappa_p},$$

and

$$M_p \equiv \kappa_p \hat{\alpha} + (1 - \gamma)\sigma_p \sigma^\top.$$

Therefore,

$$\begin{aligned}\frac{\partial y_{Y,2}^T}{\partial \hat{\alpha}_p} &= b_Y^T + 2d_Y^T \hat{\alpha}_p \\ &= \frac{y_{Y,0} b_Y^0}{y_{Y,0} + \kappa_p} + 2d_Y^T \left(\hat{\alpha}_p + \frac{M_p}{y_{Y,0} + \kappa_p} \right). \end{aligned} \tag{C.26}$$

Using the price-dividend-ratio expansion,

$$\frac{1}{p(X,\epsilon)} \frac{\partial p(X,\epsilon)}{\partial F(X,\epsilon)} = -\frac{1}{y_{Y,0}} \frac{\partial y_{Y,2}^T}{\partial \hat{\alpha}_p} \frac{\epsilon}{x_0} + O(\epsilon^2).$$

Substituting (C.26) and the expressions for b_Y^0 and d_Y^0 gives

$$\frac{1}{p(X,\epsilon)} \frac{\partial p(X,\epsilon)}{\partial F(X,\epsilon)} = -\left(1 - \psi^{-1}\right) \frac{\gamma \|\sigma\|^2}{y_{Y,0}} \left[\frac{y_{Y,0}}{y_{Y,0} + \kappa_p} (\hat{\gamma}_0 - \bar{\gamma}_a) + \frac{1}{x_a} \frac{1}{1 + \frac{2\kappa_p}{y_{Y,0}}} \left(\hat{\alpha}_p + \frac{M_p}{y_{Y,0} + \kappa_p} \right) \right] \epsilon + O(\epsilon^2).$$

Equivalently, using $\overline{\alpha}_p = 1 + \hat{\alpha}_p \epsilon$ and

$$\frac{\gamma_0 - \mathbb{E}_a[\gamma_j]}{\gamma} = (\hat{\gamma}_0 - \bar{\gamma}_a)\epsilon,$$

we can write

$$\frac{1}{p(X,\epsilon)} \frac{\partial p(X,\epsilon)}{\partial F(X,\epsilon)} = (1 - \psi^{-1}) \frac{\gamma \|\sigma\|^2}{y_{Y,0}} \left[\frac{1}{x_a} \frac{1}{1 + \frac{2\kappa_p}{y_{Y,0}}} \left(1 - \overline{\alpha}_p - \frac{\epsilon M_p}{y_{Y,0} + \kappa_p} \right) - \frac{y_{Y,0}}{y_{Y,0} + \kappa_p} \frac{\gamma_0 - \mathbb{E}_a[\gamma_j]}{\gamma} \right] + O(\epsilon^2). \tag{C.27}$$

Thus, time variation in passive demand attenuates both components of the fixed-passive-share elasticity, but with different attenuation factors. The demand-misalignment term is scaled by $(1 + 2\kappa_p / y_{Y,0})^{-1}$, while the preference-heterogeneity term is scaled by $y_{Y,0}/(y_{Y,0} + \kappa_p)$. The term involving M_p is new and captures predictable rebalancing and passive-demand risk. When $\kappa_p = 0$ and $\sigma_p = 0$, (C.27) collapses to the fixed-$\hat{\alpha}_p$ result with preference heterogeneity.

Step 9 (time-varying α_p): Consumption-wealth ratio.

We write the second-order consumption-wealth ratio as

$$c_{j,2} = c_{j,2}^0 + \Delta c_{j,2},$$

where $c^0_{j,2}$ is the fixed-$\hat{\alpha}_p$ correction from the previous subsection and $\Delta c_{j,2}$ collects only the new terms generated by the dynamics of $\overline{\alpha}_p$.

The consumption-wealth ratio satisfies

$$c_{j,t} = \psi\rho + (1-\psi)\left[r_t + \eta_t\,\sigma_{j,t}^\top - \frac{\gamma_j}{2}\|\sigma_{j,t}\|^2\right] + \mu_{c_j,t} + (1-\gamma_j)\sigma_{c_j,t}\sigma_{j,t}^\top + \frac{\psi-\gamma_j}{1-\psi}\frac{\|\sigma_{c_j,t}\|^2}{2}.$$

The last term is fourth order and does not enter the second-order expansion.

From the fixed-$\hat{\alpha}_p$ derivation, the second-order term can be written as

$$\begin{aligned} c_{j,2} = (1-\psi)&\Big[r_2 + \eta_2\sigma^\top + \eta_1\sigma_{j,1}^\top - \frac{\gamma}{2}\sigma_{j,1}\sigma_{j,1}^\top - \gamma\hat{\gamma}_j\sigma\sigma_{j,1}^\top\Big] \\ &+ \mu_{c_j,2} + (1-\gamma)\sigma_{c_j,2}\sigma^\top. \end{aligned}$$

Here the terms involving $\sigma_{j,2}$ cancel because $\eta_0 = \gamma\sigma$. Therefore, relative to the fixed-$\hat{\alpha}_p$ economy,

$$\Delta c_{j,2} = (1-\psi)\left[\Delta r_2 + \Delta\eta_2\sigma^\top\right] + \Delta\mu_{c_j,2} + (1-\gamma)\Delta\sigma_{c_j,2}\sigma^\top.$$

All first-order terms involving η_1, $\sigma_{j,1}$, and $\hat{\gamma}_j$ are already included in $c^0_{j,2}$ and therefore do not appear in $\Delta c_{j,2}$.

Using Step 1, this becomes

$$\begin{aligned} \Delta c_{j,2} = (1-\psi)&\left[\Delta r_2 + \Delta\eta_2\sigma^\top\right] \\ &+ \frac{c_{j,2,\hat{\alpha}_p}}{c_{j,0}}\kappa_p(\hat{\alpha} - \hat{\alpha}_p) + \frac{1}{2}\frac{c_{j,2,\hat{\alpha}_p\hat{\alpha}_p}}{c_{j,0}}\sigma_p\sigma_p^\top \\ &+ (1-\gamma)\frac{c_{j,2,\hat{\alpha}_p}}{c_{j,0}}\sigma_p\sigma^\top. \end{aligned} \tag{C.28}$$

This equation is the time-varying-α_p counterpart of the fixed-$\hat{\alpha}_p$ expression for $c_{j,2}$. It shows that the new correction to the consumption-wealth ratio is driven by the changes in the interest rate and market price of risk, together with the direct drift and diffusion terms induced by the dynamics of passive demand.

Equations (C.16), (C.10), and (C.28) form the closed system for the incremental second-order corrections when the passive portfolio share is time-varying.

D Derivation of the Market Elasticity

Let $p(X,\epsilon) \equiv 1/y(X,\epsilon)$ denote the price-dividend ratio. The second-order expansion of $p(X,\epsilon)$ is given by

$$p(X,\epsilon) = \frac{1}{y_0(X)} - \frac{y_1(X)}{y_0^2(X)}\epsilon + \left[\frac{y_1^2(X)}{y_0^3(X)} - \frac{y_2(X)}{y_0^2(X)}\right]\epsilon^2 + O(\epsilon^3). \tag{D.1}$$

Let

$$F(X,\epsilon) \equiv \omega_0 \frac{W_0(1+\hat{\alpha}_p\epsilon) - W_0}{P} = \hat{\alpha}_p \epsilon x_0$$

denote the flow into the risky asset relative to the benchmark economy. Note that $F = O(\epsilon)$.

In this section, we consider the case where the portfolio of passive investors is constant.

Proof. Using (D.1), the derivative of the price-dividend ratio with respect to flows can be written as

$$\frac{\partial p(X,\epsilon)}{\partial F(X,\epsilon)} = \frac{1}{\partial F/\partial \hat{\alpha}_p}\frac{\partial p(X,\epsilon)}{\partial \hat{\alpha}_p} = -\frac{1}{y_0^2(X)}\frac{\partial y_1(X)}{\partial \hat{\alpha}_p}\frac{1}{x_0} + O(\epsilon).$$

From Proposition 3, $y_1(X)$ does not depend on $\hat{\alpha}_p$, so

$$\frac{\partial y_1(X)}{\partial \hat{\alpha}_p} = 0.$$

Therefore,

$$\frac{\partial p(X,\epsilon)}{\partial F(X,\epsilon)} = 0 + O(\epsilon),$$

which implies

$$\varepsilon_M^{-1} = \frac{1}{p(X,\epsilon)}\frac{\partial p(X,\epsilon)}{\partial F(X,\epsilon)} = 0 + O(\epsilon).$$

Thus, the aggregate market elasticity is infinite to first order.

To understand this result, note that from the pricing condition

$$y_t = r_t + \pi_t - \mu_{P,t},$$

we can write the first-order term as

$$y_1(X) = r_1(X) + \pi_1(X) - \mu_{P,1}(X).$$

Since $y_1(X)$ is constant, we must have

$$\mu_{y,1}(X) = 0, \qquad \sigma_{y,1}(X) = 0,$$

which implies $\sigma_{R,1}(X) = 0$ and therefore $\mu_{P,1}(X) = 0$.

Hence,

$$\frac{\partial y_1(X)}{\partial \hat{\alpha}_p} = \frac{\partial r_1(X)}{\partial \hat{\alpha}_p} + \frac{\partial \pi_1(X)}{\partial \hat{\alpha}_p}.$$

From Proposition 3,

$$\frac{\partial r_1(X)}{\partial \hat{\alpha}_p} = -\sigma \frac{\partial \eta_1(X)}{\partial \hat{\alpha}_p} = \frac{\gamma \|\sigma\|^2}{x_a} x_0,$$
$$\frac{\partial \pi_1(X)}{\partial \hat{\alpha}_p} = \sigma \frac{\partial \eta_1(X)}{\partial \hat{\alpha}_p} = -\frac{\gamma \|\sigma\|^2}{x_a} x_0.$$

These two effects exactly cancel. Thus, at first order, portfolio flows raise the risk premium but lower the interest rate by the same amount, leaving prices unchanged.

We now turn to the second-order effect. From (D.1), we have

$$\frac{\partial p(X,\epsilon)}{\partial F(X,\epsilon)} = -\frac{1}{y_0^2(X)} \frac{\partial y_2(X)}{\partial \hat{\alpha}_p} \frac{\epsilon}{x_0} + \mathcal{O}(\epsilon^2).$$

Because $F = \mathcal{O}(\epsilon)$, second-order changes in the dividend yield generate a first-order response in prices.

At second order, the cancellation between interest rates and risk premia breaks down due to endogenous reallocation of risk across investors. In particular, from the expression for $y_2(X)$,

$$y_2(X) = (1-\psi^{-1})\gamma\|\sigma\|^2 \sum_{j=0}^{2} x_j \left(\frac{1}{2} m_{j,1}^2 + \hat{\gamma}_j m_{j,1}\right),$$

where $\sigma_{j,1} = m_{j,1}\sigma$.

Thus, $\partial y_2 / \partial \hat{\alpha}_p \neq 0$, implying a non-zero price impact and a finite aggregate elasticity.

□

D.1 Proof of Proposition 4

Proof. Consider the case with no preference heterogeneity and no leverage constraints. That is, $\hat{\gamma}_j = 0$ for $j = 0, 1, 2$ and $\nu_{2,t} = 0$. In this case, the first-order exposure coefficients are

$$m_{0,1} = \hat{\alpha}_p, \qquad m_{j,1} = -\frac{x_0}{x_a} \hat{\alpha}_p, \qquad j = 1, 2,$$

where $x_a \equiv 1 - x_0$ is the wealth share of active investors and $\sigma_{j,1} = m_{j,1}\sigma$. The second-order correction to the dividend yield therefore simplifies to

$$\begin{aligned} y_{Y,2}(X) &= (1-\psi^{-1})\gamma\|\sigma\|^2 \sum_{j=0}^{2} x_j \frac{m_{j,1}^2}{2} \\ &= (1-\psi^{-1})\gamma\|\sigma\|^2 \frac{1}{2}\left[x_0\hat{\alpha}_p^2 + x_a\left(\frac{x_0}{x_a}\hat{\alpha}_p\right)^2\right] \\ &= (1-\psi^{-1})\gamma\|\sigma\|^2 \frac{x_0}{2x_a}\hat{\alpha}_p^2. \end{aligned}$$

Hence,

$$\frac{\partial y_{Y,2}(X)}{\partial\hat{\alpha}_p} = (1-\psi^{-1})\gamma\|\sigma\|^2\frac{x_0}{x_a}\hat{\alpha}_p.$$

Using the change of variables from the previous subsection,

$$\frac{1}{p(X,\epsilon)}\frac{\partial p(X,\epsilon)}{\partial F(X,\epsilon)} = -\frac{1}{y_0(X)}\frac{\partial y_{Y,2}(X)}{\partial\hat{\alpha}_p}\frac{\epsilon}{x_0} + O(\epsilon^2).$$

Substituting the derivative above gives

$$\frac{1}{p(X,\epsilon)}\frac{\partial p(X,\epsilon)}{\partial F(X,\epsilon)} = -(1-\psi^{-1})\frac{\gamma\|\sigma\|^2}{y_0(X)}\frac{\hat{\alpha}_p\epsilon}{x_a} + O(\epsilon^2).$$

Finally, since $\overline{\alpha}_p = 1 + \hat{\alpha}_p\epsilon$, we obtain

$$\frac{1}{p(X,\epsilon)}\frac{\partial p(X,\epsilon)}{\partial F(X,\epsilon)} = (1-\psi^{-1})\frac{\gamma\|\sigma\|^2}{y_0(X)}\frac{1-\overline{\alpha}_p}{x_a} + O(\epsilon^2).$$

This proves the result. □

D.2 Proof of Proposition 5

Proof. Consider the case in which investors have heterogeneous risk aversions but the leverage constraints are slack. Then $\hat{\nu}_{j,1} = 0$ for $j = 1, 2$. Let

$$\bar{\gamma}_a \equiv \sum_{j=1}^{2}\frac{x_j}{x_a}\hat{\gamma}_j, \qquad x_a \equiv x_1 + x_2 = 1 - x_0.$$

From the first-order solution, the scalar exposure coefficients defined by $\sigma_{j,1} = m_{j,1}\sigma$ are

$$m_{0,1} = \hat{\alpha}_p, \qquad m_{j,1} = \bar{\gamma}_a - \frac{x_0}{x_a}\hat{\alpha}_p - \hat{\gamma}_j, \qquad j = 1, 2.$$

The second-order correction to the dividend yield is

$$y_{Y,2} = (1-\psi^{-1})\gamma\|\sigma\|^2 \sum_{j=0}^{2} x_j \left(\frac{1}{2}m_{j,1}^2 + \hat{\gamma}_j m_{j,1}\right).$$

We differentiate this expression with respect to $\hat{\alpha}_p$. Since

$$\frac{\partial m_{0,1}}{\partial \hat{\alpha}_p} = 1, \qquad \frac{\partial m_{j,1}}{\partial \hat{\alpha}_p} = -\frac{x_0}{x_a}, \qquad j = 1, 2,$$

we obtain

$$\begin{aligned}\frac{\partial y_{Y,2}}{\partial \hat{\alpha}_p} &= (1-\psi^{-1})\gamma\|\sigma\|^2 \sum_{j=0}^{2} x_j (m_{j,1} + \hat{\gamma}_j) \frac{\partial m_{j,1}}{\partial \hat{\alpha}_p} \\ &= (1-\psi^{-1})\gamma\|\sigma\|^2 x_0 \left[\hat{\alpha}_p + \hat{\gamma}_0 - \sum_{j=1}^{2} \frac{x_j}{x_a}(m_{j,1} + \hat{\gamma}_j)\right].\end{aligned}$$

For active investors,

$$m_{j,1} + \hat{\gamma}_j = \bar{\gamma}_a - \frac{x_0}{x_a}\hat{\alpha}_p.$$

Therefore,

$$\frac{\partial y_{Y,2}}{\partial \hat{\alpha}_p} = (1-\psi^{-1})\gamma\|\sigma\|^2 x_0 \left[\frac{\hat{\alpha}_p}{x_a} + \hat{\gamma}_0 - \bar{\gamma}_a\right].$$

Using the change of variables from the market-elasticity derivation,

$$\frac{1}{p(X,\epsilon)}\frac{\partial p(X,\epsilon)}{\partial F(X,\epsilon)} = -\frac{1}{y_0(X)}\frac{\partial y_{Y,2}(X)}{\partial \hat{\alpha}_p}\frac{\epsilon}{x_0} + O(\epsilon^2).$$

Substituting the derivative above gives

$$\frac{1}{p(X,\epsilon)}\frac{\partial p(X,\epsilon)}{\partial F(X,\epsilon)} = -(1-\psi^{-1})\frac{\gamma\|\sigma\|^2}{y_0(X)}\left[\frac{\hat{\alpha}_p\epsilon}{x_a} + (\hat{\gamma}_0 - \bar{\gamma}_a)\epsilon\right] + O(\epsilon^2).$$

Finally, using $\overline{\alpha}_p = 1 + \hat{\alpha}_p\epsilon$ and $\gamma_j = \gamma(1+\hat{\gamma}_j\epsilon)$, we can write the expression in primitives as

$$\frac{1}{p(X,\epsilon)}\frac{\partial p(X,\epsilon)}{\partial F(X,\epsilon)} = (1-\psi^{-1})\frac{\gamma\|\sigma\|^2}{y_0(X)}\left[\frac{1-\overline{\alpha}_p}{x_a} - \frac{\gamma_0 - \mathbb{E}_a[\gamma_j]}{\gamma}\right] + O(\epsilon^2),$$

where

$$\mathbb{E}_a[\gamma_j] \equiv \sum_{j=1}^{2}\frac{x_j}{x_a}\gamma_j.$$

Equivalently, this can be written in the form of Proposition 5 by setting $x_c = 0$, $x_u = x_a$, and

$$\overline{\alpha}_p^{\text{opt}} = 1 - x_a \frac{\gamma_0 - \mathbb{E}_a[\gamma_j]}{\gamma} = 1 + \left(\sum_{j=0}^{2} x_k \hat{\gamma}_k - \hat{\gamma}_0 \right) \epsilon,$$

where the expression above coincides with first-order correction for the passive investor's optimal portfolio share, that is, the portfolio they would choose if they could freely adjust their portfolio share.

Then

$$\frac{1}{p(X,\epsilon)} \frac{\partial p(X,\epsilon)}{\partial F(X,\epsilon)} = (1-\psi^{-1}) \frac{\gamma \|\sigma\|^2}{y_0(X)} \frac{\overline{\alpha}_p^{\text{opt}} - \overline{\alpha}_p}{x_a} + O(\epsilon^2).$$

This proves the version of the result without leverage constraints.

We next derive the corresponding expression when investor 2 is constrained and investor 1 is unconstrained. In this case, the first-order exposure coefficients are

$$m_{0,1} = \hat{\alpha}_p, \qquad m_{2,1} = \hat{\sigma}, \qquad m_{1,1} = -\frac{x_0 \hat{\alpha}_p + x_2 \hat{\sigma}}{x_1},$$

where the expression for $m_{1,1}$ follows from market clearing,

$$x_0 m_{0,1} + x_1 m_{1,1} + x_2 m_{2,1} = 0.$$

Using again

$$y_{Y,2} = (1-\psi^{-1}) \gamma \|\sigma\|^2 \sum_{j=0}^{2} x_j \left(\frac{1}{2} m_{j,1}^2 + \hat{\gamma}_j m_{j,1} \right),$$

we differentiate with respect to $\hat{\alpha}_p$. Since

$$\frac{\partial m_{0,1}}{\partial \hat{\alpha}_p} = 1, \qquad \frac{\partial m_{1,1}}{\partial \hat{\alpha}_p} = -\frac{x_0}{x_1}, \qquad \frac{\partial m_{2,1}}{\partial \hat{\alpha}_p} = 0,$$

we obtain

$$\begin{aligned} \frac{\partial y_{Y,2}}{\partial \hat{\alpha}_p} &= (1-\psi^{-1}) \gamma \|\sigma\|^2 \left[x_0 (m_{0,1} + \hat{\gamma}_0) - x_0 (m_{1,1} + \hat{\gamma}_1) \right] \\ &= (1-\psi^{-1}) \gamma \|\sigma\|^2 x_0 \left[\frac{x_0 + x_1}{x_1} \hat{\alpha}_p + \frac{x_2}{x_1} \hat{\sigma} + \hat{\gamma}_0 - \hat{\gamma}_1 \right]. \end{aligned}$$

Using the change of variables from the market-elasticity derivation,

$$\frac{1}{p(X,\epsilon)}\frac{\partial p(X,\epsilon)}{\partial F(X,\epsilon)} = -\frac{1}{y_0(X)}\frac{\partial y_{Y,2}(X)}{\partial \hat{\alpha}_p}\frac{\epsilon}{x_0} + O(\epsilon^2),$$

we get

$$\frac{1}{p(X,\epsilon)}\frac{\partial p(X,\epsilon)}{\partial F(X,\epsilon)} = -(1-\psi^{-1})\frac{\gamma\|\sigma\|^2}{y_0(X)}\left[\frac{x_0+x_1}{x_1}\hat{\alpha}_p\epsilon + \frac{x_2}{x_1}\hat{\sigma}\epsilon + (\hat{\gamma}_0 - \hat{\gamma}_1)\epsilon\right] + O(\epsilon^2).$$

Finally, using $\overline{\alpha}_p = 1 + \hat{\alpha}_p\epsilon$, $\overline{\sigma} = \|\sigma\|(1+\hat{\sigma}\epsilon)$, and $\gamma_j = \gamma(1+\hat{\gamma}_j\epsilon)$, this becomes

$$\frac{1}{p(X,\epsilon)}\frac{\partial p(X,\epsilon)}{\partial F(X,\epsilon)} = (1-\psi^{-1})\frac{\gamma\|\sigma\|^2}{y_0(X)}\left[\frac{x_0+x_1}{x_1}(1-\overline{\alpha}_p) - \frac{x_2}{x_1}\left(\frac{\overline{\sigma}}{\|\sigma\|} - 1\right) - \frac{\gamma_0-\gamma_1}{\gamma}\right] + O(\epsilon^2).$$

Define $x_c \equiv x_2$, $x_u \equiv x_1 = x_a - x_c$, and $\mathbb{E}^u[\gamma_j] \equiv \gamma_1$. Also define

$$\overline{\alpha}_p^{\text{opt}} = 1 - \frac{x_u}{1-x_c}\frac{\gamma_0 - \mathbb{E}^u[\gamma_j]}{\gamma} - \frac{x_c}{1-x_c}\left(\frac{\overline{\sigma}}{\|\sigma\|} - 1\right) = 1 + \left(\sum_{j=0}^{1}\frac{x_k}{1-x_c}\hat{\gamma}_k - \hat{\gamma}_0 - \frac{x_c}{1-x_c}\hat{\sigma}\right)\epsilon,$$

where the expression above coincides with first-order correction for the passive investor's optimal portfolio share when the leverage constraint of investor 2 binds.

Since $1 - x_c = x_0 + x_1$, the previous display is equivalent to

$$\frac{1}{p(X,\epsilon)}\frac{\partial p(X,\epsilon)}{\partial F(X,\epsilon)} = (1-\psi^{-1})\frac{\gamma\|\sigma\|^2}{y_0(X)}\frac{\overline{\alpha}_p^{\text{opt}} - \overline{\alpha}_p}{x_a - x_c}(1-x_c) + O(\epsilon^2).$$

This proves the version of the result in the regime in which the leverage constraint of investor 2 binds.

□

E Useful Formula

The following are useful for computing the derivatives above:

$$\gamma\sigma\frac{\partial \sigma_{y,2}(X)}{\partial \hat{\alpha}_p} = \left(1-\psi^{-1}\right)\frac{\gamma\sigma^2}{2y_0(X)}\sum_{k=1}^{J}(\hat{\gamma}_k - \hat{\gamma}_0)\,x_k\left(-\frac{\gamma\sigma^2 x_0}{1-x_0}\right).$$

Note that we can write $\sigma_{j,2}(X)$ as follows

$$\begin{aligned}
\frac{\sigma_{j,2}(X)}{\sigma} &= \frac{\theta_2(X)}{\gamma\sigma} - \frac{\theta_1(X)}{\gamma\sigma}\hat{\gamma}_j + \frac{\theta_0(X)}{\gamma\sigma}\hat{\gamma}_j^2 - (1-\gamma^{-1})\frac{\sigma_{y,2}(X)}{\sigma} \\
&= -\left(1+\frac{x_c}{x_u}\right)\frac{\sigma_{y,2}(X)}{\sigma} - \mathbb{E}^u[\hat{\gamma}_k]\frac{\hat{\alpha}_p x_0 + \frac{\hat{\sigma}}{\sigma}x_c}{1-x_0-x_c} - \mathrm{Var}^u[\hat{\gamma}_k] + \\
&\quad - \hat{\gamma}_j\left[\sum_{k\in\mathcal{J}^u}\frac{x_k}{x_u}\hat{\gamma}_j - \frac{\hat{\alpha}_p x_0}{1-x_0}\right] + \hat{\gamma}_j^2.
\end{aligned}$$

$$\begin{aligned}
\sigma_{j,1}(X) &= \sigma \min\left\{\sum_{k\in\mathcal{J}^u}\frac{x_k}{x_u}\hat{\gamma}_k - \left(\frac{\hat{\alpha}_p x_0 + \frac{\hat{\sigma}}{\|\sigma\|}x_c}{1-x_0-x_c}\right) - \hat{\gamma}_j, \frac{\hat{\sigma}}{\sigma}\right\} \\
\sigma_{j,2}(X) &= \frac{\theta_2(X)}{\gamma} - \frac{\theta_1(X)}{\gamma}\hat{\gamma}_j + \frac{\theta_0(X)}{\gamma}\hat{\gamma}_j^2 - (1-\gamma^{-1})\sigma_{y,2}(X) \\
\sigma_{y,2}(X) &= \left(1-\psi^{-1}\right)\frac{\gamma\sigma^2}{2y_0(X)}\sum_{k=1}^{J}\left(\hat{\gamma}_k - \hat{\gamma}_0\right)x_k\sigma_{k,1}(X) \\
\theta_1(X) &= \gamma\sigma\left[\sum_{j\in\mathcal{J}^u}\frac{x_j}{x_u}\hat{\gamma}_j - \frac{\hat{\alpha}_p x_0}{1-x_0}\right] \\
\theta_2(X) &= -\frac{(\gamma-1)x_c + 1 - x_0}{1-x_0-x_c}\sigma_{y,2}(X) - \gamma\sigma\mathbb{E}^u[\hat{\gamma}_j]\frac{\hat{\alpha}_p x_0 + \frac{\hat{\sigma}}{\sigma}x_c}{1-x_0-x_c} - \gamma\sigma\mathrm{Var}^u[\hat{\gamma}_j],
\end{aligned}$$

where

$$\begin{aligned}
\frac{\partial\sigma_{y,2}}{\partial\hat{\alpha}_p} &= \left(1-\psi^{-1}\right)\frac{\gamma\sigma^2}{2y_0(X)}\sum_{k=1}^{J}\left(\hat{\gamma}_k-\hat{\gamma}_0\right)x_k\left(-\frac{\sigma x_0}{1-x_0}\right) \\
&= \left(1-\psi^{-1}\right)\frac{\gamma\sigma^2}{2y_0(X)}\sigma\left(\hat{\gamma}_0 - \mathbb{E}^u[\hat{\gamma}_j]\right)x_0 \\
\sum_{j=0}^{J}x_j\gamma\sigma\hat{\gamma}_j\frac{\partial\sigma_{j,1}(X)}{\partial\hat{\alpha}_p} &= \gamma\sigma^2\sum_{j=0}^{J}x_j\hat{\gamma}_j\left(-\frac{x_0}{1-x_0}\right)
\end{aligned}$$

F Derivation of the perturbed solution

In this section, we compute the first-order and second-order correction of the equilibrium objects. It turns out that the system of equations determining the perturbed solution is block-recursive, so we are able to solve for the equilibrium objects one by one, provided we proceed in the appropriate order.

In contrast to the case considered in the text, we allow for portfolio-flow shocks. In particular, we assume that the portfolio share of the passive investor is given by $\alpha_{0,t} = 1 + \epsilon(\overline{\alpha}_{p,t} - 1)$, where $\overline{\alpha}_{p,t}$ follows the process

$$d\overline{\alpha}_{p,t} = \theta_p(\overline{\alpha} - \overline{\alpha}_{p,t})\epsilon dt + \sigma_p\sqrt{\overline{\alpha}_{p,t}}\epsilon dZ_t.$$

Notice that $\alpha_{0,t} = 1$ and $\overline{\alpha}_{p,t}$ is constant when $\epsilon = 0$. Finally, we assume that the mortality parameter is given by $\kappa = \hat{\kappa}\epsilon$.

F.1 First-order correction

Diffusion and drift terms. The diffusion term for the price-dividend ratio is given by

$$\sigma_{p,t} = \frac{p_x}{p}\sigma_x + \frac{p_{\overline{\alpha}_p}}{p}\sigma_p\sqrt{\overline{\alpha}_{p,t}}\epsilon = O(\epsilon^2).$$

Notice that $p_{x_j} = O(\epsilon)$ and $\sigma_{x_j} = O(\epsilon)$, as p and x_j are constant when $\epsilon = 0$, so the zeroth-order terms for p_{x_j}, $p_{\overline{\alpha}_p}$, and σ_{x_j} are equal to zero. This implies that the first-order correction for $\sigma_{p,t}$ is equal to zero. A similar argument shows that $\sigma_{c_j,t} = O(\epsilon^2)$.

The drift of p is given by

$$\mu_{p,t} = \frac{p_x}{p}\mu_x + \frac{p_{\overline{\alpha}_p}}{p}\theta_p(\overline{\alpha} - \overline{\alpha}_{p,t})\epsilon + \frac{1}{2}\sum_{k=1}^{d}\left[\sigma_{x,k}^{\top}\frac{p_{xx}}{p}\sigma_{x,k} + 2\sigma_{p,k}\sqrt{\overline{\alpha}_{p,t}}\epsilon\frac{p_{x\alpha_p}}{p}\sigma_{x,k} + \frac{p_{\overline{\alpha}_p\overline{\alpha}_p}}{p}\sigma_{p,k}^2\overline{\alpha}_{p,t}\epsilon^2\right],$$

where $\sigma_{x,k}$ is the k-th column of the $J \times d$ matrix σ_x. As p_x and $p_{\overline{\alpha}_p}$ are first-order in ϵ, and the same goes for μ_x and σ_x, then $\mu_{p,t} = O(\epsilon^2)$. A similar argument shows that $\mu_{c_j,t} = O(\epsilon^2)$. Notice these facts imply that $\varsigma_j = O(\epsilon^2)$ and $\xi_{j,t} = O(\epsilon^2)$.

Risk premium. The risk premium is given by

$$\pi_1(X) = \gamma||\sigma||^2\left[\sum_{j\in\mathcal{J}^u}\frac{x_j}{x_u}\hat{\gamma}_j - \frac{x_0\hat{\alpha}_p + x_c\frac{\hat{\sigma}}{||\sigma||}}{1 - x_0 - x_c}\right],$$

using the fact that $||\sigma_{R,t}|| = ||\sigma|| + O(\epsilon^2)$, and $\hat{\alpha}_{p,t} \equiv \overline{\alpha}_{p,t} - 1$.

Portfolio share. The portfolio share of an unconstrained investor is given by

$$\alpha_j(X, \epsilon) = 1 + \left[\frac{\pi_1(X)}{\gamma||\sigma||^2} - \hat{\gamma}_j\right]\epsilon + O(\epsilon^2),$$

the portfolio share of a constrained investor is given by

$$\alpha_j(X,\epsilon) = 1 + \frac{\hat{\sigma}}{||\sigma||}\epsilon + O(\epsilon^2),$$

and the portfolio share of the passive investor is given by $\alpha_0(X,\epsilon) = 1 + \hat{\alpha}_{p,t}\epsilon$. Notice that $\sum_{j=0}^{J} x_j \alpha_{j,1}(X) = 0$, consistent with market clearing.

Interest rate. The first-order correction for the interest rate is given by

$$r_1(X) = \left(1-\psi^{-1}\right)\gamma||\sigma||^2 \sum_{j=0}^{J} x_j \left[\frac{\hat{\gamma}_j}{2} + \alpha_{j,1}(X)\right] - \pi_1(X),$$

using the fact that $\xi_t = O(\epsilon^2)$, $\mu_{p,t} = O(\epsilon^2)$, and $\sigma_{p,t} = O(\epsilon^2)$. Given the market clearing for the risky asset, we can write:

$$r_1(X) = \left(1-\psi^{-1}\right)\gamma||\sigma||^2 \sum_{j=0}^{J} x_j \frac{\hat{\gamma}_j}{2} - \pi_1(X),$$

so $r_1(X) + \pi_1(X)$ is independent of $\overline{\alpha}_p$.

Price-dividend ratio. From the pricing condition, we obtain

$$-\frac{1}{p_0(X)^2} p_1(X) = r_1(X) + \pi_1(X).$$

Rearranging the expression above, and using the expression for the interest rate, we obtain

$$p_1(X) = -p_0(X)^2\left(1-\psi^{-1}\right)\gamma||\sigma||^2 \sum_{j=0}^{J} x_j \frac{\hat{\gamma}_j}{2},$$

which is independent of $\overline{\alpha}_{p,t}$.

Consumption-wealth ratio. The consumption-wealth ratio is given by

$$c_{j,1}(X) = (1-\psi)\left[r_1(X) + \pi_1(X) + \pi_0(X)\alpha_{j,1}(X) - \frac{1}{2}\gamma|||\sigma|||^2(\hat{\gamma}_j + 2\alpha_{j,1}(X))\right].$$

Using the expression for $r_1(X)$, we can write the expression above as follows:

$$c_{j,1}(X) = (1-\psi)\left[(1-\psi^{-1})\sum_{i=0}^{J} x_i \frac{\hat{\gamma}_i}{2} - \frac{\hat{\gamma}_j}{2}\right]\gamma|||\sigma|||^2.$$

Wealth dynamics. The diffusion term of x_j is given by

$$\sigma_{x_j}(X) = x_j \alpha_{j,1}(X)\epsilon\sigma + O(\epsilon^2).$$

The drift of x_j is given by

$$\mu_{x_j}(X) = x_j\left[r_1(X) + \pi_1(X) + \pi_0(X)\alpha_{j,1}(X) - c_{j,1}(X) - \alpha_{j,1}(X)||\sigma||^2 + \hat{\kappa}\frac{\omega_j - x_j}{x_j}\right]\epsilon + O(\epsilon^2).$$

We can write the first-order correction of μ_{x_j} as follows:

$$\mu_{x_j,1}(X) = x_j\left[(\psi-1)\frac{\gamma|||\sigma|||^2}{2}\left(\sum_{i=0}^{J} x_i\hat{\gamma}_i - \hat{\gamma}_j\right) + (\gamma-1)|||\sigma|||^2\alpha_{j,1}(X)\right] + \hat{\kappa}(\omega_j - x_j).$$

F.2 Second-order correction

Diffusion and drift terms. The diffusion term for the price-dividend ratio is given by

$$\sigma_{p,2}(X) = \frac{p_{x,1}(X)}{p_0(X)}\sigma_{x,1}(X) + \frac{p_{\overline{\alpha}_p,1}(X)}{p_0(X)}\sigma_p\sqrt{\overline{\alpha}_p}.$$

We can write the expression above as follows:

$$\sigma_{p,2}(X) = -p_0(X)(1-\psi^{-1})\frac{\gamma||\sigma|||^2}{2}\sum_{j=1}^{J}\left(\hat{\gamma}_j - \hat{\gamma}_0\right)x_j\alpha_{j,1}(X)\sigma,$$

where we used the fact that $p_{\overline{\alpha}_p,1}(X) = 0$.

Similarly, the diffusion for c_j is given by

$$\sigma_{c_j,2}(X) = -(\psi-1)\frac{\gamma|||\sigma|||^2}{2c_{j,0}(X)}(1-\psi^{-1})\sum_{i=1}^{J}(\hat{\gamma}_i - \hat{\gamma}_0)\,x_i\alpha_{i,1}(X)\sigma.$$

The second-order correction for the hedging demand is then given by $\varsigma_{j,2}(X) = \frac{1-\gamma^{-1}}{\psi-1}\frac{\sigma_{c_j,2}\sigma^\top}{|||\sigma|||^2}$.

The second-order correction of the drift of p and c_j are given by

$$\mu_{p,2}(X) = \frac{p_{x,1}(X)}{p_0(X)}\mu_{x,1}(X), \qquad \mu_{c_j,2}(X) = \frac{c_{j,x,1}(X)}{c_{j,0}(X)}\mu_{x,1}(X),$$

which can be written as

$$\mu_{p,2}(X) = -p_0(X)(1-\psi^{-1})\frac{\gamma\|\sigma\|\|^2}{2}\sum_{j=1}^{J}(\hat{\gamma}_j - \hat{\gamma}_0)\mu_{x_j,1}(X)$$
$$\mu_{c_i,2}(X) = (1-\psi)(1-\psi^{-1})\frac{\gamma\|\sigma\|\|^2}{2c_{i,0}(X)}\sum_{j=1}^{J}(\hat{\gamma}_j - \hat{\gamma}_0)\mu_{x_j,1}(X).$$

The second-order correction for $\xi_{j,t}$ is then given by $\xi_{j,2}(X) = \mu_{c_j,2}(X) + (1-\gamma)\sigma_{c_j,2}\sigma^\top$.

Risk premium. The risk premium is given by

$$\pi_2(X) = \gamma\|\sigma\|\|^2\left[\frac{\gamma_{u,2}(X)}{\gamma} - \frac{\gamma_{u,1}(X)}{\gamma}\frac{x_0\hat{\alpha}_p + x_c\frac{\hat{\sigma}}{\|\sigma\|\|}}{1-x_0-x_c} + 2\sum_{k=1}^{d}\frac{\sigma_k\sigma_{p,2,k}}{||\sigma||^2} - \frac{x_c}{1-x_0-x_c}\overline{\alpha}_{c,2}(X) - \varsigma_2(X)\right].$$

Notice that we can write the aggregate risk aversion as follows:

$$\gamma_u(X) = \gamma\left[1 + \mathbb{E}^u[\hat{\gamma}_j]\epsilon - \delta^u[\hat{\gamma}_j]\epsilon^2\right] + O(\epsilon^3),$$

where $\mathbb{E}^u[\hat{\gamma}_j] \equiv \sum_{j\in\mathcal{J}^u}\frac{x_j}{x_u}\hat{\gamma}_j$ and $\delta^u[\hat{\gamma}_j] \equiv \sum_{j\in\mathcal{J}^u}\frac{x_j}{x_u}\hat{\gamma}_j^2 - \left(\sum_{j\in\mathcal{J}^u}\frac{x_j}{x_u}\hat{\gamma}_j\right)^2$, so $\gamma_{u,2}(X)/\gamma = -\delta^u[\hat{\gamma}_j]$.

Combining the previous two expressions, we obtain

$$\begin{aligned}\frac{\pi_2(X)}{\gamma\|\sigma\|\|^2} &= -\delta^u[\hat{\gamma}_j] - \mathbb{E}^u[\hat{\gamma}_j]\frac{x_0\hat{\alpha}_p + x_c\frac{\hat{\sigma}}{\|\sigma\|\|}}{1-x_0-x_c} + 2\sum_{k=1}^{d}\frac{\sigma_k\sigma_{p,2,k}}{||\sigma||^2} + \frac{x_c}{1-x_0-x_c}\sum_{k=1}^{d}\frac{\sigma_k\sigma_{p,2,k}}{\|\sigma\|\|^2} - \varsigma_2(X)\\ &= -\delta^u[\hat{\gamma}_j] - \mathbb{E}^u[\hat{\gamma}_j]\frac{x_0\hat{\alpha}_p + x_c\frac{\hat{\sigma}}{\|\sigma\|\|}}{1-x_0-x_c} + \left(1+\gamma^{-1} + \frac{x_c}{1-x_0-x_c}\right)\sum_{k=1}^{d}\frac{\sigma_k\sigma_{p,2,k}}{\|\sigma\|\|^2},\end{aligned}$$

where we used the fact that $\varsigma_2(X) = \frac{1-\gamma^{-1}}{\psi-1}\frac{\sigma_{c_j,2}\sigma^\top}{\|\sigma\|\|^2}$, $\sigma_{p,2} = \frac{\sigma_{c_j,2}}{\psi-1}$, and $\overline{\alpha}_{c,2}(X) = -\sum_{k=1}^{d}\frac{\sigma_k\sigma_{p,2,k}}{\|\sigma\|\|^2}$.

Portfolio share. The portfolio share of an unconstrained investor is given by

$$\alpha_{j,2}(X) = \frac{\pi_2(X)}{\gamma\|\|\sigma\|\|^2} - \frac{\pi_1(X)}{\gamma\|\|\sigma\|\|^2}\hat{\gamma}_j + \hat{\gamma}_j^2 - 2\sum_{k=1}^{d}\frac{\sigma_k\sigma_{p,2,k}(X)}{\|\sigma\|^2} + \varsigma_{j,2}(X),$$

the portfolio share of a constrained investor is $\alpha_{j,2}(X) = -\sum_{k=1}^{d} \frac{\sigma_k \sigma_{p,2,k}}{\|\sigma\||^2}$, and the portfolio share of the passive investor satisfies $\alpha_{0,2} = 0$.

We can write the expression above as follows:

$$\alpha_{j,2}(X) = \frac{\pi_2(X)}{\gamma\||\sigma\||^2} - \frac{\pi_1(X)}{\gamma\||\sigma\||^2}\hat{\gamma}_j + \hat{\gamma}_j^2 - (1+\gamma^{-1})\sum_{k=1}^{d} \frac{\sigma_k \sigma_{p,2,k}(X)}{\|\sigma\|^2}.$$

Notice that $\sum_{j=0}^{J} x_j \alpha_{j,2}(X) = 0$, consistent with market clearing.

Interest rate. The interest rate is given by

$$r_2(X) = \psi^{-1}(\mu_{p,2}(X) + \sigma\sigma_{p,2}(X)^\top) + (1-\psi^{-1}).\frac{\gamma\|\sigma\||^2}{2}\sum_{j=0}^{J} x_j \left[2\hat{\gamma}_j \alpha_{j,1}(X) + \alpha_{j,1}^2(X) + 2\alpha_{j,2}(X)\right]$$
$$+ (1-\psi^{-1})\gamma\|\sigma\|^2 \sum_{k=1}^{d} \frac{\sigma_k \sigma_{p,2,k}(X)}{\|\sigma\|^2} - \pi_2(X) + \psi^{-1}\xi_2(X),$$

where $\xi_2(X) = (\psi - 1)\left[\mu_{p,2}(X) + (1-\gamma)\sigma_{p,2}\sigma^\top\right]$

We can write the expression for the portfolio of the unconstrained investor as follows:

$$r_2(X) = \mu_{p,2}(X) + \sigma\sigma_{p,2}(X)^\top + (1-\psi^{-1})\gamma\|\sigma\||^2 \sum_{j=0}^{J} x_j \left[\hat{\gamma}_j \alpha_{j,1}(X) + \frac{\alpha_{j,1}^2(X)}{2}\right] - \pi_2(X),$$

Price-dividend ratio. The price-dividend ratio is given by

$$\frac{p_1^2(X)}{p_0^3(X)} - \frac{p_2(X)}{p_0^2(X)} = r_2(X) + \pi_2(X) - \mu_{p,2}(X) - \sigma\sigma_{p,2}^\top.$$

Rearranging the expression above, and using the expression for $r_2(X)$, we obtain

$$\frac{p_2(X)}{p_0(X)} = -p_0(X)(1-\psi^{-1})\gamma\|\sigma\||^2 \sum_{j=0}^{J} x_j \left[\hat{\gamma}_j \alpha_{j,1}(X) + \frac{\alpha_{j,1}^2(X)}{2}\right] + \left(\frac{p_1(X)}{p_0(X)}\right)^2.$$

Consumption-wealth ratio. The second-order correction for the consumption-wealth ratio is given by

$$c_{j,2}(X) = (1-\psi)\left[r_2(X) + \pi_2(X) + \pi_1(X)\alpha_{j,1}(X) + \pi_0(X)\alpha_{j,2}(X)\right] + \xi_{j,2}(X)$$
$$- (1-\psi)\gamma\|\sigma\||^2 \left[\alpha_{j,1}(X)\hat{\gamma}_j + \alpha_{j,2}(X) + \frac{\alpha_{j,1}^2(X)}{2} + \sum_{k=1}^{d} \frac{\sigma_k \sigma_{p,k,2}(X)}{\|\sigma\||^2}\right].$$

Wealth dynamics. The diffusion term of x_j is given by

$$\sigma_{x_j,2}(X) = x_j \alpha_{j,2} \sigma.$$

The drift of x_j is given by

$$\mu_{x_j,2} = x_j \left[r_2(X) + \pi_2(X) + \pi_1(X)\alpha_{j,1}(X) + \pi_0(X)\alpha_{j,2}(X) - c_{j,2}(X) - \mu_{p,2}(X) - \sigma\sigma_{p,2}(X)^\top - \alpha_{j,2}(X)\|\sigma\|^2 \right].$$

Aggregate market elasticity. The derivative of p with respect to $\overline{\alpha}_p$ is given by

$$\frac{1}{p(X,\epsilon)} \frac{\partial p(X,\epsilon)}{\partial \overline{\alpha}_p} = \frac{1}{p_0(X)} \frac{\partial p_2(X)}{\partial \overline{\alpha}_p} \epsilon^2 + O(\epsilon^3).$$

The market elasticity satisfies the condition

$$\frac{1}{p_0(X)} \frac{\partial p_2(X)}{\partial \overline{\alpha}_p} = -p_0(X)(1-\psi^{-1})\gamma\|\sigma\||^2 \left[x_0(\hat{\gamma}_0 + \hat{\alpha}_p) + \sum_{j \in \mathcal{J}^u} x_j \left(\hat{\gamma}_j + \alpha_{j,1}(X) \right) \left(-\frac{x_0}{x_u} \right) \right] \epsilon^2.$$

From the market clearing for the risky asset, we have $x_0 \hat{\alpha}_p + \sum_{j \in \mathcal{J}^u} x_j \alpha_{j,1}(X) + x_c \frac{\hat{\sigma}}{\|\sigma\|} = 0$, so we can write the expression above as follows:

$$\frac{1}{p_0(X)} \frac{\partial p_2(X)}{\partial \overline{\alpha}_p} = p_0(X)(1-\psi^{-1})\gamma\|\sigma\||^2 \left[\hat{\gamma}_u(X) - \hat{\gamma}_0 + \frac{1-x_c}{1-x_0-x_c}(1-\overline{\alpha}_p) - \frac{x_c \frac{\hat{\sigma}}{\|\sigma\||}}{1-x_0-x_c} \right] x_0 \epsilon^2.$$

F.3 Third-order approximation

F.3.1 Passive demand

Suppose there is no preference heterogeneity and no leverage constraint. Without loss of generality, set $J = 1$. In this case, the price-dividend ratio is given by

$$
\begin{aligned}
p(X,\epsilon) &= p^* - (p^*)^2(1-\psi^{-1})\frac{\gamma\|\sigma\|^2}{2}\frac{x_0\hat{\alpha}_p^2}{x_1}\epsilon^2 + O(\epsilon^3) \\
\pi(X,\epsilon) &= \pi_0(X) - \gamma\|\sigma\|^2\frac{x_0\hat{\alpha}_p}{x_1}\epsilon + O(\epsilon^3) \\
r(X,\epsilon) &= r_0(X) + \gamma\|\sigma\|^2\frac{x_0\hat{\alpha}_p}{x_1}\epsilon + (1-\psi^{-1})\frac{\gamma\|\sigma\|^2}{2}\frac{x_0\hat{\alpha}_p^2}{x_1}\epsilon^2 + O(\epsilon^3) \\
\alpha_0(X,\epsilon) &= 1 + \hat{\alpha}_p\epsilon + O(\epsilon^3) \\
\alpha_1(X,\epsilon) &= 1 - \frac{x_0}{x_1}\hat{\alpha}_p\epsilon + O(\epsilon^3) \\
c_j(X,\epsilon) &= c_{j,0}(X) + (1-\psi^{-1})\frac{\gamma\|\sigma\|^2}{2}\frac{x_0\hat{\alpha}_p^2}{x_1}\epsilon^2 + O(\epsilon^3) \\
\sigma_{x_1}(X,\epsilon) &= -\frac{x_0}{x_1}\hat{\alpha}_p\sigma\epsilon + O(\epsilon^3) \\
\mu_{x_1}(X,\epsilon) &= \left[(1-\gamma)\|\sigma\|^2(1-x_1)\hat{\alpha}_p + \hat{\kappa}(\omega_j - x_1)\right]\epsilon + O(\epsilon^3),
\end{aligned}
$$

where $\mu_p, \sigma_p, \mu_{c_j}$, and σ_{c_j} are all equal to zero up to second order, and $x_0 = 1 - x_1$.

The law of motion of $x_{1,t}$ can be written as

$$
\frac{dx_{1,t}}{x_{1,t}} = \left[x_{0,t}(c_{0,t} - c_{1,t}) + x_{0,t}(\alpha_{0,t} - \alpha_{1,t})\left(\|\sigma_{R,t}\|^2 - \pi_t\right) + \kappa\frac{\omega_1 - x_{1,t}}{x_{1,t}}\right]dt + (\alpha_{1,t} - 1)\sigma_{R,t}dZ_t.
$$

Diffusion and drift terms. The derivatives of $p(X,\epsilon)$ with respect to x_1 and $\overline{\alpha}_p$ are given by

$$
\frac{p_{x_1}(X,\epsilon)}{p^*} = (1-\psi^{-1})\frac{\gamma\|\sigma\|^2}{2y^*}\frac{1}{x_1^2}\hat{\alpha}_p^2\epsilon^2 + O(\epsilon^3), \qquad \frac{p_{\overline{\alpha}_p}(X,\epsilon)}{p^*} = -(1-\psi^{-1})\frac{\gamma\|\sigma\|^2}{y^*}\frac{x_0}{x_1}\hat{\alpha}_p\epsilon^2 + O(\epsilon^3)
$$

The diffusion term for $p(X,\epsilon)$ is then given by

$$
\sigma_{p,3}(X) = -(1-\psi^{-1})\frac{\gamma\|\sigma\|^2}{y^*}\frac{x_0}{x_1}\left[\frac{\hat{\alpha}_p^3}{2x_1^2}\sigma + \hat{\alpha}_p\sigma_p\sqrt{\overline{\alpha}_p}\right].
$$

and $\sigma_{c_j,3}(X) = \sigma_{p,3}(X)$. Notice that excess volatility depends on the market elasticity times the volatility of portfolio flows.

The drift of p is given by

$$\mu_{p,3}(X) = (1-\psi^{-1})\frac{\gamma\|\sigma\|^2}{2y^*}\frac{1}{x_1^2}\hat{\alpha}_p^2\mu_{x_1,1} - (1-\psi^{-1})\frac{\gamma\|\sigma\|^2}{y^*}\frac{x_0}{x_1}\hat{\alpha}_p\theta_p(\overline{\alpha}-\overline{\alpha}_{p,t}),$$

and $\mu_{c_j,3}(X) = \mu_{p,3}(X)$.

Risk premium. The risk premium is given by

$$\pi(X) = \gamma\|\sigma\|^2\left[1 - \frac{x_0(\overline{\alpha}_p - 1)}{x_1} - \frac{1-\gamma^{-1}}{\psi - 1}\frac{\gamma\|\sigma\|^2}{y^*}\frac{x_0}{x_1}\left[\frac{(1-\overline{\alpha}_p)^3}{2x_1^2}\sigma + \hat{\alpha}_p\sigma_p\sqrt{\overline{\alpha}_p}\right]\right]$$

G Higher-order perturbations

Suppose we have the $(n-1)$-th order perturbation of $c_j(X,\epsilon) = \sum_{k=0}^{n-1} c_{j,k}(X)\epsilon^k$ and the law of motion of X. Let $lc_j(X,\epsilon) = \sum_{k=0}^{n-1} lc_{j,k}(X)\epsilon^k$ denote the expansion of $\log c_j(X,\epsilon)$. Then we can compute $\sigma_j(X)$ up to order n:

$$\sigma_{c_j,n}(X) = \sum_{k=1}^{n-1} lc_{j,k,X}(X)\sigma_{X,n-k}(X),$$

which is independent of the n-th order term in $c_j(X,\epsilon)$ and $\sigma_X(X,\epsilon)$, as $lc_{j,0,X}(X) = 0$. Similarly, we can compute $\sigma_{y,n}(X)$. A similar argument gives $\mu_{p,n}(X)$ and $\mu_{c_j,n}(X)$. We can then compute $\pi_n(X)$ and $\alpha_{j,n}(X)$. The n-th term of the consumption-wealth ratio satisfies the condition

$$c_{j,t} = \psi\rho + (1-\psi)\left[\pi_t(\alpha_{j,t}-1) + \mu + \mu_{p,t} + \sigma\sigma_{p,t}^\top - \frac{\gamma_j}{2}\|\sigma_{R,t}\|^2\alpha_{j,t}^2 + \sum_{j=0}^{J} x_j c_{j,t}\right] + \xi_{j,t}$$

We can rewrite the system above in matrix form as follows:

$$[I - (1-\psi)\mathbf{1}_{J+1}x_t^\top]c_t = \zeta_t,$$

where $c_t = [c_{0,t},\ldots,c_{J,t}]^\top$, $x_t = [x_{0,t},\ldots,x_{J,t}]^\top$, $\mathbf{1}_{J+1}$ is a $(J+1)$-th dimensional vector of ones, and $\zeta_{j,t} \equiv \psi\rho + (1-\psi)\left[\pi_t(\alpha_{j,t}-1) + \mu + \mu_{p,t} + \sigma\sigma_{p,t}^\top - \frac{\gamma_j}{2}\|\sigma_{R,t}\|^2\alpha_{j,t}^2\right] + \xi_{j,t}$. Applying the Sherman-Morrison formula, we obtain

$$c_t = [I - (1-\psi^{-1})\mathbf{1}_{J+1}x_t^\top]\zeta_t,$$

or $c_{j,t} = \zeta_{j,t} - (1-\psi^{-1})x_t^\top\zeta_t$. Notice that ζ_t can be computed at order n based on the coefficients of order $n-1$ and their derivatives.

Computing the derivatives. The derivation above shows that, given the order $n-1$ expansion of $\zeta_j(X,\epsilon)$ and its derivatives, we can compute the expansion of order n. Suppose the expansion of $\zeta_j(X,\epsilon)$ is given by

$$\zeta_j(X,\epsilon) = \sum_{k=0}^{n-1} \zeta_{j,k}(X)\epsilon^k,$$

where $\zeta_{j,k}(X)$ takes the form:

$$\zeta_{j,k}(X) = A_{j,k} + B_{j,k}^\top(X-\overline{X}) + \frac{1}{2}(X-\overline{X})^\top C_{j,k}(X-\overline{X}),$$

where $\overline{X}$ is a reference point, $A_{j,k}$ is a scalar, $B_{j,k}$ is a vector, and $C_{j,k}$ is a matrix. Notice that . $\zeta_{j,k}(\overline{X}) = A_{j,k}$, $\zeta_{j,k,X}(\overline{X}) = B_{j,k}$ and $\zeta_{j,k,XX} = C_{j,k}$. Given this expansion, we can compute $c_j(X,\epsilon) = \sum_{k=0}^{n-1} c_{j,k}(X)\epsilon^k$.

G.1 Inner region

Consider the case of no preference heterogeneity and no leverage constraints. Consider the following change of variables: $x_1 = \epsilon\tilde{x}_1$. Define $\tilde{c}_j(\tilde{x}_1,\overline{\alpha}_p) = c_j\left(x_1,\overline{\alpha}_p\right)$, so $c_{j,x_1} = \frac{1}{\epsilon}\tilde{c}_{j,\tilde{x}_1}$ and $c_{j,x_1x_1} = \frac{1}{\epsilon^2}\tilde{c}_{j,\tilde{x}_1\tilde{x}_1}$. This implies the following is true:

$$\sigma_{c_j} = \frac{1}{\epsilon}\frac{\tilde{c}_{j,\tilde{x}_1}}{\tilde{c}_j}\sigma_{x_1},$$

where $\sigma_{x_1} = -\frac{1-\epsilon\tilde{x}_1}{\tilde{x}_1}\hat{\alpha}_p\sigma_R$. Similarly, we can write σ_y

$$\sigma_y = \frac{1}{\epsilon}\frac{\tilde{y}_{x_1}}{y}\sigma_{x_1}.$$

The drift of y is given by

$$\mu_y = \frac{1}{\epsilon}\frac{\tilde{y}_{\tilde{x}_1}}{y}\mu_{\tilde{x}_1} + \frac{1}{2\epsilon^2}\frac{\tilde{y}_{\tilde{x}_1\tilde{x}_1}}{y}\sigma^2_{\tilde{x}_1}.$$

The term of order 0 is the same as before.

H Volatility

Consider the case without preference heterogeneity or leverage constraints. The price-dividend ratio is given by:

$$\hat{p}_t = -(1-\psi^{-1})\frac{\gamma\|\sigma\|^2}{2y^*}\frac{(1-x_{a,t})(1-\overline{\alpha}_{p,t})^2}{x_{a,t}}.$$

where, given $\overline{\alpha}_{a,t} \equiv \frac{1-(1-x_{a,t})\overline{\alpha}_{p,t}}{x_{a,t}}$, we have

$$\sigma_{x_a,t} = x_{a,t}(\overline{\alpha}_{a,t}-1)\sigma, \qquad \sigma_{\overline{\alpha}_p,t} = \sigma_{\overline{\alpha}_p}$$

The diffusion of $\hat{p}_t$ is given

$$\sigma_{p,t} = \hat{p}_{x_a,t}\sigma_{x_a,t} + \hat{p}_{\overline{\alpha}_p,t}\sigma_{\overline{\alpha}_p}.$$

$$\hat{c}_t \;\equiv\; c_t - c^* \;=\; \chi_{0,t} \;+\; \chi_p\,\hat{p}_t \;+\; O(\epsilon^3),$$

with coefficients

$$\chi_p = (\psi-1)\,\frac{1}{p^*}, \qquad \chi_{0,t} = (\psi-1)\,\frac{\gamma\|\sigma\|^2}{2}\,\frac{f_t^2}{x_{0,t}(1-x_{0,t})}.$$

I Flow of Funds aggregation

This appendix describes the procedure used in Section 5.1 to aggregate Flow of Funds (FoF) sectors into the three ultimate sectors of the model: passive, constrained active, and unconstrained active. We follow the aggregation in Koijen et al. (2024), drop sectors with no or minimal equity holdings, and route pass-through vehicles to their ultimate end investors. ETFs (L.124) and closed-end funds (L.123) are aggregated onto the household balance sheet (L.101); mutual-fund holdings (L.122) are split across end investors using the FoF's breakdown, with the household portion kept separate. Defined-contribution pensions (L.118c, L.119c, L.120c) are routed to households. Hedge funds are separated out using Table B.101.f for domestic funds and the Enhanced Financial Accounts for foreign funds, and the corresponding holdings are subtracted from the household (L.101) and rest-of-the-world (L.133) balance sheets. The remaining insurance and pension sectors (L.115, L.116, L.118b, L.119b, L.120b) are aggregated into a single long-horizon sector. The three ultimate sectors correspond, respectively, to *passive* (households and non-profits net of hedge funds, government, monetary authority, and money-market funds), *constrained* (hedge funds, broker-dealers, and other short-horizon intermediaries), and *unconstrained* (insurance, pensions, and rest-of-the-world net of hedge funds).

J Numerical methodology

This appendix describes the numerical procedure used to solve the dynamic model in Section 3. We follow a two-step strategy where we first solve a 'simplified' model with an aggregate claim in positive net supply and a derivative security in zero net supply. In the second step (valuation step), we solve the valuation equation for the dividend and risky debt claims. This ordering is justified since, once the equilibrium is computed in the first step, the market prices of risk and interest rates are known functions of the simplified model, and the individual valuation ratios can be recovered from the respective pricing equations.

J.1 Simplified economy

The state variables in the simplified economy are

$$X = (x_c, x_u, \alpha_p) \in \mathcal{S}, \qquad x_p = 1 - x_c - x_u,$$

where x_c and x_u are the wealth shares of the constrained and unconstrained active investors, and α_p is the exogenous risky-share demand of the passive sector. The aggregate claim is in positive net supply and the derivative is in zero net supply. Denote the active portfolio exposures by $(\alpha_{j,Y}, \theta_j)$, where $\alpha_{j,Y}$ is the exposure to the aggregate claim and $\alpha_{j,F}$ is the exposure to the derivative. The passive investor has exposure $(\alpha_{p,t}, 0)$. We approximate

$$\xi_c(X), \quad \xi_u(X), \quad \xi_p(X), \quad \alpha_{c,Y}(X), \quad \alpha_{c,F}(X), \quad \alpha_{u,Y}(X), \quad \alpha_{u,F}(X)$$

with neural networks parameterized by Θ:

$$\begin{aligned}
\hat{\xi}_j &: \mathcal{S} \to \mathbb{R}_+, \qquad j \in \{c, u, p\}, \\
\hat{\alpha}_{j,Y} &: \mathcal{S} \to \mathbb{R}, \qquad j \in \{c, u\}, \\
\hat{\alpha}_{j,F} &: \mathcal{S} \to \mathbb{R}, \qquad j \in \{c, u\}.
\end{aligned}$$

and derive for all other equilibrium quantities as functions of the state variables and the approximated objects. Note that the passive investor does not trade the derivative, so $\alpha_{p,F}(X) \equiv 0$, while the passive exposure to the aggregate claim is the state variable itself, $\alpha_{p,Y}(X) = \alpha_p$. Thus the passive portfolio is imposed directly rather than approximated. We use fully connected feed-forward neural networks with 8 hidden layers and 10 neurons in each layer. Each neuron in the hidden and final layer is represented by a softplus activation function for the consumption-wealth ratios $\hat{\xi}_j$'s, and a linear function for the others. The state space is sampled globally using Latin-hypercube points on the simplex

$$\{(x_c, x_u, \alpha_p) : x_c > 0,\ x_u > 0, \alpha_p > 0,\ x_c + x_u < 1\}$$

Given a candidate collection of network outputs at X, the remaining equilibrium objects are recovered analytically.

First, goods-market clearing implies the aggregate claim dividend yield

$$\hat{y}(X) = x_c\hat{\xi}_c(X) + x_u\hat{\xi}_u(X) + x_p\hat{\xi}_p(X).$$

Second, the volatility of the dividend yield is recovered from the Itô consistency condition. Third, once $\sigma_y(X)$ is known, the excess-return volatilities of the aggregate claim and the derivative, denoted $(\sigma_{R,Y}(X), \sigma_{R,F}(X))$, are recovered. The unconstrained active investor's first-order conditions then pin down the two risk premia $\pi_Y(X), \pi_F(X)$. Fourth, conditional on $(\hat{y}, \pi_Y, \pi_F, \sigma_y)$, the risk-free rate $r(X)$, the state drift $\mu_X(X)$, the state diffusion matrix $\Sigma_X(X)$, the consumption-wealth ratio drifts $\mu_{c_j}(X)$, and the market price of risk vector $\eta(X)$ are all recovered from the equilibrium identities and Itô's lemma. These calculations are explicit once the neural-network outputs and their first and second derivatives are known. The problem reduces to minimizing the following loss functions

1. **HJB residuals.** For each investor type $j \in \{c, u, p\}$, we construct the residual from the investor's Hamilton–Jacobi–Bellman equation after replacing all equilibrium objects by either neural-network outputs or the derived quantities defined above. We denote these residuals by

$$\mathcal{L}_j^{\text{HJB}}(X; \Theta).$$

2. **Portfolio FOCs.** The constrained investor's first-order conditions for the aggregate claim and derivative lead to two residuals denoted as:

$$\mathcal{L}_{c,Y}^{\text{FOC}}(X; \Theta), \qquad \mathcal{L}_{c,F}^{\text{FOC}}(X; \Theta).$$

   Note that the unconstrained investor's first-order conditions are used to recover (π_Y, π_F) exactly and therefore do not appear separately in the loss.

3. **Market-clearing residuals.** We impose clearing in the aggregate-claim and derivative markets as residuals:

$$\mathcal{L}_Y^{\text{MC}}(X; \Theta), \qquad \mathcal{L}_F^{\text{MC}}(X; \Theta).$$

The resulting loss is

$$\mathcal{L}(X;\Theta) = \sum_{j\in\{c,u,p\}} \left(\mathcal{L}_j^{\mathrm{HJB}}(X;\Theta)\right)^2 + \left(\mathcal{L}_{c,Y}^{\mathrm{FOC}}(X;\Theta)\right)^2 + \left(\mathcal{L}_{c,F}^{\mathrm{FOC}}(X;\Theta)\right)^2 + \left(\mathcal{L}_{Y}^{\mathrm{MC}}(X;\Theta)\right)^2 + \left(\mathcal{L}_{F}^{\mathrm{MC}}(X;\Theta)\right)^2.$$

For a training batch $\{X^n\}_{n=1}^N$, the training objective is the empirical mean

$$\widehat{\mathcal{L}}(\Theta) = \frac{1}{N}\sum_{n=1}^{N} \mathcal{L}(X^n;\Theta). \tag{J.1}$$

The full algorithm is given in Algorithm (1).

Algorithm 1 First-stage training.

Initialize NNs $\{\hat{\xi}_c, \hat{\xi}_u, \hat{\xi}_p, \hat{\alpha}_{c,Y}, \hat{\alpha}_{c,F}, \hat{\alpha}_{u,Y}, \hat{\alpha}_{u,F}\}$ with parameters Θ.
for $t = 1, 2, \dots, T_{\max}$ **do**
    Sample a batch of aggregate states $\{X^n = (x_c^n, x_u^n, \alpha_p^n)\}_{n=1}^N$.
    Evaluate the seven network outputs at each sampled state.
    Recover the derived equilibrium objects state-by-state:

    (a) compute $\hat{y}(X^n)$ from goods-market clearing;

    (b) compute $\sigma_y(X^n)$ using Itô's lemma;

    (c) recover $(\sigma_{R,Y}(X^n), \sigma_{R,F}(X^n))$;

    (d) compute the hedging terms from the gradients of $(\hat{\xi}_c, \hat{\xi}_u, \hat{\xi}_p)$;

    (e) recover $(\pi_Y(X^n), \pi_F(X^n))$ from the unconstrained investor's first-order conditions;

    (f) get $(r(X^n), \mu_X(X^n), \Sigma_X(X^n), \mu_{\xi_j}(X^n))$ from equilibrium identities and Itô's lemma.

    Construct the residual blocks $\mathcal{L}_j^{\mathrm{HJB}}$, $\mathcal{L}_{c,Y}^{\mathrm{FOC}}$, $\mathcal{L}_{c,F}^{\mathrm{FOC}}$, $\mathcal{L}_Y^{\mathrm{MC}}$, and $\mathcal{L}_F^{\mathrm{MC}}$.
    Form the batch loss $\widehat{\mathcal{L}}(\Theta)$ in (J.1).
    Update Θ using ADAM, and update learning-rate scheduler using the current training loss.
end for

J.2 Valuation ratios

After solving the simplified economy, we price the dividend and risky-debt claims as a function of the state variables (X, s), where $s \in [0, 1]$ denote the dividend share of the aggregate claim. We solve for the valuation ratio of the dividend claim, denoted by $p_D(X, s)$, and the valuation ratio of risky-debt claim, denoted by $p_B(X, s)$. We approximate these functions by two additional neural networks parameterized by Θ_D and Θ_B, respectively:

$$\hat{p}_D : \mathcal{S} \times [0,1] \to \mathbb{R}_+,$$
$$\hat{p}_B : \mathcal{S} \times [0,1] \to \mathbb{R}_+,$$

At this stage the first-stage simplified equilibrium is held fixed. For each state (X, s), the first-stage solution supplies the coefficients entering the claim-pricing PDEs: the risk-free rate $r(X)$, the market price of risk $\eta(X)$, the state drift $\mu_X(X)$, the state diffusion matrix $\Sigma_X(X)$, and the aggregate-claim return volatility. The remaining coefficients associated with the s-state are known analytically from the decomposition of the aggregate claim into dividend and risky-debt components. Given a candidate valuation ratio $\hat{p}_k(X, s)$ for claim $k \in \{D, B\}$, we compute its first and second derivatives with respect to (X, s) using automatic differentiation, leading to the residual

$$\mathcal{L}_k^{\text{val}}(X, s; \Theta_k), \qquad k \in \{D, B\}.$$

The training objective is simply the empirical mean squared PDE residual,

$$\widehat{\mathcal{L}}_k^{\text{val}}(\Theta_k) = \frac{1}{N}\sum_{n=1}^{N}\left(\mathcal{L}_k^{\text{val}}(X^n, s^n; \Theta_k)\right)^2, \qquad k \in \{D, B\}.$$

The full algorithm is given in Algorithm (2).

Algorithm 2 Valuation-ratio step.

1: Load the trained first-stage networks.
2: Initialize the valuation-ratio network $\hat{p}_k$ for claim $k \in \{D, B\}$ with parameters Θ_k.
3: **for** $t = 1, 2, \ldots, T_{\max}^{\text{val}}$ **do**
4: Sample a batch of states $\{(X^n, s^n)\}_{n=1}^{N}$.
5: Evaluate $\hat{p}_k(X^n, s^n)$ and compute its first and second derivatives with respect to (X, s).
6: Recover the drift and volatility of the valuation ratio from Itô's lemma.
7: Use the fixed baseline solution to evaluate the coefficients entering the pricing PDE.
8: Construct the claim-pricing residual $\mathcal{L}_k^{\text{val}}(X^n, s^n; \Theta_k)$.
9: Compute batch loss $\widehat{\mathcal{L}}_k^{\text{val}}(\Theta_k)$.
10: Update Θ_k using ADAM step, update the learning-rate scheduler using current training loss.
11: **end for**

We follow Gopalakrishna and Wu (2026) and use vectorized automatic differentiation to compute the gradients that speeds up the computation. The total time to run the baseline model on a single-threaded computer with 128GB RAM and NVIDIA Ada 4500 GPU is 8 hours. The HJB loss in the first-stage training problem converges to a mean-squared loss of 3.7×10^{-8}. The corresponding valuation-ratio training problems converge to a normalized mean-squared training loss of 1.4×10^{-5}. These losses are computed from the batch objectives in (J.1) and in the valuation-ratio objective, respectively.